Global minimality of controlled-rail brachistochrones in conformal Kerr and Vaidya spacetimes

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A controlled-rail brachistochrone is the fastest trajectory that keeps a prescribed energy relative to a selector field. We study global minimality in conformal Kerr (Thakurta–Kerr) spacetimes with a time-dependent factor and in ingoing Vaidya, for arrival and proper time. On the arrival branch the rails are the null curves of an auxiliary Lorentzian metric, so first arrival is a Fermat problem: where this metric is globally hyperbolic, minimality among all three-dimensional competitors holds exactly up to the null cut point. For conformal Schwarzschild the cut point lies at swept angle π for any factor when $\hat{E}^2/A^2 \geq 3/2$; below $3/2$ this holds within an explicit radius, and the threshold is sharp. The rule extends to Vaidya with non-decreasing mass, below $3/2$ under radius and accretion-rate bounds. For conformal Kerr we prove a three-dimensional excursion barrier, a small-spin result, and equatorial results at every spin for constant data and in certified bands otherwise. On the proper-time branch minimality is lost not at a conjugate point but by escape toward $r = 2M$, or in Vaidya toward the return frontier. We compute the escape price, select constant-data minimizers by a convex-envelope rule, and prove selection near constant data. Under explicit gap conditions, a late drop of the factor destroys attainment, late accretion preserves the minimizer, and early accretion beyond the critical separation prevents it. Computer-assisted steps are explicit inequalities certified in exact or interval arithmetic.

I. INTRODUCTION

A controlled-rail brachistochrone is the fastest way for a massive body, held by a steering control at a prescribed value $\hat{E}$ of its energy relative to a selector field W, to reach a given place. In Paper I[1] the problem was set up as a time-optimal control problem on a non-stationary spacetime, with existence and normality of minimizers on compact regular domains (Theorems I.1 and I.2) and a Hamilton–Jacobi *verification* criterion for global optimality (Proposition I.3); it treated FLRW, where every clock is a function of conformal time and the brachistochrone is the spatial geodesic of the slice, and Vaidya. Paper II[2] specialized it to the conformal Kerr (Thakurta–Kerr) geometry $g = A(\eta)^2 g_{\mathrm{Kerr}}$ with $W = \partial_\eta$[3,4]: it located the control domain, which ends at $r = 2M$, classified the exterior extremals, and computed their adiabatic response to a slowly varying A.

What neither paper provides is a statement of *global* minimality that holds without constructing a value function by hand. Paper I's criterion certifies a candidate once a solution of the Hamilton–Jacobi equation saturated by it is exhibited; short-time certificates can be built explicitly, but they cover durations that are small compared with the orbital time scales of interest, and they do not by themselves compare a candidate with competitors that wind the other way, reverse radially, or leave the equatorial plane. This paper supplies such statements for the two costs that matter physically: the arrival time (conformal time η, equivalently coordinate time) and the proper time τ of the traveling body.

a. Two branches, two mechanisms. The two costs behave in structurally different ways, and the paper is organized around this difference.

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TABLE I. Main results. $q = \hat{E}^2/A^2$ is the effective energy ratio; for Vaidya $q = \hat{E}^2$.

Geometry, branch	Regime	Result
Conformal Schwarzschild, arrival	$q \geq 3/2$, any $A(\eta)$	cut at swept angle π (Theorem 2)
	$q < 3/2$	cut at π within $R(q_{\min}) > 8M$ (Theorem 3); sharp for constant A (Proposition V.8)
Vaidya, arrival	$\hat{E}^2 \geq 3/2$, $m' \geq 0$	cut at the antipode (Theorem 8)
	$\hat{E}^2 < 3/2$, $0 \leq m' < \mu_*$	cut at π within $\widehat{R}\, m_0$ (Theorem 9)
	$m' < 0$	conditional (Theorem 10)
Conformal Kerr, arrival	3D, fast arrivals	cut-point rule within a barrier (Theorem 11)
	3D, small spin	continuation of the $a = 0$ minimum (Theorem 12)
	equator, bands in q, $\lvert a\rvert \leq M$	one ray per winding class; two classes compete (Theorem 14)
Conformal Schwarzschild, proper time	constant A	escape beyond δ_c; Maxwell rule (Theorems 18 and 19)
	variable A	escape price (Theorem 21); selection near constant A (Theorem 22); late drops (Theorem 25); examples (section XI E)
Conformal Kerr, proper time	equator	ring and escape price; selection; example (section XII)
Vaidya, proper time	$m' \geq 0$	escape price (Theorem 35); late and early accretion, last contact (Theorems 36–38)

- On the *arrival branch* the rails are exactly the future null curves of an auxiliary Lorentzian metric

$$g_{\text{opt}} = -(\mathrm{d}\eta - \beta)^2 + N\, h_0, \qquad N = \Big(1 - \frac{A(\eta)^2 f}{\hat{E}^2}\Big)^{-1}, \tag{1}$$

  built from the threading data (f, β, h) of the seed and its optical metric $h_0 = h/f$ (section II). First arrival is a Fermat problem, and the global causal theory of Lorentzian manifolds applies as it stands: where g_{opt} is globally hyperbolic, first arrival is attained, and a null geodesic arrives first exactly up to its null cut point, the first conjugate point or the first point reached by a second null geodesic (Beem and Ehrlich[5]). Minimality is lost beyond the cut point.

- On the *proper-time branch* the cost is the integral in conformal time of a positive potential under a speed limit, and it is not a Fermat problem. The surface $r = 2M$, or in Vaidya the frontier from which the target can still be reached, lies at finite cost, and angular motion becomes cheap near it. Minimality is lost by *escape*: beyond a critical separation the cheapest way is to dive toward the boundary, turn there and climb out, and the infimum, the *escape price*, is not attained. Interior candidates are selected by comparing their cost with this price.

Both mechanisms rest on two elementary bounds: waiting above a radius costs at least the elapsed time times the potential there, and every radial passage costs at least a fixed weight per unit radius. Table I summarizes what is proved.

b. The arrival branch. For conformal Schwarzschild ($a = 0$) with $\sup A < \hat{E}$, so that the rail never freezes, g_{opt} is globally hyperbolic and first arrival is characterized by the cut-point rule in three dimensions, with no condition on A' or A'' (Theorem 1). The cut point lies at swept angle π whenever $q \geq 3/2$ on the whole future (Theorem 2). Two tools do the work. In a conformal representative with static slices the difference of the two Jacobi modes contains no time derivative, which excludes conjugate points before π; and the rays that reach a receiver are counted in the universal cover of a plane with the Morse theory of light rays of Giannoni, Masiello and Piccione[6] (Proposition III.11). Below $3/2$ the comparison fails only at large radii, and rays that matter never return from beyond $3M$:

the rule holds for launch and target radii below an explicit $R(q_{\min}) > 8M$ (Theorem 3). It genuinely fails beyond: to first order the Fermat geometry of a constant factor carries a mass with the sign of $2q-3$, whose repulsion below $3/2$ produces a radial caustic, so $3/2$ is sharp (Proposition V.8). For ingoing Vaidya with non-decreasing mass the arrival branch is again a Fermat problem, globally hyperbolic on the whole regular exterior (Theorem 5); for $\hat{E}^2 \geq 3/2$ the cut point is at the antipode (Theorem 8), and below $3/2$ the same radius argument works once the accretion rate is bounded, because accretion can turn an outgoing ray back (Theorem 9). For conformal Kerr the exterior $r > 2M$ is not globally hyperbolic for g_{opt}, but fast competitors cannot reach $r = 2M$: a radial excursion costs an explicit amount of conformal time, uniformly in latitude, and below that cost the cut-point rule holds in three dimensions (Theorem 11); for small spin the zero-spin minimum persists (Theorem 12). On the equatorial plane, in the regimes of Theorem 14 (constant data with $q \geq 5/3$ at every spin; variable factors with $q \geq 5/2$ for $|a| \leq 33M/50$ or in certified bands), each winding class contains one ray, which we prove with exact polynomial certificates, and a first variation in the angle of the receiver shows that only the two classes adjacent to the ray of zero angular momentum compete (Theorem 14).

c. The proper-time branch. For $a = 0$ and a constant factor the boundary $r = 2M$ is a single point of the metric completion, at finite cost. For equal radii there is a critical separation δ_c beyond which the infimum is the dive and is not attained (Theorem 16, Proposition X.5); below it, the minimizer is selected by the lower convex envelope of a function of the Clairaut constant, a Maxwell rule (Theorem 18), which can select two interior dips at once (Proposition X.8); for $q \geq 3/2$ the classification extends to unequal radii (Theorem 19). For a time-dependent factor every relaxed competitor has a rail of equal cost, every minimizer saturates its angular budget (Proposition XI.3), and when A tends to a positive limit minimizers exist below an explicit escape price: the cost of the optimal descent, which pays A along its own history, plus the ascent at the asymptotic value of A (Theorem 21). Near a constant factor the minimizer is unique and continues the frozen one (Theorem 22). For finite variations the question is decided by comparing the frozen minimum with an escape price: a sufficiently late drop of A leaves no minimizer under the cost-gap and transition-duration conditions of Theorem 25: waiting costs, so a candidate must be close to $r = 2M$ when the factor drops, with a small angular momentum, and the total angle it can then sweep is too small; explicit examples show uniqueness for a pulse and the three regimes of a drop (Theorems 23 and 24). On the equator with spin the boundary becomes a ring with a finite cost per radian, and the same program is carried out there (section XII). For ingoing Vaidya the velocity body is a translated ellipsoid and the escape price involves the frontier from which the target can still be reached (Theorem 35); near constant mass the minimizer continues the frozen one (Theorem 33), late accretion leaves the frozen minimizer in place (Theorem 36), early accretion beyond the frozen critical separation prevents a minimizer (Theorem 37), and in between there is a last contact with the escape price (Theorem 38). On fixed time windows the selected minimizers admit a uniform first-order adiabatic expansion, which connects with Papers I–II (section XI F).

d. Computer-assisted steps. Some statements concern specific transcendental quantities: the positivity of curvature polynomials on the equator of Kerr, the critical separation for given data, and the constants of the examples. In each case the text reduces the claim analytically to finitely many explicit inequalities, which are then certified in exact rational or outward-rounded interval arithmetic, with independent verifiers; the certificates and their descriptions are in the supplementary material and in the accompanying repository. Where the reduction is itself a complete proof, as for the closed forms of the Vaidya example, the certificate serves as an independent check. Floating-point computations reported in the text are labeled as diagnostics and are not used in any proof.

e. Verification methods and use of AI tools. Three AI models were used in this work: OpenAI GPT-6 Astra and GPT-6.1 (Sol), through OpenAI Codex, and Anthropic Claude Opus 5.5, through Claude Code. They assisted with symbolic-algebra checks, with writing the scripts that generate the certificates and the independent verifiers, with drafting and auditing derivations and proofs, with bibliographic verification against the primary sources,

and with language and structural editing. They were used because the proofs involve long case distinctions and many explicit constants, for which independent cross-checks reduce the risk of error; work produced through one interface was audited through the other. No result rests on AI output alone. The author checked the derivations used in proofs; symbolic identities and explicit inequalities were also cross-checked in MATHEMATICA. The 33 portable curvature certificates are checked by an independent verifier that imports no code from their generators. The remaining computer-assisted bounds are verified by the exact-arithmetic and interval procedures documented in the supplementary material.

f. What is not covered. Conformal Kerr in three dimensions at finite spin beyond the barrier, which leads to a state-constrained problem at $r = 2M$; on the equator, a time-dependent factor with $2 \le q < 5/2$ or $|a| > \sqrt{2451}/50\,M$, and the physical reading of the rail inside the ergoregion; the regime outside the uniform bound $\sup A < \hat{E}$, including moving freezing fronts when they occur (Remark IV.2); on the proper-time branch, the identification of the minimizer with a specific extremal between the regimes covered by the theorems; for Vaidya, decreasing masses outside the stated bounds and the necessity of the accretion-rate bound below 3/2. The weak-field regime of a star that loses mass, relevant for the solar system, is a different problem, with the stellar surface in place of the surface $r = 2M$. The Kodama selector is required to remain timelike throughout.

g. Relation to earlier work. Fermat's principle for arbitrary, including non-stationary, metrics was formulated by Kovner[7] and proved in an arbitrary Lorentzian manifold by Perlick[8]; the travel-time and arrival-time brachistochrones of stationary spacetimes were treated variationally by Perlick[9], who also reduces the proper-time problem with prescribed energy to a Riemannian metric (his Proposition 3.1) in a stationary spacetime. Giannoni and Piccione[10] formulate an arrival-time variational problem for a general observer field and establish variational principles, existence and multiplicity results in the stationary case. Giannoni, Piccione and Verderesi[11] gave a sub-Riemannian formulation whose admissible region $|Y| < k$ is the analogue of our control domain, and Caponio, Corona, Giambò and Piccione[12] treat fixed-energy solutions of a Lagrangian with an infinitesimal symmetry joining a point to a flow line of that symmetry. The Morse theory of light rays from an event to a timelike curve, in the form we use for stably causal spacetimes, is that of Giannoni, Masiello and Piccione[6]; Hasse and Perlick[13] applied Morse theory to lensing by a Kerr–Newman black hole, and their spin bound $m/\sqrt{2}$, at which the photon region reaches the ergosphere, is the limit $q \to \infty$ of the threshold $a^*(q)$ of section IX. Earlier, Goldstein and Bender[14] posed the special-relativistic brachistochrone in uniform fields, distinguishing laboratory and proper time, and Kamath[15] extended it to the weak static field of a spherical source; Wang and Jia[16] treat a traveler at constant local speed in static spherically symmetric spacetimes, a different constraint from the maintained energy of the rail. Taş[17] builds ruled timelike surfaces from arrival-time extremals; in the static Schwarzschild exterior his reduction is our arrival-time optical metric at frozen factor, and he leaves rotating and non-stationary backgrounds open. At fixed η the rail arrival integrand is a Randers[18] deformation of the null Fermat metric of Kerr[19]: its quadratic part is Nh_0 and its one-form is β. The stationary null correspondence is described by Gibbons, Herdeiro, Warnick and Werner[20], and the associated Randers–Zermelo correspondence by Bao, Robles and Shen[21]; Paper II found the rail integrand at the level of the cost functional. Beyond the boundary of the control domain the geometry continues as a wind Finslerian structure in the sense of Caponio, Javaloyes and Sánchez[22], while the control problem ends there. What is new here is that the rail, including its non-stationarity through $A(\eta)$, is itself the null cone of a Lorentzian metric in one more dimension, so that the non-stationary arrival problem needs no separate treatment; and that on the proper-time branch, where no such reduction exists, the escape price organizes the global theory.

h. Plan. Section II derives g_{opt} and identifies its null geodesics with the extremals of Papers I and II; section III collects the causal tools, proves an abstract first-arrival theorem and counts rays in the universal cover of a plane. Sections IV and V treat conformal Schwarzschild, section VI persistence near constant data, section VII the Vaidya arrival branch, sections VIII and IX conformal Kerr in three dimensions and on the equator. The

proper-time branch follows: constant factor (section X), time-dependent factor (section XI), equator with spin (section XII) and ingoing Vaidya (section XIII); section XIV concludes with the open problems. The supplementary material contains an alternative proof by convex fronts, the computer-assisted steps and the constants of the examples.

II. THE RAIL AS THE NULL CONE OF AN AUXILIARY METRIC

A. Setting

Let g_{Kerr} be the Kerr metric of mass $M > 0$ and spin a, $|a| \le M$, in Boyer–Lindquist coordinates, with $\Sigma = r^2 + a^2\cos^2\vartheta$ and $\Delta = r^2 - 2Mr + a^2$. Write it in threading form with respect to the stationary field,

$$g_{\mathrm{Kerr}} = -f\,(\mathrm{d}t - \beta)^2 + h, \qquad f = 1 - \frac{2Mr}{\Sigma}, \qquad \beta = \frac{g_{t\varphi}}{f}\,\mathrm{d}\varphi = -\frac{2Mar\sin^2\vartheta}{\Sigma f}\,\mathrm{d}\varphi, \tag{2}$$

$$h = \frac{\Sigma}{\Delta}\,\mathrm{d}r^2 + \Sigma\,\mathrm{d}\vartheta^2 + \frac{\Delta\sin^2\vartheta}{f}\,\mathrm{d}\varphi^2, \tag{3}$$

valid where $f > 0$, i.e. outside the ergosurface. The value of $h_{\varphi\varphi}$ follows from the identity $f g_{\varphi\varphi} + g_{t\varphi}^2 = \Delta\sin^2\vartheta$. The Thakurta–Kerr metric is

$$g = A(\eta)^2\, g_{\mathrm{Kerr}}\big|_{t=\eta}, \tag{4}$$

with A a smooth positive function on an open interval $I \ni \eta$. The selector is $W = \partial_\eta$. A *rail curve* of charge $\hat{E} > 0$ is a future-directed timelike curve with unit tangent u, $g(u,u) = -1$, on which

$$-g(u, W) = \hat{E}. \tag{5}$$

This is the rail of Paper II, $\hat{E} = -u_\eta$.

Define

$$Z = \frac{A(\eta)^2 f}{\hat{E}^2}, \qquad N = \frac{1}{1 - Z}, \qquad h_0 = \frac{h}{f}, \tag{6}$$

and the *regular domain*

$$D = \{(\eta, x) :\ \eta \in I,\ f(x) > 0,\ Z(\eta, x) < 1\}. \tag{7}$$

On D one has $N \ge 1$; the surface $Z = 1$ is the freezing surface of Paper II, where the rail admits no motion. The metric h_0 is the optical metric of the seed; at the equator

$$h_0\big|_{\vartheta=\pi/2} = \frac{r^2}{f\Delta}\,\mathrm{d}r^2 + \frac{\Delta}{f^2}\,\mathrm{d}\varphi^2, \qquad \beta\big|_{\vartheta=\pi/2} = -\frac{2Ma}{rf}\,\mathrm{d}\varphi, \tag{8}$$

which are exactly the data α_K, β_K of Paper II, equation (randers-data).

B. The auxiliary metric

Definition II.1. On D the *auxiliary metric* is

$$g_{\mathrm{opt}} = -(\mathrm{d}\eta - \beta)^2 + N(\eta, x)\, h_0, \tag{9}$$

time-oriented by ∂_η.

Lemma II.2. *On D:*

(1) $|\beta|_{h_0} < 1$; more precisely

$$1 - |\beta|^2_{h_0} = \frac{f g_{\varphi\varphi}}{\Delta \sin^2\vartheta}, \qquad |\beta|_{h_0} = \frac{2M|a|r\sin\vartheta}{\Sigma\sqrt{\Delta}};$$

(2) g_{opt} is Lorentzian, ∂_η is unit timelike and η is a time function: $g^{-1}_{\mathrm{opt}}(\mathrm{d}\eta, \mathrm{d}\eta) = -1 + |\beta|^2_{h_0}/N < 0$;

(3) a vector $v = v^\eta \partial_\eta + v_x$ is future-directed null if and only if

$$v^\eta = \beta(v_x) + \sqrt{N}\,|v_x|_{h_0}, \qquad v_x \neq 0, \tag{10}$$

and future-directed causal if and only if $v^\eta \geq \beta(v_x) + \sqrt{N}|v_x|_{h_0}$, $v \neq 0$.

Proof. (1) $|\beta|^2_{h_0} = \beta^2_\varphi / h_{0,\varphi\varphi} = g^2_{t\varphi}/(f g_{\varphi\varphi} + g^2_{t\varphi})$ by (2)–(3); the identity quoted after (3) gives both formulas, and $f, g_{\varphi\varphi} > 0$ where $f > 0$ off the axis. (2) The metric is $-(\mathrm{d}\eta - \beta)^2$ plus a positive-definite form in x, so it has one negative direction; the inverse component follows by block inversion, and it is negative since $N \geq 1 > |\beta|^2_{h_0}$. (3) $g_{\mathrm{opt}}(v,v) = -(v^\eta - \beta(v_x))^2 + N|v_x|^2_{h_0}$, and future-directed means $-g_{\mathrm{opt}}(v, \partial_\eta) = v^\eta - \beta(v_x) > 0$. If $v_x = 0$ the vector is a positive multiple of ∂_η, which is timelike. □

Proposition II.3 (the rail is the null cone of g_{opt})**.** *On D the following hold.*

(1) The tangent u of a rail curve of charge $\hat{E}$ is a future-directed null vector of g_{opt}.

(2) Conversely, every future-directed null vector v of g_{opt} has a unique positive multiple λv that is g-unit and satisfies (5)*.*

Hence the rail curves of charge $\hat{E}$, as unparametrized curves, are exactly the future-directed null curves of g_{opt} in D. Reparametrized by η, their elapsed conformal time is

$$\mathrm{d}\eta = \sqrt{N}\,|\mathrm{d}x|_{h_0} + \beta(\mathrm{d}x), \tag{11}$$

which at the equator is equation (randers) of Paper II, with index $n = \sqrt{N}$.

Proof. Write $w = u^\eta - \beta(u)$. By (2) and (4), $g(u, W) = -A^2 f w$ and $g(u,u) = A^2\big(-f w^2 + h(u,u)\big)$.

(1) The rail gives $w = \hat{E}/(A^2 f) > 0$, and $g(u,u) = -1$ then gives $h(u,u) = f w^2 - A^{-2}$. Hence

$$\frac{h(u,u)}{f\,w^2} = 1 - \frac{A^2 f}{\hat{E}^2} = 1 - Z = \frac{1}{N},$$

that is $w^2 = N\,h_0(u,u)$ with $w > 0$: u is future-directed null for g_{opt}.

(2) For v future-directed null, $w_v := v^\eta - \beta(v) = \sqrt{N}|v_x|_{h_0} > 0$. The choice $\lambda = \hat{E}/(A^2 f w_v)$ enforces (5), and then

$$g(\lambda v, \lambda v) = A^2\lambda^2 f\big(-w_v^2 + h_0(v,v)\big) = -A^2\lambda^2 f\, w_v^2\, Z = -\frac{Z\hat{E}^2}{A^2 f} = -1.$$

Uniqueness is clear because (5) is linear in λ. Equation (11) is (10) written for $\mathrm{d}x/\mathrm{d}\eta$, and $N = n^2$ with $n = \hat{E}/\sqrt{\hat{E}^2 - A^2 f}$ by (6). □

Remark II.4 (admissible classes)**.** Paper I works with absolutely continuous trajectories whose velocity lies a.e. in the indicatrix $\mathcal{I}_x$ of the rail, and, for existence, with the relaxed class in which it lies in the convex body $\mathcal{K}_x$ bounded by $\mathcal{I}_x$ (Paper II, remark on the indicatrix). Through (10) these are, respectively, the locally Lipschitz future-directed curves of g_{opt} that are null almost everywhere, and the future-directed causal curves. The first-arrival problems over the two classes have the same value and the same minimizers (Proposition III.6 below): the minimizer is a null geodesic in both. This is the Lorentzian form of the remark in Paper II that an arrival-time optimum never uses an interior velocity.

C. Extremals are null geodesics

Proposition II.5. *The normal Pontryagin extremals*[23] *(section 4.1) of the arrival-time problem of Papers I and II are, up to reparametrization, the null geodesics of g_{opt} in D.*

Proof. Extend the state by η. With costate (p_η, p) and the cost normalized as in Paper II, the extended Hamiltonian is $\mathcal{H} = p_\eta + H_\eta(\eta, x, p)$, where $H_\eta + 1 = \max_{v\in\mathcal{I}_x}\langle p, v\rangle =: \sigma(\eta, x, p)$ is the support function of the indicatrix, and extremals lie on $\mathcal{H} = 0$. By Lemma II.2(3) the indicatrix is the section $v^\eta = 1$ of the future null cone of g_{opt}, so the spacetime covector $P = (P_\eta, p) = (p_\eta - 1, p)$ satisfies $P_\eta = -\sigma(p)$ on $\mathcal{H} = 0$: its kernel is a hyperplane supporting the cone along a null generator. A covector supports the null cone of a Lorentzian metric along a generator if and only if it is null, so, on the non-zero covectors, the level set $\{\mathcal{H} = 0\}$ coincides with the future-null part of $\{\mathcal{G} = 0\}$, $\mathcal{G} = \frac{1}{2}g_{\mathrm{opt}}^{-1}(P, P)$. On a common regular level set the Hamiltonian vector fields of $\mathcal{H}$ and $\mathcal{G}$ are parallel, since $\mathrm{d}\mathcal{G} = \mu\,\mathrm{d}\mathcal{H}$ there for some function $\mu \neq 0$; their integral curves therefore coincide up to reparametrization, and those of $\mathcal{G}$ on $\{\mathcal{G} = 0\}$ are the null geodesics.

For the equatorial problem this can be checked against Paper II directly: with H_η of Paper II, equation (Heta), the covector $(-(H_\eta + 1), p_r, p_\varphi)$ is null for g_{opt} identically (`auxiliary_metric_null_cone.py` in the accompanying repository). □

Remark II.6 (which clocks are covered)**.** Since $\mathrm{d}t = A\,\mathrm{d}\eta$ with $A > 0$, first arrival in t and in η are the same problem (Paper II), and everything below applies to both. The proper-time branch minimizes $\int A^2 f(\mathrm{d}\eta - \beta(\mathrm{d}x))/\hat{E}$, a different cost. The arrival-time Fermat certificates do not apply to it; its boundary behavior is studied separately in section X.

III. CAUSAL TOOLS AND AN ABSTRACT FIRST-ARRIVAL THEOREM

Throughout this section (D, g) is a time-oriented Lorentzian manifold of the form $D = \{(\eta, x)\}$ in which η is a time function and every line $L_x = \{(\eta, x) : \eta \in I_x\}$ is timelike and future-directed in increasing η. For $o \in D$, $I^+(o)$ and $J^+(o)$ denote its chronological and causal futures and $E^+(o) = J^+(o) \setminus I^+(o)$ its future horismos. We use the standard conventions of Minguzzi's review[24].

Definition III.1. Let $o = (\eta_0, x_0) \in D$ and let x_f be a spatial point. The *first-arrival time* at x_f is $\eta_*(o, x_f) = \inf\{\eta : (\eta, x_f) \in J^+(o)\}$. A *first-arrival curve* is a future-directed causal curve from o to (η_*, x_f).

The facts we use are the following.

Fact III.2 (Choquet-Bruhat and Cotsakis[25], Theorem 2.1)**.** *Let $D = M \times I$ with $g = -\mathcal{N}^2\mathrm{d}\eta^2 + g_{ij}(\mathrm{d}x^i + \beta^i\mathrm{d}\eta)(\mathrm{d}x^j + \beta^j\mathrm{d}\eta)$. Suppose that the lapse is bounded, $0 < \mathcal{N}_m \le \mathcal{N} \le \mathcal{N}_M$; that the slice metrics g_η are complete and uniformly bounded below by one of them, $g_\eta \ge c\,g_{\eta_1}$ for all η, with $c > 0$; and that the g_η-norm of the shift is bounded. Then every slice $\{\eta = \mathrm{const}\}$ is a Cauchy surface and (D, g) is globally hyperbolic. The authors state it for $I = (t_0, \infty)$ and note that it applies to any interval.*

Fact III.3 (Minguzzi[24])**.** *In a globally hyperbolic spacetime:*

(1) $J^+(o)$ is closed (global hyperbolicity implies causal simplicity, Proposition 4.118);

(2) any two causally related points are joined by a maximizing causal geodesic (Avez–Seifert, Theorem 4.123).

In any spacetime, $J^+ \circ I^+ \cup I^+ \circ J^+ \subset I^+$ (push-up, Theorem 2.24), and every causal curve from o to a point of $E^+(o)$ is an achronal null pregeodesic (Theorem 2.28).

Fact III.4 (Beem and Ehrlich[5,26])**.** *Let $\gamma : [0, b) \to D$ be a future-directed null geodesic with $\gamma(0) = o$, and let $t_0 = \sup\{t : \gamma(t) \notin I^+(o)\}$; if $t_0 < b$, $\gamma(t_0)$ is the* future null cut point *of γ. Then:*

(1) $\gamma(t) \notin I^+(o)$ *for* $t \le t_0$ *and* $\gamma(t) \in I^+(o)$ *for* $t > t_0$ *([5], section 5);*

(2) the cut point is not later than the first conjugate point ([26], section 1);

(3) if two distinct null geodesic segments join o *to* q*, then* q *is at or beyond the cut point of each ([5], Lemma 5.1);*

(4) if (D, g) *is globally hyperbolic,* $\gamma(t_0)$ *is the first conjugate point of* o *along* γ*, or it is joined to* o *by at least two maximal null geodesics, or both ([5], Theorem 5.3).*

Lemma III.5 (first arrival is the horismos)**.** *Let* γ *be a future-directed causal curve from* o *to* $q = (\eta_q, x_f)$*. Then* q *is a first arrival at* x_f*, i.e.* $\eta_q = \eta_*(o, x_f)$ *and no causal curve from* o *reaches* x_f *earlier, if and only if* $q \notin I^+(o)$*.*

Proof. If $q \in I^+(o)$, then, since $I^+(o)$ is open, $(\eta_q - \epsilon, x_f) \in I^+(o)$ for small $\epsilon > 0$, an earlier arrival. Conversely, if some $q' = (\eta', x_f) \in J^+(o)$ has $\eta' < \eta_q$, the segment of L_{x_f} from q' to q is future-directed timelike, so $q \in I^+(o)$ by push-up. □

Proposition III.6 (abstract first-arrival theorem)**.** *Let* (D, g) *be globally hyperbolic, with* η *and the lines* L_x *as above, and let* $o \in D$ *and* x_f*, different from the spatial point of* o*, be such that* $L_{x_f} \cap J^+(o) \neq \emptyset$*. Then:*

(1) the first-arrival time η_* *is attained, and* $q_* = (\eta_*, x_f) \in E^+(o)$*;*

(2) every first-arrival curve is a null pregeodesic, and it has no point conjugate to o *in its interior;*

(3) a null geodesic γ *from* o *is a first-arrival curve to the spatial point of* $\gamma(t)$ *if and only if* $t \le t_0$*, where* $\gamma(t_0)$ *is its null cut point (*$t_0 = +\infty$ *if* γ *has no cut point in* D*); the cut point is the first conjugate point or the first point reached by a second maximal null geodesic from* o*;*

(4) for $t < t_0$ *the first-arrival curve to* $\gamma(t)$ *is unique up to reparametrization.*

Proof. (1) Fix a known arrival $q \in L_{x_f} \cap J^+(o)$. The arrivals of a minimizing sequence on L_{x_f} before q lie in the causal diamond $J^+(o) \cap J^-(q)$, which is compact by global hyperbolicity; a subsequence converges to a point of the manifold, on L_{x_f} and in $J^+(o)$ (which is closed, Fact III.3(1)), and it realizes the infimum. This does not need the line L_{x_f} to be defined for all η. Call it q_*; then $q_* \notin I^+(o)$ by Lemma III.5.

(2) By Fact III.3(2) a causal geodesic joins o to q_*, and by the last part of Fact III.3 every causal curve from o to $q_* \in E^+(o)$ is an achronal null pregeodesic. Let γ be one, with $\gamma(t_q) = q_*$. No point of γ before q_* lies in $I^+(o)$, otherwise $q_* \in I^+(o)$ by push-up; hence $t_q \le t_0$, and by Fact III.4(2) every interior point $t < t_q$ precedes the first conjugate point.

(3) By Lemma III.5, $\gamma(t)$ is a first arrival iff $\gamma(t) \notin I^+(o)$, which by Fact III.4(1) holds iff $t \le t_0$. The description of the cut point is Fact III.4(4).

(4) If a second first-arrival curve reached $\gamma(t)$, $t < t_0$, it would be a null pregeodesic by (2), distinct from γ, and Fact III.4(3) would place $\gamma(t)$ at or beyond the cut point of γ. □

Remark III.7 (regularity)**.** The facts above are stated for smooth metrics. With $A \in C^\infty$ the metric g_{opt} is smooth on D. Existence of minimizers (Paper I) needs less; we have not tried to lower the regularity of the cut-point statements.

Remark III.8 (what the characterization says in the rail language)**.** By Propositions II.3 and II.5, Proposition III.6 says that the first arrival at x_f is attained by a rail extremal, that an extremal launched from o is globally minimal exactly as long as it has not passed its cut point, and that it ceases to be so for one of two reasons: a conjugate point, detected by the magnetic Jacobi equation of Paper II, or a *Maxwell point*, where another extremal from o, possibly winding the other way or leaving the plane, arrives at the same event. The competitors are all causal curves: every direction, every winding number, every radial reversal, and, in three dimensions, every out-of-plane motion.

A. Counting rays in the universal cover of a plane

In the spherically symmetric problems of this paper the extremals are planar, and a first arrival is identified by counting the null geodesics that reach a receiver in the universal cover of a plane. We count them once, with the Morse theory of light rays of Giannoni, Masiello and Piccione[6].

a. Setting (P). Let $J = (t_-, \infty)$ be an open interval, $b : J \to [0, \infty)$ continuous, X either $\mathbb{R}$ or the unit sphere, and $\mathcal{D} = \{(t, r, x) : \ t \in J, \ r > b(t), \ x \in X\}$ with a smooth Lorentzian metric for which t is a time function. Write future causal vectors as positive multiples of $(1, s, \zeta)$, $\zeta \in T_xX$, or as vectors with $\mathrm{d}t = 0$, which vanish. Assume smooth functions $s_- < 0 < s_+ \leq 1$ and $a > 0$ of (t, r) such that

(i) every future causal vector $(1, s, \zeta)$ has $s_- \leq s \leq s_+$, and $s = s_\pm$ only if $\zeta = 0$;

(ii) $(1, s_\pm, 0)$ is null, and, for $X = \mathbb{R}$, so is $(1, 0, \pm a)$.

The outgoing and ingoing radial null curves are the solutions of $\mathrm{d}r/\mathrm{d}t = s_\pm(t, r)$, and $(1, 0, 0)$ is timelike.

Lemma III.9 (outermost ray)**.** *Assume (P). Let $z : [0, 1] \to \mathcal{D}$ be an absolutely continuous curve from an event e whose velocity is future causal or zero almost everywhere, and let ρ_+ be the outgoing radial null curve from e. Then ρ_+ is defined in $\mathcal{D}$ at least up to $t(z(1))$, and $r(z(s)) \leq \rho_+(t(z(s)))$ for every s. If equality holds at $s = 1$, then z is the radial curve ρ_+ up to reparametrization and rests. The same holds for the ingoing curve ρ_- with the reverse inequality, as long as ρ_- stays in $\mathcal{D}$.*

Proof. Along z, t is non-decreasing and $\dot r \leq s_+ \dot t$ almost everywhere. Let $[t_e, t_{\max})$ be the maximal interval of ρ_+ in $\mathcal{D}$ and s_1 with $t(z(s_1)) < t_{\max}$. For $s \in [0, s_1]$ the radial segments between $z(s)$ and ρ_+ lie in $\mathcal{D}$, since both end points do, and form a compact set on which s_+ is Lipschitz in r with a constant Λ. Put $w = \rho_+ \circ t \circ z - r \circ z$ and $d = s_+\dot t - \dot r \geq 0$. The function $y = (-w)_+$ has $\dot y \leq \Lambda \dot t\, y$ almost everywhere and $y(0) = 0$, so $y \equiv 0$ by Grönwall and $w \geq 0$. Then $\dot w \geq -\Lambda \dot t\, w + d$, so $(e^{\Lambda t \circ z} w)^{\cdot} \geq e^{\Lambda t \circ z} d \geq 0$ and $w(s) \geq e^{-\Lambda t(z(s))} \int_0^s e^{\Lambda t \circ z} d$. If $t_{\max} \leq t(z(1))$, then as $t \to t_{\max}$ the curve ρ_+ stays above the compact image of z, at a positive distance from the boundary, and below $r_e + t_{\max} - t_e$ since $s_+ \leq 1$; it therefore extends, a contradiction. Equality at $s = 1$ forces $d = 0$ almost everywhere: where $\dot t > 0$ the velocity is radial by (i), and where $\dot t = 0$ it vanishes. The ingoing case is the same with the signs reversed. □

Corollary III.10 (radial extremals)**.** *Assume (P) with X the sphere. The outgoing radial null geodesic from o is, at each of its events, the unique earliest arrival at the spatial point of that event, up to reparametrization, among all future causal curves from o; so is the ingoing one, while it stays in $\mathcal{D}$. Radial extremals have no cut point.*

Proof. Let a causal curve from o reach the spatial point of $\rho_+(t_1)$ at time $t' \leq t_1$. Since $s_+ > 0$, ρ_+ increases, and Lemma III.9 gives $r_1 \leq \rho_+(t') \leq \rho_+(t_1) = r_1$: so $t' = t_1$, and the equality case makes the curve radial. The ingoing case is symmetric. □

Proposition III.11 (one ray per short class)**.** *Assume (P) with $X = \mathbb{R}$, $\mathcal{D}$ globally hyperbolic, $o = (t_0, r_0, 0)$, $\delta \neq 0$, and the receiver $\gamma(t) = (t, r_T, \delta)$, $t \in J$, with the uniform margin*

$$\sup_J b < r_T, \qquad \inf_{t \geq t_0} a(t, r_T) > 0, \qquad \sup_{t \geq t_0, \ r \geq r_T} s_-(t, r) < 0.$$

If every future null geodesic of $\mathcal{D}$ from o to γ has index zero and an end point not conjugate to o, there is at most one such geodesic up to reparametrization, and exactly one if some future causal curve from o meets γ.

Proof. The hypotheses of[6] *(Theorem 1.7).* $\mathcal{D}$ is globally hyperbolic, hence stably causal; we take the open set $\Lambda = \mathcal{D}$, which has no boundary, so the conditions on the boundary are void. The receiver is timelike, it is a closed embedding after an increasing reparametrization by $\mathbb{R}$, since it leaves every compact set as $t \to t_-$ or $t \to \infty$, and o is not on it because $\delta \neq 0$. Since the reparametrization is increasing, it suffices to bound sublevels of the arrival coordinate t: the curves arriving at $t \leq c$ lie in the compact diamond $J^+(o) \cap J^-(\gamma(c))$, which gives the c-precompactness of[6] (Definition 1.2). Non-conjugacy is assumed. Let $\mathcal{L}^+$ be the set of H^1 curves $z : [0,1] \to \mathcal{D}$ from o to γ whose velocity is null and future-pointing, or zero, almost everywhere[6] ((1.4)); we show that it is contractible when it is not empty.

The tail. Let e be an event on a curve of $\mathcal{L}^+$, so $t_e \geq t_0$. If $r_e \leq r_T$, follow the outgoing radial null curve from e to its first crossing of $r = r_T$, which exists by Lemma III.9 applied to the rest of that curve. If $r_e \geq r_T$, follow the ingoing one; it stays in $r \geq r_T > \sup b$ and reaches r_T in finite time. Both crossings are transversal. Then follow the null curve at $r = r_T$ with $\mathrm{d}\varphi/\mathrm{d}t = \pm a$, the sign pointing toward δ, until $\varphi = \delta$, which happens in finite time because $a(t, r_T) \geq a_* > 0$. Call $T(e)$ this curve, parametrized on $[0,1]$ affinely in t over its total duration $\Delta(e)$; it is constant if $e \in \gamma$.

Continuity of T in H^1. If $\Delta(e_0) > 0$, the pieces depend continuously on e, the junction times are continuous, and the velocities are locally bounded; a piece that disappears in the limit has measure tending to zero, so $T(e) \to T(e_0)$ uniformly with derivatives in L^2. If $e_0 \in \gamma$, then near e_0 one has $\Delta(e) \leq C(|r_e - r_T| + |\varphi_e - \delta|)$ and $|\mathrm{d}T/\mathrm{d}t| \leq K$, so the energy of $T(e)$ is at most $K^2\Delta(e)^2 \to 0$, whatever the ratio of the durations of the pieces.

The contraction. Set $H(s,z) = z$ on $[0,s]$ and $H(s,z) = T(z(s))$, rescaled to $[s,1]$, on $[s,1]$, with $H(1,z) = z$. Each $H(s,z)$ lies in $\mathcal{L}^+$, $H(0,z) = T(o)$ and $H(1,\cdot)$ is the identity. Continuity for $s < 1$ follows from the continuity of evaluation, concatenation and rescaling in H^1. Let $s_n \to 1$ and $z_n \to z$ in H^1; then $z_n(s_n) \to z(1) \in \gamma$ and, by the bounds above and Cauchy–Schwarz, the energy of the rescaled tail is at most

$$\frac{K^2\Delta(z_n(s_n))^2}{1-s_n} \leq C\int_{s_n}^1 |\dot{z}_n|^2 \leq 2C\|\dot{z}_n - \dot{z}\|_{L^2}^2 + 2C\int_{s_n}^1 |\dot{z}|^2 \to 0.$$

Hence $H(s_n, z_n) \to z$ in H^1, H is continuous on $[0,1] \times \mathcal{L}^+$, and it contracts $\mathcal{L}^+$.

Counting. If $\mathcal{L}^+ \neq \emptyset$, by[6] (Theorem 1.7), for any field the number N of geodesics satisfies

$$N = 1 + (1+\xi)S(\xi), \qquad S(\xi) = \sum_k s_k \xi^k,\ s_k \in \mathbb{N} \cup \{+\infty\},$$

since all indices vanish and the Poincaré series of a contractible space is 1. The coefficient of ξ gives $s_0 + s_1 = 0$, so $s_0 = 0$, and the constant term gives $N = 1$. If a causal curve from o meets γ, Proposition III.6(1)–(2) applied to the receiver line gives a null pregeodesic from o to γ, so $\mathcal{L}^+ \neq \emptyset$. □

The sign of the time derivative of the metric is not used. The uniform margin on a is: in a Vaidya spacetime whose mass increases to $r_T/2$, the angular speed on the receiver can be integrable, and the tail at $r = r_T$ then does not reach the angle δ.

IV. CONFORMAL SCHWARZSCHILD

For $a = 0$ one has $\beta = 0$, $f = 1 - 2M/r$, and

$$g_{\mathrm{opt}} = -\mathrm{d}\eta^2 + N(\eta, r)\, h_0, \qquad h_0 = \frac{\mathrm{d}r^2}{f^2} + \frac{r^2}{f}\,\mathrm{d}\Omega^2, \tag{12}$$

where h_0 is the optical metric of Schwarzschild on $\Omega = \{r > 2M\} \times S^2$.

a. Hypothesis (H1). $a = 0$, $M > 0$, $\hat{E} > 0$, $A \in C^\infty(I)$ positive on an open interval I, and

$$\sup_I A < \hat{E}. \tag{13}$$

No assumption is made on A' or A'': neither slow expansion nor monotonicity nor convexity. Under (13), $Z \le \sup A^2/\hat{E}^2 < 1$ on all of $r > 2M$, so there is no freezing surface and $D = I \times \Omega$. Moreover

$$1 \le N \le N_{\max} := \Big(1 - \frac{\sup_I A^2}{\hat{E}^2}\Big)^{-1} < \infty. \tag{14}$$

Lemma IV.1. (Ω, h_0) *is complete.*

Proof. Let $\rho(r) = \int_{3M}^r \mathrm{d}s/f(s) = r - 3M + 2M \ln \frac{r-2M}{M}$, which maps $(2M, \infty)$ onto $\mathbb{R}$. Since $|\mathrm{d}x|_{h_0} \ge |\mathrm{d}r|/f = |\mathrm{d}\rho|$, $d_{h_0}(x, y) \ge |\rho(x) - \rho(y)|$. A closed h_0-bounded set therefore has ρ bounded, hence r confined to a compact interval $[r_-, r_+] \subset (2M, \infty)$, and it is a closed subset of the compact set $[r_-, r_+] \times S^2$. Bounded closed sets being compact, h_0 is complete by Hopf–Rinow. □

Theorem 1 (conformal Schwarzschild)**.** *Assume (H1). Then:*

(1) (D, g_{opt}) *is globally hyperbolic, and every slice* $\eta = \mathrm{const}$ *is a Cauchy surface;*

(2) for every launch event $o = (\eta_0, x_0)$ *and every* $x_f \in \Omega$ *with* $x_f \ne x_0$ *and* $\eta_0 + \sqrt{N_{\max}}\, d_{h_0}(x_0, x_f) \in I$*, a first-arrival rail curve from* o *to* x_f *exists;*

(3) every first-arrival curve is a rail extremal (a null pregeodesic of g_{opt}*) with no conjugate point in its interior;*

(4) a rail extremal γ *from* o *is the global minimizer of the arrival time to the spatial point of* $\gamma(t)$ *if and only if* $\gamma(t)$ *is not beyond its null cut point, which is the first conjugate point or the first point reached by a second extremal from* o*; before the cut point the minimizer is unique.*

The comparison is with all rail curves in the three-dimensional exterior. The same statements hold for the coordinate-time arrival problem.

Proof. (1) In the form of Fact III.2, (12) has lapse $\mathcal{N} = 1$, shift zero, and slice metrics $g_\eta = N(\eta, \cdot)h_0$ with $h_0 \le g_\eta \le N_{\max} h_0$ by (14). By Lemma IV.1 each g_η is complete, and $g_\eta \ge h_0 \ge N_{\max}^{-1} g_{\eta_1}$ for any fixed $\eta_1 \in I$. Fact III.2 applies.

(2) By Lemma IV.1 and Hopf–Rinow there is a minimizing h_0-geodesic $\sigma : [0, d] \to \Omega$ from x_0 to x_f, parametrized by h_0-arclength, $d = d_{h_0}(x_0, x_f)$. The initial value problem $\mathrm{d}\eta/\mathrm{d}s = \sqrt{N(\eta, \sigma(s))}$, $\eta(0) = \eta_0$, has a right-hand side in $[1, \sqrt{N_{\max}}]$, so its solution exists while $\eta \in I$ and reaches $s = d$ at some $\eta \le \eta_0 + \sqrt{N_{\max}}\, d \in I$. The curve $s \mapsto (\eta(s), \sigma(s))$ is future-directed null by (10), so $L_{x_f} \cap J^+(o) \ne \emptyset$. Proposition III.6(1)–(2) gives a first-arrival curve, which is a null pregeodesic, hence a rail curve by Proposition II.3.

(3) and (4) are Proposition III.6(2)–(4), read through Propositions II.3 and II.5. The last sentence is Remark II.6. □

Remark IV.2 (necessity of the hypothesis on A)**.** If A exceeds $\hat{E}$ somewhere, the freezing surface $r_F(\eta) = 2M/(1 - \hat{E}^2/A(\eta)^2)$ appears, and the regular domain is $r < r_F(\eta)$. By (10) the radial speed obeys $|\mathrm{d}r/\mathrm{d}\eta| \le f/\sqrt{N}$, which vanishes on $r = r_F$. If A grows, the front moves inwards, and a line L_x with $r(x)$ fixed is cut off in finite η. If a fixed spatial line terminates at the inward-moving front before clock t, its inextendible segment misses the slice $\eta = t$, which is therefore not a Cauchy surface, and the product structure used for Fact III.2 is lost. Global hyperbolicity may still hold, since a front advancing with positive speed, where the admissible radial speed vanishes, acts as a barrier for causal

curves; it then has to be proved separately, together with the existence and compactness of minimizers up to the moving boundary. Condition (13) is the hypothesis under which the product structure, and hence Fact III.2, applies; the regime outside the uniform bound $\sup A < \hat{E}$ is not treated here.

Remark IV.3 (the long arc)**.** Under the hypotheses of Theorem 2, an extremal whose swept angle exceeds π has passed its first cut point and is not globally minimal. The conjugate point at π is generated by rotations of its orbital plane, so an analysis restricted to in-plane variations does not detect this obstruction.

Remark IV.4 (relation to Paper I's verification criterion)**.** Under (H1), Theorem 1 gives a geometric criterion for global minimality of an arrival extremal. The arrival-time value function is saturated along every minimizing segment. A smooth Hamilton–Jacobi verification can be used locally where the minimizer is unique and its endpoint shooting map is non-degenerate; no general identification of the cut locus with the set of non-differentiability points is needed here.

V. WHERE THE CUT POINT LIES: THE SWEPT-ANGLE THEOREM

Theorem 1 reduces global minimality to the location of the null cut point. This section locates it for conformal Schwarzschild. Throughout, (H1) holds, and we write

$$q(\eta) = \frac{\hat{E}^2}{A(\eta)^2} > 1, \qquad V^2 = \frac{1}{N} = 1 - \frac{f}{q(\eta)}. \tag{15}$$

For a non-radial extremal the *swept angle* $\Delta\varphi$ is the total angle described about the center in its plane, not reduced modulo 2π, with the plane oriented so that φ increases; for a radial extremal we set $\Delta\varphi = 0$. "Global minimum" means first arrival at the extremal's own spatial point from its launch event, among all rail curves in D.

A. The plane, and the mode out of it

An extremal is planar: the reflection in its plane is an isometry of g_{opt}, which depends on η only through N, so the plane is totally geodesic and the reflection splits the Jacobi fields along the extremal into an in-plane and an out-of-plane mode. Null geodesics, as unparametrized curves, and null conjugate points are conformally invariant; we compute them in the representative

$$\bar{g}_\infty = N^{-1} g_{\mathrm{opt}} = -V^2\,\mathrm{d}\eta^2 + h_0, \qquad h_0 = \frac{\mathrm{d}r^2}{f^2} + \frac{r^2}{f}\,\mathrm{d}\Omega^2, \tag{16}$$

whose slices carry the static metric h_0: the factor A enters only through $V^2 = 1 - f/q(\eta)$. In the plane the auxiliary spacetime is three-dimensional, its screen is one-dimensional, and the in-plane Jacobi equation is $y'' + \mathrm{Ric}(k,k)\,y = 0$ in an affine parameter, with the Ricci tensor of the induced three-dimensional metric.

Lemma V.1 (the out-of-plane mode)**.** *Along a non-radial extremal from o the first point conjugate to o in the out-of-plane mode is at $\Delta\varphi = \pi$, if that angle is reached in I.*

Proof. Rotations about the axis through the center and the launch point fix o and are isometries of g_{opt} for every A. They give a Jacobi field normal to the plane, vanishing at o, with non-zero initial derivative for non-zero angular momentum. Work in the conformal representative $\bar{g}_\infty$ of (16): there the rotation field has orthonormal screen component proportional to $\rho \sin\Delta\varphi$, $\rho = r/\sqrt{f}$ (in g_{opt} itself the factor is $\sqrt{N}\rho$), and its zeros are unchanged under conformal rescaling. The out-of-plane mode is scalar, so every Jacobi field of that mode vanishing at o is a multiple of it. Null conjugate points are conformally invariant, so the conclusion holds in any conformal frame. □

B. The theorem

Theorem 2 (swept-angle theorem)**.** *Assume (H1) on an open interval $I \supset [\eta_0, \infty)$ and*

$$q(\eta) = \frac{\hat{E}^2}{A(\eta)^2} \;\geq\; \frac{3}{2} \qquad (\eta \in I). \tag{17}$$

Let γ be a rail extremal of the arrival branch launched at $o = (\eta_0, x_0)$ in $r > 2M$. For every event of γ:

(1) γ is the global minimum of the arrival time to its spatial point, among all rail curves in three dimensions, if and only if $\Delta\varphi \leq \pi$;

(2) if $\Delta\varphi < \pi$ the minimum is unique up to reparametrization;

(3) if $\Delta\varphi = \pi$ the event is the first null cut point and the minimum is not unique;

(4) beyond $\Delta\varphi = \pi$, γ is not a minimum.

Radial extremals have no cut point at finite time. No condition is placed on A' or A'', and no upper bound on q: A may tend to zero. The cut point is at the first event with $\Delta\varphi = \pi$ if such an event is reached; an extremal that never reaches it has no finite cut point.

The proof is in section V C. Its analytic input is a comparison of the two Jacobi modes, in which the threshold $3/2$ appears and no derivative of A enters; the rays are counted by Proposition III.11. On narrower ranges of q an alternative proof through convex wavefronts on a Hadamard surface, in static conformal frames, is given in sections S2–S5 of the supplementary material. Where it applies it also excludes meetings of distinct null rays in the universal cover of the plane at every swept angle; it works directly on a time interval that need not contain the whole future, and finite-window minimality is also established here by Corollary V.7.

C. Proof of the swept-angle theorem

We work in the static frame $\bar{g}_\infty = -V^2 \mathrm{d}\eta^2 + h_0$, which has the null geodesics, conjugate points and cut points of g_{opt}, and in the plane of the extremal, with the orientation in which the angular momentum L is positive. A null direction at (η, r) is described by an angle $\alpha \in [0, \pi]$ through

$$\frac{\mathrm{d}r}{\mathrm{d}\eta} = fV\cos\alpha, \qquad \frac{\mathrm{d}\varphi}{\mathrm{d}\eta} = \frac{\sqrt{f}\,V}{r}\sin\alpha, \tag{18}$$

$\alpha = 0$ being the outgoing and $\alpha = \pi$ the ingoing radial direction. Put

$$B(q, r) = (q-1)r^2 + (6-3q)Mr - 8M^2, \qquad C(r, q) = -\frac{B(q, r)}{r^2\,(2M + (q-1)r)}.$$

Lemma V.2 (screen comparison)**.** *Let $q \geq q_{\min} > 1$. Along a non-radial null geodesic, with an affine parameter λ and parallel orthonormal screen vectors $e_\perp$ (normal to the plane) and $e_\parallel$, the tidal coefficients $K_i = R(e_i, k, e_i, k)$ satisfy*

$$K_\perp - K_\parallel = \Big(\frac{\mathrm{d}\varphi}{\mathrm{d}\lambda}\Big)^2 \frac{M\,P_q(r)}{r\,(2M + (q-1)r)^2},$$

$$P_q = 3(2q-3)(q-1)r^2 + (33q-38)Mr + 40M^2,$$

with $q = q(\eta)$ evaluated at the point and no derivative of q. If $q \geq 3/2$, then $K_\perp > K_\parallel$, and the in-plane mode has no zero before the first zero of the out-of-plane mode; in particular there is no conjugate point with $0 < \Delta\varphi < \pi$, and at $\Delta\varphi = \pi$ only the out-of-plane mode vanishes.

Proof. We perform the comparison in $\bar{g}_\infty$, with its affine parameter λ, its tangent $k = \mathrm{d}\gamma/\mathrm{d}\lambda$ and parallel orthonormal screens; $L = \rho^2\,\mathrm{d}\varphi/\mathrm{d}\lambda$. The Ricci contributions cancel in $K_\perp - K_\parallel$, leaving a difference of Weyl contractions. Under $\tilde{g} = \Omega^2 g$ this difference is unchanged for the same null vector and rescaled orthonormal screens (screen representatives may differ by multiples of the null tangent); affine tangents must also be rescaled, and both the tidal difference and $(\mathrm{d}\varphi/\mathrm{d}\lambda)^2$ acquire the same tangent weight, so their ratio, displayed above, is the same in g_{opt} and $\bar{g}_\infty$. It is a direct computation. For $q \geq 3/2$ every coefficient of P_q is non-negative and the last two are positive. The out-of-plane mode is $u = (\rho_0/L)\,\rho \sin\varphi$ with $\rho = r/\sqrt{f}$ (Lemma V.1), normalized by $u(0) = 0$, $u'(0) = 1$; let z be the in-plane mode with the same data. While $u, z > 0$, $W = z'u - zu'$ satisfies $W' = (K_\perp - K_\parallel)uz > 0$, $W(0) = 0$, so $(z/u)' = W/u^2 > 0$ and $z > u > 0$. If $\varphi = \pi$ is reached, u vanishes there with $u' < 0$ while $W > 0$, so $z > 0$. This is Sturm's comparison theorem[27] (Theorem 5.20), applied with $u_0 = z$ and $u_1 = u$. □

Lemma V.3 (direction equation)**.** *Along a future null geodesic,* $\mathrm{d}\alpha/\mathrm{d}\eta = V \sin\alpha\, C(r, q(\eta))$*, with no derivative of* q*. For* $q \geq q_{\min} > 1$*:* $B(q, 2M) < 0 < B(q, 3M)$*,* B *has a single positive zero* $r_c(q) \in (2M, 3M)$*, and* r_c *increases with* q*; moreover* $C(r, q) \leq 3M/r^2 - 1/r$*, so* $C \leq -1/(2r)$ *for* $r \geq 6M$*.*

Proof. Differentiate $\cos\alpha$ along the Hamiltonian flow of $\bar{g}_\infty$, including the evolution of the energy momentum, which carries q'; the terms in q' cancel. $B(q, 2M) = -2qM^2$, $B(q, 3M) = M^2$, and at the zero $B_q = r(r - 3M) < 0 < B_r$, so $\mathrm{d}r_c/\mathrm{d}q > 0$. The bound on C follows from $C = f_r/2 - ff_r/(2(q - f)) - f/r$: the middle term is non-positive, since $q > 1 > f > 0$ and $f_r > 0$, and $f_r/2 - f/r = 3M/r^2 - 1/r$. □

At constant q, $r_c(q)$ is the radius of the circular null orbit of $\bar{g}_\infty$ (where $\alpha = \pi/2$ is stationary); it is not the photon sphere $r = 3M$ of the physical metric.

Proof of Theorem 2. By Theorem 1 a first arrival exists (for $q \geq 3/2$, $N = (1 - f/q)^{-1} < (1 - 1/q)^{-1} \leq 3$, and the time requirement holds since I contains the future) and is a null geodesic; by spherical symmetry it is planar, in the plane through the center, the launch and the target when these are not antipodal. In that plane its swept angle is δ in one sense or $2\pi - \delta$ in the other, plus windings. An angle above π contains the conjugate point of Lemma V.1 in its interior, so it is not a first arrival; an angle equal to π means an antipodal target. The first arrival therefore lifts to a ray from $\tilde{o}$ with lifted angle δ in the universal cover of the plane.

In $\bar{g}_\infty$, restricted to that cover, Setting (P) holds with $t = \eta$, $b = 2M$, $J = I$, $s_\pm = \pm fV$ and $a = \sqrt{f}V/r$: the null cone is $-V^2 + s^2/f^2 + (r^2/f)\zeta^2 = 0$, and $fV \leq 1$. Since $q \geq 3/2$, $V \geq V_{\min} = 1/\sqrt{3}$, and f increases in r, so the margin of Proposition III.11 holds: $a(\eta, r_T) \geq \sqrt{f(r_T)}V_{\min}/r_T$ and $s_- \leq -f(r_T)V_{\min}$ on $r \geq r_T$. The slices $\eta = \mathrm{const}$ are Cauchy surfaces of D (Theorem 1); an inextendible causal curve of the closed, totally geodesic plane is inextendible in D; and the projection of an inextendible causal curve of the cover is inextendible in the plane, since a tail near an end point would lie in an evenly covered neighborhood, and its lift in a single sheet. So the slices and their lifts are Cauchy surfaces of the plane and of its cover. Along every future null geodesic of the cover from $\tilde{o}$ to the receiver $L = \rho^2\,\mathrm{d}\varphi/\mathrm{d}\lambda$ is conserved; $L = 0$ would keep $\varphi \equiv 0 \neq \delta$, so φ is strictly monotone and the swept angle is exactly $\delta \in (0, \pi)$. In the plane, which is totally geodesic, the screen is the in-plane mode, and Lemma V.2 gives $z > 0$ on $(0, \delta]$: every such geodesic has index zero and a non-conjugate end point. Proposition III.11 then identifies the first arrival with γ: items 1–2 for $\Delta\varphi < \pi$. Item 3: if the antipodal event were in $I^+(o)$, so would be earlier points of γ; the rotations give distinct minimizers. Item 4: beyond π the conjugate point is interior. Radial extremals: Corollary III.10, in $\bar{g}_\infty$ with X the sphere. □

The proof uses (H1) only through regularity, global hyperbolicity and the margin of the receiver; condition (17) enters through Lemma V.2. It is sufficient, not shown to be necessary: below 3/2 the comparison of the two modes fails at large r already for constant

A, which does not by itself produce conjugate points; section V D recovers the conclusions below 3/2 for launch and target radii up to an explicit bound.

D. Below $q = 3/2$: a radius bound

For $q < 3/2$ the comparison of Lemma V.2 fails only at large radii, and the rays that matter for a first arrival never go far out.

Lemma V.4 (the radius of the comparison)**.** *For $1 < q < 3/2$ let $R(q)$ be the positive zero of P_q,*

$$R(q) = M\,\frac{33q - 38 + \sqrt{(33q-38)^2 + 480(3-2q)(q-1)}}{6(3-2q)(q-1)},$$

and $R(q) = \infty$ for $q \geq 3/2$. Then R is strictly increasing on $(1, 3/2)$, $R(1^+) = 8M$, $R(q) \sim 23M/(6(3/2-q))$ as $q \to 3/2^-$, and $R(q) > 8M > 3M$. For $q_{\min} > 1$, every $q \geq q_{\min}$ and $0 < r < R(q_{\min})$, $P_q(r) \geq P_{q_{\min}}(r) > 0$.

Proof. Write $P_q = -ar^2 + bMr + 40M^2$ with $a = 3(3-2q)(q-1) > 0$ and $b = 33q - 38$: one positive zero, $P_q > 0$ below it. $\partial_q^2 P = 12r^2 > 0$ and $\partial_q P = 3r[(4q-5)r + 11M]$. For $q_{\min} \geq 5/4$, $\partial_q P > 0$. For $1 < q_{\min} < 5/4$, $P_q(11M/(5-4q)) = -M^2(86q^2 - 72q + 1)/(4q-5)^2 < 0$, and $86q^2 - 72q + 1 > 0$ on $(1, 5/4)$, its zeros $(36 \pm \sqrt{1210})/86$ being below 1, so $R(q_{\min}) < 11M/(5 - 4q_{\min})$ and $\partial_q P > 0$ at $q_{\min}$ for $r < R(q_{\min})$, hence for all larger q by convexity. At the zero, $\partial_r P = -M\sqrt{b^2 + 160a} < 0$ and $\partial_q P > 0$, so $R' > 0$. The limits follow from the formula. □

Lemma V.5 (no return beyond $3M$)**.** *Let $q > 1$ on $[\eta_0, \infty)$. A ray from o that meets $r = r_T$ at η_T satisfies $r(\eta) \leq \max(r_0, r_T, 3M)$ for $\eta_0 \leq \eta \leq \eta_T$.*

Proof. For $r > 3M$, $B(q, r) > 0$ for every $q > 1$, since $r_c(q) < 3M$; so $C < 0$ there, and in fact $C \leq -(r - 3M)/r^2$ (Lemma V.3) and $V > \sqrt{2M/r}$ uniformly in $q > 1$, since $V^2 = 1 - f/q > 1 - f$. If the ray exceeded $\max(r_0, r_T, 3M)$, it would cross a level ℓ above that value upwards, with $\alpha \leq \pi/2$, at $r = \ell > 3M$. The region $\{r > 3M,\ \alpha \leq \pi/2\}$ is invariant: inside it $\mathrm{d}r/\mathrm{d}\eta \geq 0$ and $\mathrm{d}\alpha/\mathrm{d}\eta \leq 0$, and at $\alpha = \pi/2$, $\mathrm{d}\alpha/\mathrm{d}\eta < 0$. So $r \geq \ell > r_T$ afterwards. □

Theorem 3 (swept-angle theorem with a radius bound)**.** *Assume (H1) on an open interval $I \supset [\eta_0, \infty)$ and $q(\eta) \geq q_{\min} > 1$. Let γ be a non-radial arrival extremal launched at radius $r_0 < R(q_{\min})$. At every event of γ with radius $r_T < R(q_{\min})$, the segment is a global arrival-time minimizer among all three-dimensional rail competitors if and only if $\Delta\varphi \leq \pi$; it is unique up to reparametrization for $\Delta\varphi < \pi$; an event with $\Delta\varphi = \pi$ is the first null cut point; beyond π the segment is not minimizing. The bound is sufficient, not asserted to be sharp.*

Proof. As for Theorem 2, with $V_{\min} = \sqrt{1 - 1/q_{\min}}$ in the margin of the receiver. Every future null geodesic of the cover from $\tilde{o}$ to the receiver meets $r = r_T$ at its end, so by Lemma V.5 it stays in $r \leq \max(r_0, r_T, 3M) < R(q_{\min})$, where $K_\perp > K_\parallel$ by Lemma V.4; the Wronskian argument of Lemma V.2 then gives index zero and a non-conjugate end point, and Proposition III.11 identifies the first arrival. For an event with $\Delta\varphi > \pi$ the out-of-plane conjugate point is interior regardless of the comparison. □

Corollary V.6 (within $8M$)**.** *Under (H1) on $I \supset [\eta_0, \infty)$, with no lower bound on q beyond the one implied by (H1), the conclusions of Theorem 2 hold for every launch and target with $2M < r_0, r_T \leq 8M$.*

Proof. With $S = \sup_I A < \hat{E}$, (H1) gives $q \geq q_{\min} = \hat{E}^2/S^2 > 1$ on the whole future, whether or not the supremum is attained; Lemma V.4 gives $R(q_{\min}) > 8M$. □

The hypothesis on the whole future serves a statement about all events of an extremal. A single segment needs it only up to its arrival time.

Corollary V.7 (finite window)**.** *Let A be smooth and positive on an open interval containing $[\eta_0, T]$, with $q \geq q_{\min} > 1$ on $[\eta_0, T]$. Let γ be a non-radial arrival extremal launched at radius r_0 and reaching radius r_T at time T, with $r_0, r_T < R(q_{\min})$. Then the conclusions of Theorem 3 hold at that event, with no assumption on A after T.*

Proof. Since $\max(r_0, r_T, 3M) < R(q_{\min})$ and R is continuous, with $R(q) \to \infty$ as $q \to 3/2^-$, there is $\bar{q} \in (1, q_{\min})$ with $\max(r_0, r_T, 3M) < R(\bar{q})$. By continuity $q > \bar{q}$ on $[T, T+\epsilon]$ for some $\epsilon > 0$. Let χ be smooth, equal to 1 near $(-\infty, T]$ and to 0 on $[T+\epsilon, \infty)$, and set $q_{\text{ext}} = \chi q + (1-\chi) q_{\min}$ after T, $q_{\text{ext}} = q$ up to T (extended in the same way slightly before η_0 if needed). It is smooth, $q_{\text{ext}} \geq \bar{q} > 1$, and $A_{\text{ext}} = \hat{E}/\sqrt{q_{\text{ext}}}$ satisfies (H1). Theorem 3 with $\bar{q}$ applies to A_{ext}. Every competitor that arrives no later than T lies in $\eta \leq T$, where the two metrics coincide; so minimality, uniqueness and the cut-point statement at the event transfer to A. The extension is fixed by the given time T, not by a first arrival, so the argument is not circular. The margin $\bar{q} < q_{\min}$ is needed: if $q(T) = q_{\min}$ and $q'(T) < 0$, no smooth extension keeps $q \geq q_{\min}$. Statements about the cut point of γ after T refer to the extended problem, that is, to one admissible continuation of A. □

The bound $R(q_{\min})$ is sufficient and far from optimal. Far enough beyond it the swept-angle rule fails for constant factors, and $3/2$ is the sharp threshold for a rule valid at all radii.

For constant A the auxiliary metric is static, and its null geodesics are the geodesics of the Fermat metric h_0/V^2, $V^2 = 1 - f/q$, with the same conjugate points; arrival time is Fermat length. In the equatorial plane, with the circumferential radius $\rho = r/(\sqrt{f}V)$, comparing the direction equation with Clairaut's relation $\rho \sin\alpha = b$ gives $C = -f\rho_r/\rho$. Hence, by Lemma V.3, ρ increases exactly on the outer branch $r > r_c(q)$, where it is a coordinate with minimum $b_c = \rho(r_c)$, and there the Fermat plane is

$$F(\rho)^2 \mathrm{d}\rho^2 + \rho^2 \mathrm{d}\varphi^2, \qquad F = \frac{\mathrm{d}r/\mathrm{d}\rho}{fV} = 1 + \frac{A_1 M}{\rho} + G(\rho), \qquad A_1 = \frac{\sqrt{q}\,(2q-3)}{(q-1)^{3/2}}. \tag{19}$$

In $x = M/r$ the inverse radius $w = M/\rho = x\sqrt{(1-2x)(q-1+2x)}/\sqrt{q}$ has $w'(0) > 0$ and F is smooth in x with $F(0) = 1$; so F is smooth in w, $G(0) = G_w(0) = 0$, and there are K, ρ_* (depending on q) with

$$|G(\rho)| \leq \frac{KM^2}{\rho^2}, \qquad |G'(\rho)| \leq \frac{2KM^2}{\rho^3} \qquad (\rho \geq \rho_*). \tag{20}$$

Proposition V.8 (sharpness of the all-radii threshold)**.** *Let $a = 0$ and let A be constant with $1 < q = \hat{E}^2/A^2 < 3/2$, so that $c := -A_1 > 0$. There is $d_0(q) > r_c(q)$ such that for every $d \geq d_0(q)$ some ingoing ray launched at radius d returns to radius d at an in-plane conjugate point, with swept angle $\Phi < \pi$. Its continuation just beyond that point is not minimizing, although its swept angle is still below π. With $D = \rho(d)$, suitable such rays have impact parameter and angle*

$$b_0 = \sqrt{cMD}\,(1 + o(1)), \qquad \pi - \Phi = 4\sqrt{cM/D}\,(1 + o(1)) \qquad (d \to \infty).$$

For $q = 6/5$ one can take $d_0 = 8200M$.

Proof. Return angle. For $d > r_c$ and $b_c < b < D$, the ray launched inwards at $\rho = D$ with Clairaut constant b turns at $\rho = b$ on the outer branch and returns to $\rho = D$ after the angle

$$\Phi(b) = 2\int_b^D \frac{bF(\rho)\,\mathrm{d}\rho}{\rho\sqrt{\rho^2 - b^2}} = 2\int_1^{D/b} \frac{F(bt)\,\mathrm{d}t}{t\sqrt{t^2-1}}.$$

The first two terms of (19) integrate in closed form, and (20) with $\int_1^\infty \mathrm{d}t/(t^3\sqrt{t^2-1}) = \pi/4$ bounds the rest: for $b \geq \rho_*$,

$$\Phi = 2\arccos\frac{b}{D} - \frac{2cM\sqrt{D^2-b^2}}{bD} + R_0, \qquad |R_0| \leq \frac{\pi K M^2}{2b^2},$$
$$\Phi_b = -\frac{2F(D)}{\sqrt{D^2-b^2}} + \frac{2cM\sqrt{D^2-b^2}}{b^2 D} + R_1, \qquad |R_1| \leq \frac{\pi K M^2}{b^3},$$

the first term of Φ_b coming from the moving end point D/b, and

$$R_1 = 2\int_1^{D/b} \frac{G'(bt)\,\mathrm{d}t}{\sqrt{t^2-1}}.$$

A zero of Φ_b. Fix $0 < \lambda_- < \sqrt{c} < \lambda_+$ and put $b = \lambda\sqrt{MD}$. As $d \to \infty$, $D\Phi_b \to -2+2c/\lambda^2$ uniformly for $\lambda \in [\lambda_-, \lambda_+]$, while $b \geq \max(\rho_*, b_c)$ and $b/D \to 0$. So for d large Φ_b changes sign on $[\lambda_-, \lambda_+]\sqrt{MD}$ and has a zero b_0 there; letting $\lambda_\pm \to \sqrt{c}$ gives $b_0/\sqrt{MD} \to \sqrt{c}$. For $b \leq D/2$, $\Phi \leq \pi - 2\arcsin(b/D) - \sqrt{3}\,cM/b + \pi K M^2/(2b^2) < \pi$ once $b > \pi K M/(2\sqrt{3}\,c)$, and the expansion gives the deficit $2b_0/D + 2cM/b_0 + o(\sqrt{M/D}) = 4\sqrt{cM/D}(1+o(1))$.

Conjugate point. The rays from the launch event form the unit-speed family $\gamma(s, \alpha_0)$ with $b = D\sin\alpha_0$, $\mathrm{d}b/\mathrm{d}\alpha_0 \neq 0$; its Jacobi field $J = \partial_{\alpha_0}\gamma$ is normal to the ray. At the return, $\rho_s > 0$, and differentiating $\rho(\gamma(s(\alpha_0), \alpha_0)) = D$ gives $\partial_{\alpha_0}\Phi = (J^\varphi \rho_s - J^\rho \varphi_s)/\rho_s$, which vanishes exactly when the normal component of J does. So $\Phi_b(b_0) = 0$ is an in-plane conjugate point; the out-of-plane mode $\rho\sin\varphi$ has no zero before π (Lemma V.1).

Not minimizing beyond. Let s_c be the conjugate parameter and $S > s_c$ close to it, so that the swept angle at S is still below π. Let $Y = J$ on $[0, s_c]$ and $Y = 0$ on $[s_c, S]$. The index form of the Fermat length satisfies $I(Y, Y) = 0$ and $I(Y, Z) = \langle D_s J(s_c), Z(s_c)\rangle$ for Z vanishing at the ends, and $D_s J(s_c) \neq 0$. With $Z(s_c) = -D_s J(s_c)$, $I(Y + \epsilon Z, Y + \epsilon Z) < 0$ for small $\epsilon > 0$; after smoothing, this gives nearby curves in the plane with the same end points and shorter Fermat length, that is, earlier arrivals.

The case $q = 6/5$. Here (20) holds with $K = 465$ and $\rho_* = \rho(10M)$, by exact Bernstein certificates after a rational parametrization of $r \geq 10M$, and the bounds above give a zero of Φ_b for every $D = \rho(d) \geq 2 \times 10^4 M$, which holds for every $d \geq 8200M$; the verification is in section S6 of the supplementary material. □

The proposition identifies a conjugate point, not the first cut point, which may come earlier through a crossing of distinct rays. The threshold is sharp in the family of constant factors; it does not follow that every varying factor that dips below 3/2 fails.

Remark V.9 (the Fermat geometry below 3/2)**.** The next coefficient is $A_2 = q(15q^2 - 42q + 35)/(2(q-1)^3) > 0$, and since the turning point is at $\rho = b$ the deflection is linear in F: $\delta(b) = 2A_1 M/b + \pi A_2 M^2/(2b^2) + O(b^{-3})$; for $q \to \infty$ this is the deflection of light, $4M/b + 15\pi M^2/(4b^2)$. To first order the Fermat plane is a spatial Schwarzschild metric whose mass has the sign of $2q - 3$: attractive above 3/2, where the only caustic is the axis $\Delta\varphi = \pi$, and repulsive at large impact parameter below it, which produces the radial caustic of Proposition V.8. Since $\delta \to +\infty$ as $b \to b_c^+$ and $\delta < 0$, $\delta' > 0$ for large b, the deflection also has an interior minimum, so a radial caustic exists even for a source at infinity. Numerically, conjugate points of this kind appear at much smaller radii than those the proposition provides. For $A = 1$, $\hat{E}^2 = 6/5$, $r_0 = 10^4 M$ and impact parameter $10b_c$, three independent integrations of the screen Jacobi equation give an in-plane zero at $\Delta\varphi \approx 0.9451\pi$, $r \approx 530.09M$. For $q = 1.2$ and $r_0 = r_T = 1280M$ one is found at $\Phi \approx 0.949\pi$. Selected examples for $1.05 \leq q \leq 1.45$ are shown in Figure 1. These examples are not certified and do not locate the onset of failure. The earlier searches at $r_0 = r_T = 20M$ detected no additional ray and no in-plane screen zero.

Figure 1 summarizes the coverage.

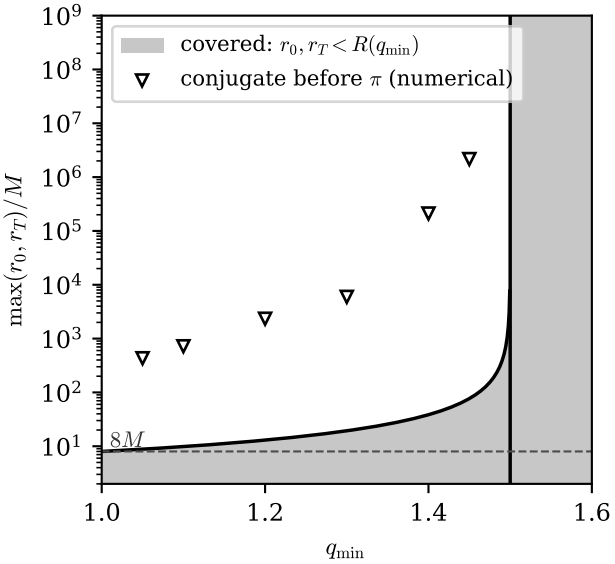


FIG. 1. Arrival branch, $a = 0$. Theorem 2 covers $q_{\min} \geq 3/2$ for an arbitrary smooth factor. Below 3/2, Theorem 3 covers launch and target radii below $R(q_{\min})$ (shaded, logarithmic scale), which exceeds $8M$ (dashed) for every $q_{\min} > 1$ and diverges at 3/2. Triangles: selected numerical examples, for constant A, of in-plane conjugate points before π (Remark V.9), plotted at $\max(r_0, r_T)$ with r_T the radius of the zero; they do not mark the onset of failure. The unshaded region is not covered by the radius estimate; failure at large radii is proved for constant factors (Proposition V.8), and no sharp boundary is asserted.

Remark V.10 (what is new, and what is not covered)**.** Theorem 2 holds for an arbitrary smooth conformal factor with $q \geq 3/2$, with no adiabatic assumption and no upper bound on q; for constant A it strengthens the frozen global-minimality statement of Paper II to all competitors in three dimensions. The convex-front argument in static conformal frames (supplementary material, sections S2–S5) is an alternative proof on narrower ranges of q, where it also gives a stronger no-crossing property. For $q < 3/2$ somewhere, Theorem 3 covers launch and target radii below $R(q_{\min})$, in particular everything within $8M$ (Corollary V.6). Not covered by these swept-angle theorems: $q < 3/2$ with launch or target beyond $R(q_{\min})$, where the pointwise comparison of the two modes fails and, for constant A at large radii, the rule itself fails (Proposition V.8); small spin (Theorem 12) and persistence (Theorem 4) cover some cases outside both theorems. Not covered at all: freezing; the spin $a \neq 0$, where the drag of the seed adds a wind to the front motion and the rotations about the launch axis are no longer isometries; and the proper-time branch.

VI. PERSISTENCE NEAR CONSTANT DATA

Where the frozen problem is understood, its global minima survive a small change of the conformal factor. The statement is qualitative.

Theorem 4 (persistence, arrival branch)**.** *Fix a launch event, two distinct spatial points and a compact time cylinder. Consider a family of smooth, strongly convex rail indicatrices containing the origin uniformly, depending on a parameter $A(\eta)$, and assume the uniform existence hypotheses of Paper I, Theorem I.2, with a common competitor arriving with a strict time margin inside the cylinder. Let the problem with $A \equiv A_0$ have a unique global minimum γ_0, interior to the spatial region, such that the shooting map (initial direction, duration) $\mapsto$ end point has invertible differential at γ_0. Then there is $\delta > 0$ such that for $\|A - A_0\|_{C^1} < \delta$ on the cylinder the arrival problem has a unique global minimum in the*

region, the continuation of γ_0. The threshold δ depends on the problem and on the cylinder.

Proof. Parametrize by η. With $B_A(\eta, x) = \{v : F_A(x, v, \eta) \le 1\}$ the Hamiltonian is $H_A = \max_{v \in B_A} \langle p, v\rangle$, for conformal Schwarzschild $H_A = V_A|p|_{h_0^*}$; the flow $\dot{x} = \partial_p H_A$, $\dot{p} = -\partial_x H_A$ involves A only through its value, and by homogeneity $|p(\eta_0)| = 1$ may be imposed. Suppose $A_k \to A_0$ with minimizers γ_k different from the continued extremals. Theorem I.2 gives existence; the common competitor and uniform coercivity give uniform bounds; a subsequence converges in the convex body of velocities. For each fixed $\lambda < 1$ sufficiently close to one, slow the reference spatial trajectory by the factor λ: it lies strictly inside the frozen velocity body, so it is admissible for the perturbed relaxed problem for large k, and it arrives at $\eta_0 + (T_0 - \eta_0)/\lambda$; taking first $k \to \infty$ and then $\lambda \uparrow 1$ gives $\limsup T_k \le T_0$, while the limit reaches the target at a time $\ge T_0$. Hence $T_k \to T_0$ and, by uniqueness, $\gamma_k \to \gamma_0$. For large k the γ_k are interior, hence normal extremals (Theorem I.2(ii)); their initial directions lie on a compact sphere and every limit, together with T_0, generates γ_0, so by strong convexity the limit direction is that of γ_0. The variational equations give convergence of the shooting map and of its Jacobian, and the implicit function theorem gives a unique zero near that of γ_0, which must be the data of γ_k: a contradiction. □

Lemma VI.1 (persistence for a parameter family)**.** *Fix the launch, the spatial target and a compact regular time cylinder. Let $B_s(\eta, x)$ be a uniformly coercive family of smooth, uniformly strongly convex velocity bodies, with the origin uniformly in their interior. Assume a common competitor with a strict time margin, existence and compactness of minimizers in a common region, and continuous dependence on s of the bodies and of their support functions $H_s(\eta, x, p)$, including the (x, p) derivatives needed for the Hamiltonian flow and its first variation away from $p = 0$. If the reference problem $s = 0$, possibly non-autonomous, has a unique interior minimum with invertible end-point shooting map, then for s sufficiently close to 0 there is a unique global minimum in the common region, its continuation.*

Proof. As for Theorem 4, except for the upper bound on the arrival times, since slowing down a non-autonomous reference need not be admissible. Let z_0 be the shooting data of the reference. The perturbed shooting maps S_s are C^1 in z and, together with their z-derivative, depend continuously on s uniformly near z_0, by the assumptions on H_s and the first-variation equations; since the reference Jacobian is invertible, the contraction $z \mapsto z - DS_0(z_0)^{-1} S_s(z)$ has a fixed point $z_s \to z_0$. The corresponding extremal is a rail curve of the perturbed problem, interior for small s, reaching the target at a time $T_s \to T_0$: this gives $\limsup T \le T_0$ for the minimal arrival times. The common competitor gives the uniform bounds; closure in the convex body of velocities gives $\liminf$; uniqueness gives convergence to the reference; interiority gives normality and normalized shooting data in a compact set; the flow and strong convexity give convergence of those data; the invertible Jacobian gives a local uniqueness that contains every perturbed minimum. □

Remark VI.2 (quantifiers and the proper-time branch)**.** The order of quantifiers is "for each frozen arc there is δ"; no uniform δ over all arcs is claimed, and at $\Delta\varphi = \pi$ the frozen minimum is neither unique nor non-degenerate. A slowly varying $A(\eta) = \bar{A}(\epsilon\eta)$ satisfies the C^1 condition on a fixed window, not on windows of length $O(\epsilon^{-1})$. The proper-time problem requires a separate argument, because its cost differs from the arrival-time functional. Section XI establishes existence, absence of a relaxation gap and perturbative global selection for $a = 0$; finite variations of A remain outside the global-selection theorem.

Corollary VI.3 (near a constant factor)**.** *Let $a = 0$ and let γ_0 be an arrival extremal for a constant factor A_0 with $q_0 \ge 3/2$ and $\Delta\varphi < \pi$. Then there is a C^1 neighborhood of A_0, depending on γ_0, in which the continuation of γ_0 to the same target is the unique global minimum among all rail curves.*

Proof. Theorem 2 (for constant A and $q \ge 3/2$) gives uniqueness, and Lemmas V.2 and V.1 give non-degeneracy before π for a non-radial reference. For a radial reference, Corollary III.10 gives earliest arrival at every later event and no finite cut point, so no finite

conjugate point occurs. In the frozen problem the end-point shooting map is the exponential map of the Fermat metric, parametrized by initial unit direction and length, and non-conjugacy makes its differential invertible. Competitors that can beat an arc of duration at most $T_{\rm up}$ satisfy $|\mathrm{d}x/\mathrm{d}\eta|_{h_0} \le 1$ and stay in the closed h_0-ball of radius $T_{\rm up}$, which is compact; take a regular region containing it strictly. Theorem 4 applies. □

a. Vaidya. For ingoing Vaidya, the time reverse of Vaidya's radiating metric[28], with frozen mass m_0, $g = -f\,\mathrm{d}v^2 + 2\,\mathrm{d}v\,\mathrm{d}r + r^2\mathrm{d}\Omega^2$, $f = 1 - 2m_0/r$, the Kodama selector[29,30] is $\partial_v = \partial_t$ with $v = t + r_*(r)$, $r_*' = 1/f$, and the rail charge is the Schwarzschild one. For a spatial tangent ξ the arrival costs are

$$F_t(\xi) = \frac{\hat{E}}{\sqrt{\hat{E}^2 - f}}\sqrt{\frac{\xi_r^2}{f^2} + \frac{r^2}{f}|\xi_\Omega|^2}, \qquad F_v(\xi) = F_t(\xi) + \frac{\xi_r}{f}, \tag{21}$$

so $\Delta v = \Delta t + r_*(r_f) - r_*(r_0)$: a fixed boundary term, which preserves the order of all competitors. The frozen v-branch is therefore the Schwarzschild t-branch with $q = \hat{E}^2$, and its indicatrix is that of Paper I. For a varying mass this is not a static change of variables, and Theorem 4 is applied directly to the Vaidya support function with clock v and parameter $m(v)$.

Corollary VI.4 (Vaidya, arrival branch)**.** *Let $\hat{E}^2 \ge 3/2$, $r_0, r_f > 2m_0$, and let γ_0 be a frozen arrival extremal with $\Delta\varphi < \pi$. Then every positive mass function $m(v)$ sufficiently close to m_0 in C^1 on a common window admits a unique minimum of v to the same target, the continuation of γ_0, among all regular rail curves in $r > 2m(v)$.*

Proof. Frozen uniqueness and non-degeneracy come from Theorem 2 at constant mass, $q = \hat{E}^2$. It remains to confine the relevant competitors. With $w = \hat{E}^2 - f$, along every competitor $\mathrm{d}r/\mathrm{d}v \le -w + \hat{E}\sqrt{w} = f\sqrt{w}/(\hat{E} + \sqrt{w}) < f/2 \le 1/2$, so none of duration at most $T_{\rm up}$ goes beyond $r_0 + T_{\rm up}/2$. For the inner side take $2m_0 < \rho < \min(r_0, r_f)$ and $|m - m_0| \le \epsilon_m$, $m_\pm = m_0 \pm \epsilon_m$; in $r \ge \rho$, $\mathrm{d}r/\mathrm{d}v \le \frac{1}{2}(1 - 2m_-/r)$, so a competitor that dips below ρ needs at least

$$L(\rho, r_f) = 2(r_f - \rho) + 4m_- \log\frac{r_f - 2m_-}{\rho - 2m_-}$$

to climb back to r_f. At $\epsilon_m = 0$ this bound diverges as $\rho \to 2m_0$: choose first ρ with $L > T_{\rm up}$ at $\epsilon_m = 0$, then ϵ_m small enough to keep $L > T_{\rm up}$ and $\rho > 2m_+$ (at fixed $\epsilon_m > 0$ the bound does not diverge). All relevant competitors then lie in a compact shell strictly outside $2m(v)$, where $f > 0$ and $w \ge \hat{E}^2 - 1 > 0$ uniformly, and Theorem 4 applies. □

Remark VI.5. The radius of the neighborhood depends on the end points and the duration; it is not uniform over Vaidya problems, and the result does not reach the region where the Kodama selector stops being timelike. The arrival-time Fermat criterion does not certify minimality of the proper-time functional, which is treated separately in section XIII.

VII. INGOING VAIDYA: THE ARRIVAL BRANCH

Corollary VI.4 covers masses close to a constant. For every non-decreasing mass the arrival branch is again a Fermat problem, and the results of sections IV and V carry over, with the mass in place of the conformal factor. With $w = \hat{E}^2 - f$, $f = 1 - 2m(v)/r$ and $\hat{E} > 1$, define on $D = \{(v, r, \Omega) : v \in I,\ r > 2m(v)\}$ the Lorentzian metric

$$G = -w\,\mathrm{d}v^2 + \frac{(\mathrm{d}r + w\,\mathrm{d}v)^2}{\hat{E}^2} + r^2\mathrm{d}\Omega^2. \tag{22}$$

For a curve parametrized by v, $G(1,s,\zeta) = -w+(s+w)^2/\hat{E}^2+r^2|\zeta|^2$, so the future null cone of G is exactly the rail and its causal curves are the relaxed curves (the velocity body K of section XIII); the (v,r) block has determinant $-w/\hat{E}^2 < 0$, $G(\partial_v,\partial_v) = -wf/\hat{E}^2 < 0$ in D, and $G^{-1}(\mathrm{d}v,\mathrm{d}v) = -1/w < 0$, so v is a temporal function. The future null geodesics of G are the arrival extremals, with $P_v = -h$ for the Vaidya support function h. For constant mass, $v = t + r_*(r)$ gives $G = (wf/\hat{E}^2)\,[-\mathrm{d}t^2 + Nh_0]$ with $N = \hat{E}^2/w$, the auxiliary metric of section IV with $A = 1$, up to a conformal factor. Equivalently, the normalized causal velocities satisfy $(s+w)^2/\hat{E}^2 + r^2|\zeta|^2 \le w$; the proper-time cost on this same velocity body is treated separately in section XIII.

Theorem 5 (Vaidya arrival branch, non-decreasing mass)**.** *Let $\hat{E} > 1$ and let m be smooth, positive and non-decreasing on an open interval I. Then (D,G) is globally hyperbolic. For every launch event o and every spatial target $x_f \neq x_o$ reached by some future causal curve from o, the first arrival in v is attained; every first-arrival curve is a null pregeodesic of G; and a null geodesic γ from o is a first arrival to the spatial point of $\gamma(\lambda)$, among all relaxed three-dimensional competitors in D, exactly for $\lambda \le \lambda_c = \sup\{\lambda : \gamma(\lambda) \notin I^+(o)\}$. If $\gamma(\lambda_c)$ is an interior event of γ, it is the first conjugate point or the first point reached by a second maximal null geodesic, and before it the first arrival is unique up to reparametrization.*

Proof. A causal vector with $\mathrm{d}v = 0$ has $G = \mathrm{d}r^2/\hat{E}^2 + r^2|\mathrm{d}\Omega|^2 \ge 0$, so it vanishes: v increases strictly along future causal curves and (D,G) is causal. Let $q \in J^+(o)$ and let a causal curve from o to q be parametrized by v. With $x = r - 2m(v)$ and $s_+ = -w + \hat{E}\sqrt{w}$, almost everywhere $x' = s - 2m' \le s_+ \le f/2 = x/(2r) \le x/(4m(v_o))$, using $m' \ge 0$ twice and $\sqrt{w} \le \hat{E}$; backward integration gives $x \ge x_q\, e^{-(v_q - v)/(4m(v_o))}$, and $s \le 1/2$ gives $r \le r_o + (v_q - v_o)/2$. Hence $J^+(o) \cap J^-(q)$ lies in a compact subset of D. It is closed: for $z_n \to z$ in the diamond, concatenate causal curves $o \to z_n \to q$; parametrized on $[v_o, v_q]$ they have uniformly bounded velocities in that compact set, a subsequence converges uniformly to a causal curve from o to q (closure of the cone through support functions), and it passes through z. Causality and compact diamonds give global hyperbolicity[31]. If an arrival at x_f occurs at v_q, then $r_f > 2m(v_q) \ge 2m(v)$ for $v \le v_q$, so the line of x_f is timelike on $[v_o, v_q]$; Proposition III.6 applies with η replaced by v. □

No closeness to constant mass and no limit of m are assumed; the bound $m' \ge 0$ is the null energy condition of the ingoing dust. The theorem characterizes minimality by the cut point without locating it: Corollary VI.4 places a prescribed target before the cut for masses close to constant, and Theorem 8 below locates it at the antipode for $\hat{E}^2 \ge 3/2$. The construction is a specific instance of Fermat's principle in a Lorentzian manifold[32].

Lemma VII.1 (antipodal bound)**.** *Rotations about the axis through the launch point are isometries of G. Along an equatorial non-radial null geodesic they generate the Jacobi field $\varepsilon\, r \sin\varphi$ normal to the plane, which vanishes at the launch and at $\varphi = \pi$. If a null geodesic reaches $\varphi = \pi$ in D, that event is conjugate to o, and the cut point is not later.*

Theorem 6 (screen comparison)**.** *Let $\hat{E}^2 \ge 3/2$ and let m be smooth, positive and non-decreasing. Let γ be a future-directed non-radial null geodesic of G starting at o, with positive swept angle φ. There are no points conjugate to o with $0 < \varphi < \pi$. If γ reaches $\varphi = \pi$ in D, that point is its first conjugate point and has screen multiplicity one.*

Proof. Choose an affine parameter λ, an equatorial plane, and $L = r^2\varphi' > 0$. The reflection across the plane diagonalizes the tidal operator in a parallel screen frame $(e_\perp, e_\parallel)$, $e_\perp = r^{-1}\partial_\theta$. With $K_i = R(e_i,k,e_i,k)$ and $k = \gamma'$, $K_\perp + K_\parallel = \mathrm{Ric}(k,k)$, $K_\perp = L^2/r^4 - r''/r$, and a direct computation gives

$$K_\perp - K_\parallel = \frac{L^2}{r^4(\hat{E}^2 r + 2m - r)^2}\Big[\frac{mP(r)}{r} + m'\,r\big(3(\hat{E}^2-1)r + 4m\big)\Big],$$

$$P(r) = 3(2\hat{E}^2-3)(\hat{E}^2-1)r^2 + (33\hat{E}^2 - 38)mr + 40m^2.$$

For $\hat{E}^2 \geq 3/2$ every coefficient of P is non-negative and the constant term is positive, and $m' \geq 0$: the right-hand side is positive. The normalized out-of-plane solution is $u = (r_0/L)\, r \sin\varphi$, $u(0) = 0$, $u'(0) = 1$; let $z'' + K_\parallel z = 0$, $z(0) = 0$, $z'(0) = 1$. While $u, z > 0$, $(z'u - zu')' = (K_\perp - K_\parallel)uz > 0$ and $(z/u)' = (z'u - zu')/u^2 > 0$; since $z/u \to 1$ at the launch, $z > u > 0$ for $0 < \varphi < \pi$, so the screen Jacobi determinant uz does not vanish there. If $\varphi = \pi$ is reached at λ_π in D, then $u(\lambda_\pi) = 0$, $u'(\lambda_\pi) = -r_0/r_\pi < 0$, and integrating the Wronskian identity gives $z(\lambda_\pi) > 0$: only the out-of-plane mode vanishes. □

The threshold $\hat{E}^2 = 3/2$ is that of Theorem 2, where the same polynomial appears (Lemma V.2): the difference of the two modes is a Weyl component, and at constant mass the two metrics are conformal. Below it the uniform comparison fails already for constant mass at large radii, and there, since at constant mass the problem is conformal Schwarzschild with $q = \hat{E}^2$, for $1 < \hat{E}^2 < 3/2$ there are in-plane conjugate points before the antipode at large radii (Proposition V.8); the m' term helps the sign of the comparison, which is not a statement that accretion moves the cut point.

Corollary VII.2 (strict local arrival minimality)**.** *Under the hypotheses of Theorem 6, every compact segment with $0 < \varphi_T < \pi$ is a strict local minimizer of the arrival coordinate v at the timelike receiver of fixed spatial position through its endpoint, modulo orientation-preserving reparametrization, among sufficiently close future-directed causal curves, with locality understood for example in a fixed C^1 parametrization gauge.*

Proof. On the screen the Dirichlet index form $I(\eta, \eta) = \int_0^{\lambda_T} \sum_i (\eta_i'^2 - K_i \eta_i^2)\, \mathrm{d}\lambda$ is positive definite and coercive on H_0^1 by Picone's identity, both fundamental solutions being positive on $(0, \lambda_T]$. The arrival Hessian at the timelike receiver is $-I/G(k_T, \partial_v)$, with a positive prefactor, and the local Fermat principle with a fixed reparametrization gauge gives the claim for null competitors[33]; by spherical symmetry this includes out-of-plane perturbations. Competitors are causal curves in a sufficiently small tube about the segment, written as graphs over the clock. Away from a small neighborhood of the launch, treated in a convex normal neighborhood, the null exponential map is a regular branch on the tube, since there are no conjugate points; the null hypersurface it generates is then a local barrier for causal curves in the tube. Since ∂_v is timelike, the hypersurface is locally a graph $v = b(x)$ with future side $F = v - b(x) \geq 0$, and $\mathrm{d}F \geq 0$ on future causal directions on it, with equality only along the generator; the velocity bodies vary in a locally Lipschitz way on the compact tube, so along a competitor parametrized by v, $F' \geq -C|F|$ after shrinking the tube. Grönwall on the negative part keeps $F \geq 0$, and once $F > 0$ the exponential lower bound keeps it positive; equality at the arrival therefore keeps the competitor on the hypersurface, along a generator. So nearby timelike competitors cannot arrive earlier either. □

a. The cut at the antipode. Conjugate points before π are excluded; it remains to exclude intersections of distinct earliest rays, with Proposition III.11. Write $q = \hat{E}^2$, $w = q - 1 + 2m(v)/r \in (q-1, q)$ in D, and

$$s_+(v, r) = -w + \sqrt{qw} = \frac{wf}{\sqrt{qw} + w}, \qquad 0 < s_+ \leq \frac{f}{2} \leq \frac{1}{2} \quad \text{in } D.$$

For a future causal curve parametrized by v the cone gives $(\mathrm{d}r/\mathrm{d}v + w)^2/q \leq w$, hence $\mathrm{d}r/\mathrm{d}v \leq s_+$. The radial null curves are $\mathrm{d}r/\mathrm{d}v = s_+$ (outgoing) and $\mathrm{d}r/\mathrm{d}v = -w - \sqrt{qw} \leq -(q-1)$ (ingoing), and at constant r the curve with $r^2(\mathrm{d}\varphi/\mathrm{d}v)^2 = wf/q$ is null. A future causal vector with $\mathrm{d}v = 0$ vanishes, so along an absolutely continuous curve whose velocity is future causal or zero almost everywhere, v is non-decreasing and $\dot{r} \leq s_+ \dot{v}$ almost everywhere.

Theorem 7 (uniqueness in the planar cover)**.** *Let $\hat{E}^2 \geq 3/2$, let m be smooth, positive and non-decreasing on an open interval I, and let $o = (v_0, r_0, 0)$ lie in the universal cover $\widetilde{D}$ of the equatorial exterior. For every $0 < \delta < \pi$ there is at most one future null geodesic, up to reparametrization, from o to the partial receiver*

$$\widetilde{\Gamma} = \{(v, r_T, \delta) :\ v \in I,\ r_T > 2m(v)\}.$$

Proof. Suppose there are two, with arrivals $v_1 \le v_2$. Then $r_T > 2m(v_2)$; choose $b > v_2$ in I with $2m(b) < r_T$ and a smooth cutoff $0 \le \chi \le 1$ equal to one until after v_2 and zero for $v \ge b$. Keep the mass before v_2 and set afterwards $\widetilde{m}(v) = m(v_2) + \int_{v_2}^{v} \chi(t) m'(t)\,\mathrm{d}t$, for all $v \ge v_2$, with $\chi m'$ extended by zero beyond b: a smooth, positive, non-decreasing mass on $(\inf I, \infty)$, constant and equal to $m_\infty < r_T/2$ for $v \ge b$. Both rays are unchanged up to their arrivals, and $\widetilde{m} \le m_\infty$ also before v_0, so the receiver is defined on all of $(\inf I, \infty)$.

With $\widetilde{m}$, Setting (P) holds on $\widetilde{D}$ with $t = v$, $b = 2\widetilde{m}$, $J = (\inf I, \infty)$, $s_\pm = -w \pm \sqrt{qw}$ and $a = \sqrt{wf/q}/r$: the null cone is $(s+w)^2/q + r^2\zeta^2 = w$, and $s_+ \le 1/2$. The margin holds: $\sup b = 2m_\infty < r_T$, $a(v, r_T) \ge \sqrt{(q-1)(1-2m_\infty/r_T)/q}/r_T$, and $s_- \le -(q-1)$. The proof of Theorem 5 applies to $\widetilde{m}$; in the cover the angle is bounded on diamonds as well, since $|\mathrm{d}\varphi/\mathrm{d}v| \le \sqrt{w}/r \le \hat{E}/(2\widetilde{m}(v_a))$ along causal curves from any event at $v = v_a$, so $\widetilde{D}$ is globally hyperbolic. The equatorial plane is totally geodesic, being the fixed set of a reflection, so the null geodesics of $\widetilde{D}$ are the equatorial ones and their screen curvature is $K_\parallel$. Along each of them $L = r^2\,\mathrm{d}\varphi/\mathrm{d}\lambda$ is conserved; $L = 0$ gives $\varphi \equiv 0 \ne \delta$, so $L \ne 0$, φ is strictly monotone and the swept angle is exactly $\delta \in (0, \pi)$. The proof of Theorem 6 gives $z > 0$ on $(0, \delta]$: every ray has index zero and a non-conjugate end point. Proposition III.11 allows at most one, a contradiction. □

b. The direction of a ray. The uniqueness proof does not need the following description of the rays, which we record because it holds for every smooth positive mass. In a fixed plane, with the orientation chosen so that $L > 0$, a future null direction is described by an angle $\alpha \in [0, \pi]$ through

$$\frac{\mathrm{d}r}{\mathrm{d}v} = -w + \sqrt{qw}\cos\alpha, \qquad \frac{\mathrm{d}\varphi}{\mathrm{d}v} = \frac{\sqrt{w}}{r}\sin\alpha, \tag{23}$$

$\alpha = 0$ being the outgoing and $\alpha = \pi$ the ingoing radial direction. Here $q = \hat{E}^2$ is the case $A = 1$; it is constant, unlike $\hat{E}^2/A(\eta)^2$ in the conformal problem.

Lemma VII.3 (direction equation)**.** *Along a future non-radial null geodesic of G,*

$$\frac{\mathrm{d}\alpha}{\mathrm{d}v} = \frac{\sin\alpha}{r}\,B(w, \cos\alpha), \qquad B(w, c) = c\,(2w - q + 1) - \tfrac{1}{2}\sqrt{q/w}\,(3w - q + 1).$$

The equation contains no m' term; $B(q, 1) = \frac{1}{2}$ and $B(q-1, c) = (q-1)c - \sqrt{q(q-1)} < 0$ for $|c| \le 1$.

Proof. Put $s = \mathrm{d}r/\mathrm{d}v$, $h = \mathrm{d}v/\mathrm{d}\lambda$ and $\dot{w} = 2m'/r - 2ms/r^2$, the total v-derivative of w along the ray, and $\bar{k} = (1, s, \mathrm{d}\varphi/\mathrm{d}v)$. The connection of G gives

$$\Gamma^v(\bar{k}, \bar{k}) = \frac{s}{r} + \frac{\dot{w}}{2w} + \frac{\cos\alpha}{r}\,B(w, \cos\alpha).$$

Differentiating the conserved angular momentum $L = r\sqrt{w}\,h\sin\alpha$ and using $h_v/h = -\Gamma^v(\bar{k}, \bar{k})$, the term $\dot{w}/(2w)$, which carries m', cancels against the derivative of $\sqrt{w}$, and the stated equation follows where $\cos\alpha \ne 0$; continuity gives it also at $\cos\alpha = 0$. (Equivalently, differentiate $\cos\alpha$ along the Hamiltonian flow of G on the null cone.) □

The accretion rate enters the direction only through the position of the ray, not explicitly; in particular the outgoing radial direction is repelling near the boundary, where $B \to \frac{1}{2}$, and attracting at large radii, where $B < 0$, for every mass profile.

Theorem 8 (the arrival cut is at the antipode)**.** *Let $\hat{E}^2 \ge 3/2$ and let m be smooth, positive and non-decreasing. Every compact segment of a future non-radial null geodesic of G with swept angle in $(0, \pi)$ is the unique global earliest arrival at its spatial end point, up to reparametrization, among all future causal competitors in D, in three dimensions; in particular it is the unique minimum of v among all rail curves. If the geodesic reaches swept angle π in D, that point is its first null cut point.*

Proof. By Theorem 5 an earliest arrival exists and is a null geodesic. By spherical symmetry it lies in the plane through the center, the launch and the target; it cannot have swept angle above π, since the antipodal conjugate point of Lemma VII.1 would be interior, nor equal to π; so it is in the same short lifted class as the given segment, and Theorem 7 identifies the two. If the antipodal event, when reached in D, were in $I^+(o)$, openness of $I^+(o)$ would put preceding points of the same ray there as well, contradicting their earliest-arrival property; the rotations of the plane give distinct earliest rays, and beyond it the conjugate point is interior. □

This is the analogue of Theorem 2 for every non-decreasing mass, with no closeness to constant mass, no bound on the accretion rate and no limit of m. The step that does not transfer from the static case, the convexity of the fronts, is replaced by the count of Proposition III.11, in which the moving boundary enters only through the truncation of the mass and the margin of the receiver. As an illustration only, non-certified root searches with $\hat{E}^2 = 2$, $r_0 = 6$ and $r_T \in \{6, 10\}$ found one same-sense ray for each sampled lifted angle in $[0.02\pi, 1.9\pi]$ for constant mass, $m = 1 + \frac{1}{2}\tanh(v/5)$ and a jump of m from 1 to 2 within $\Delta v \approx 0.6$; the search does not certify completeness, and the observations beyond π are not conclusions of the theorem.

c. Below $\hat{E}^2 = 3/2$: a radius bound. As for conformal Schwarzschild (Theorem 3), the comparison of Theorem 6 survives near the center, and it suffices that the rays reaching a receiver do not go far out. Here, unlike the conformal case, accretion can turn an outgoing ray back, because the velocity body moves with the mass; a bound on the accretion rate excludes this. Write $q = \hat{E}^2$, $f = 1 - 2m/r$ and $\widehat{R}(q) = R(q)/M$, with R from Lemma V.4.

Lemma VII.4 (no return beyond $3m$)**.** *Let $1 < q < 3/2$ and let m be smooth, positive and non-decreasing with*

$$m' < \mu_*(q) := \min\Big\{\frac{1}{27q}, \frac{q-1}{q}\Big\}.$$

At every point of a future null geodesic of G where $\mathrm{d}r/\mathrm{d}v = 0$ and $r \geq 3m(v)$, one has $\mathrm{d}^2r/\mathrm{d}v^2 > 0$. Consequently a segment with radial end points r_0, r_T satisfies $r \leq \max\{r_0, r_T, 3\sup m\}$.

Proof. By (23) and Lemma VII.3, at a turning point $\cos\alpha = \sqrt{w/q}$ and the total derivative of w along the ray is $2m'/r$, so

$$\frac{\mathrm{d}^2r}{\mathrm{d}v^2} = -\frac{1}{r}\Big[m' + \Big(1 - \frac{w}{q}\Big)Q(w)\Big], \qquad Q(w) = 2w^2 - \Big(\frac{5q}{2} - 1\Big)w + \frac{q(q-1)}{2},$$

since $\sqrt{qw}\,B(w, \sqrt{w/q}) = Q(w)$. With $w = q - f$ and $1/3 \leq f < 1$ for $r \geq 3m$, the function $g_q(f) = -(1 - w/q)Q(w) = -2f^3/q + (3/2 + 1/q)f^2 - f/2$ satisfies

$$g_q(f) = \tfrac{3}{2}(1-f)\frac{1}{27q} + \tfrac{3}{2}\Big(f - \frac{1}{3}\Big)\frac{q-1}{q} + \frac{(f - \frac{1}{3})(1-f)(12f + 10 - 9q)}{6q} \;\geq\; \mu_*(q),$$

because the first two weights are non-negative with sum one and $12f + 10 - 9q > 1/2$. Hence $\mathrm{d}^2r/\mathrm{d}v^2 = (g_q(f) - m')/r > 0$. If a segment exceeded $\max\{r_0, r_T, 3\sup m\}$, the radius would have an interior maximum there, with $\mathrm{d}r/\mathrm{d}v = 0$, $\mathrm{d}^2r/\mathrm{d}v^2 \leq 0$ and $r > 3m(v)$: a contradiction. No assumption on the number of turning points is needed. □

Theorem 9 (radius bound below 3/2)**.** *Let $1 < \hat{E}^2 < 3/2$ and let m be smooth and non-decreasing on an open interval I, with $0 < m_0 \leq m \leq m_{\max}$ and $m' < \mu_*(\hat{E}^2)$. Let a non-radial future null geodesic of G be launched at $o = (v_0, r_0, \Omega_0)$, and consider its events at radius r_T, with the segment in D. If*

$$\max\{r_0, r_T, 3m_{\max}\} < \widehat{R}(\hat{E}^2)\, m_0,$$

the segment is a global earliest arrival in v among all relaxed three-dimensional competitors, in particular among all rail curves, if and only if its swept angle is at most π; for an angle below π it is unique up to reparametrization; if the angle π is reached in D, that event is the first null cut point and the first conjugate point, of screen multiplicity one; beyond it the segment is not minimizing. The bound on m' is sufficient; it is not asserted to be necessary or sharp.

Proof. Comparison. The identity for $K_\perp - K_\parallel$ in the proof of Theorem 6 holds for every $q > 1$, with $P(r; m) = m^2 P_q(r/m)$ and P_q the polynomial of Lemma V.4 at $M = 1$. Along every ray with radial end points r_0, r_T, Lemma VII.4 and $m \geq m_0$ give $r \leq \max\{r_0, r_T, 3m_{\max}\} < \widehat{R}(q)m_0 \leq \widehat{R}(q)m(v)$, so $P(r; m(v)) > 0$ pointwise; the m' term is non-negative. The Wronskian argument of Theorem 6 then excludes conjugate points before π and gives multiplicity one at π.

Uniqueness in the cover. Suppose two rays reach the receiver (v, r_T, δ), $0 < \delta < \pi$, at $v_1 \leq v_2$. Truncate the mass after v_2 as in Theorem 7; the new mass $\widetilde{m}$ keeps $m_0 \leq \widetilde{m} \leq m_{\max}$ and $0 \leq \widetilde{m}' = \chi m' < \mu_*(q)$. Every ray of the cover from the launch to the receiver, including those arriving after v_2, has radial end points r_0, r_T, so the previous step applies to all of them: index zero and non-conjugate end point. The other hypotheses of Setting (P), the margin of the receiver and global hyperbolicity use only $q > 1$, as in the proof of Theorem 7; Proposition III.11 allows at most one ray.

Three dimensions and the cut. As in Theorem 8: an earliest arrival exists, is planar and lies in the short class, so it is the given segment. Before an event at angle π all points have radius at most $\max\{r_0, r_T, 3m_{\max}\}$ and are earliest arrivals; openness of $I^+(o)$ and the rotations of the plane make the antipodal event the first cut point, and beyond it the out-of-plane conjugate point is interior. □

Corollary VII.5 (within $8m_0$)**.** *Under the regularity, mass and accretion-rate assumptions of Theorem 9, if $m_{\max} \leq \frac{8}{3}m_0$, the conclusions hold for every non-radial segment with launch and target radii $r_0, r_T \leq 8m_0$ in D.*

Proof. $\widehat{R}(q) > 8$ for every $q > 1$, so $\max\{r_0, r_T, 3m_{\max}\} \leq 8m_0 < \widehat{R}(q)m_0$. □

The same holds on a finite window $[v_0, T]$, provided m is smooth and non-decreasing on an open neighborhood of it, with the bounds there and a slightly larger $\bar{m}_{\max}$ still satisfying the radius condition. Choose a short interval after T on which $0 \leq m' < \mu_*(q)$, $m \leq \bar{m}_{\max}$ and $2m < r_T$, and a smooth cutoff χ, equal to one near T and to zero afterwards; set $\widetilde{m}(v) = m(T) + \int_T^v \chi(s)m'(s)\,\mathrm{d}s$ for $v \geq T$, keeping m unchanged up to T, as in the truncation of Theorem 7. This preserves monotonicity and the rate bound, and competitors arriving no later than T meet only the unchanged profile. One-sided monotonicity up to T is not enough for this extension. The threshold μ_* equals $(q-1)/q$ for $q \leq 28/27$ and $1/(27q)$ above, with maximum $1/28$ and $\mu_* \to 0$ as $q \to 1^+$. It is the best uniform lower bound of g_q on $r \geq 3m$; above it the equations admit local apoastra. Whether the cut rule still holds under the stated radius restriction when the accretion-rate bound is removed remains open.

d. Decreasing mass. In the ingoing Vaidya convention, a decreasing $m(v)$ corresponds to an ingoing negative-energy flux and violates the null energy condition; the profiles considered here may be viewed as effective evaporation models, rather than classical positive-energy accretion. The argument above used $m' \geq 0$ only for global hyperbolicity and for the sign of the accretion term in the comparison. Both have quantitative replacements, at the price of a domain with a timelike boundary. The metric (22) is Lorentzian on the whole ambient $\{r > 0\}$, where $w = q - 1 + 2m/r > 0$, the (v, r) block has determinant $-w/q$ and v remains a time function; the competitors live in $\Lambda = \{r > 2m(v)\}$. Set

$$F(v) = -2m'' + \frac{m' + (4 + 5/q)\,m'^2 + 4(1+q)\,m'^3/q^2}{m},$$

$$H_q(z) = \frac{z\,[3(2q-3)(q-1) + (33q-38)z + 40z^2]}{3(q-1) + 4z},$$

and, for a launch $o = (v_0, r_0, \Omega_0)$, $R(v) = r_0 + (v - v_0)/2$.

e. Hypothesis (H_{dec}). $q = \hat{E}^2 \geq 3/2$; $m \in C^\infty(\mathbb{R})$ is positive with $-q < m' < 0$ and $F \leq 0$ everywhere; $-m'(v) \leq H_q\big(m(v)/R(v)\big)$ for $v \geq v_0$; the launch has $r_0 > 2m(v_0)$, and the receiver radius satisfies $r_T > 2 \sup m$.

Lemma VII.6 (the boundary)**.** *Under (H_{dec}) the boundary $r = 2m(v)$ of Λ is smooth and timelike, and Λ is light-convex: a null geodesic of the ambient metric contained in $\bar{\Lambda}$ with end points in Λ lies in Λ.*

Proof. With $X = \partial_v + 2m'\partial_r$, tangent to the boundary, $G(X, X) = 4m'(1+m'/q) < 0$. For a null vector k tangent to the boundary, $k^r = 2m'k^v$ and $(2m)^2|k^\Omega|^2 = -4m'(1+m'/q)(k^v)^2$, and a direct computation gives $\mathrm{Hess}_G(r-2m)(k,k) = (k^v)^2 F(v) \leq 0$. Let a null geodesic in $\bar{\Lambda}$ touch the boundary at an interior parameter; parametrize it by v and put $u = r - 2m(v) \geq 0$, with $u = u' = 0$ at the contact. Eliminating the angular velocity with the null condition, which is regular there because $-4m'(1 + m'/q) > 0$, the radial equation reads $u'' = F + E$ with $|E| \leq C(|u| + |u'|)$, so $u'' \leq C(u + |u'|)$. Grönwall applied to $(u')_+$ and u, forwards and backwards, gives $u \equiv 0$ near the contact; the set of contacts is open and closed along the geodesic, which would then lie on the boundary, contradicting its end points. □

Lemma VII.7 (strict comparison)**.** *Under (H_{dec}), along every future causal curve from o, $r(v) < R(v)$ for $v > v_0$, and along every non-radial null geodesic from o in Λ, $K_\perp - K_\parallel > 0$. In particular, in the plane of such a geodesic the in-plane mode has no zero on $(0, \pi]$: within the plane, no point with swept angle in $(0, \pi]$ is conjugate to o.*

Proof. Along a causal curve, $\mathrm{d}r/\mathrm{d}v \leq s_+ = wf/(\sqrt{qw} + w) < f/2 < \frac{1}{2}$, since $w < q$ and $0 < f < 1$; so $r(v) < R(v)$. For $q \geq 3/2$, H_q is strictly increasing on $z > 0$: with $c_0 = 3(2q-3)(q-1) \geq 0$, $c_1 = 33q - 38 > 0$ and $c_2 = 3(q-1) > 0$, the numerator of H_q' is $c_0c_2 + 2c_1c_2z + (4c_1 + 120c_2)z^2 + 320z^3 > 0$; hence $m' + H_q(m/r) > m' + H_q(m/R) \geq 0$, and the formula in the proof of Theorem 6 writes $K_\perp - K_\parallel$ as a positive multiple of $m' + H_q(m/r)$: indeed $mP(r)/r + m'r(3(q-1)r + 4m) = r(3(q-1)r + 4m)\,[m' + H_q(m/r)]$. With u and z as there, $W = z'u - zu'$ has $W' = (K_\perp - K_\parallel)uz > 0$ while $u, z > 0$, so $z > u > 0$ before $\varphi = \pi$; at $\varphi = \pi$, $u = 0$ and $W > 0$ give $z > 0$. □

Lemma VII.8 (null polygons)**.** *Let Λ be an open subset of a stably causal Lorentzian manifold of dimension at least two and $c : [0,1] \to \Lambda$ a future causal curve. There is a future null broken geodesic in Λ with the same end points; in particular it admits an H^1 parametrization.*

Proof. For each point of the compact image of c choose a convex normal neighborhood W with closure in Λ, carrying a local time function t, and a smaller neighborhood U with closure in W such that every future null geodesic from a point of U can be continued inside W at least up to the largest value of t on U; this is possible because null vectors normalized by $\mathrm{d}t = 1$ have uniformly bounded spatial components on compact charts. By compactness and continuity, split c at points $p_0, \ldots, p_n$ so that each subarc is contained in a common U; its end points are then causally related within W. Below, I^-_W and J^-_W denote the chronological and causal pasts relative to W. If $p_i = p_{i+1}$, drop the piece; if they are null-related in W, use the null segment. If $p_i \ll p_{i+1}$ in W, follow a future null geodesic from p_i in W: it starts in $I^-_W(p_{i+1})$, it does not pass through p_{i+1} (otherwise the two points would be null-related in W), and at the level $t(p_{i+1})$ it is at a point different from p_{i+1}, hence outside $J^-_W(p_{i+1})$. Its first crossing of $\partial I^-_W(p_{i+1})$ is a point e null-related to p_{i+1} in W, and p_ie, ep_{i+1} are null segments in W. Concatenate. □

Proposition VII.9 (one ray per class, decreasing mass)**.** *Assume (H_{dec}) and work in the universal cover $\tilde{\Lambda}$ of the equatorial domain, with the receiver $\gamma(v) = (v, r_T, \Theta)$, $v \in \mathbb{R}$, $0 < |\Theta| \leq \pi$. Then there is exactly one future null geodesic from o to γ in $\tilde{\Lambda}$, and it is the unique earliest arrival among all future causal curves in $\tilde{\Lambda}$ from o to γ.*

Proof. Hypotheses of[6] *(Theorem 1.7).* The ambient $\mathbb{R}_v \times (0,\infty) \times \mathbb{R}_\varphi$ is stably causal, v being a time function. By Lemma VII.6 the boundary of $\tilde{\Lambda}$ is smooth, timelike and light-convex. The receiver is timelike, since $G(\partial_v, \partial_v) = -wf/q < 0$ at $r_T > 2\sup m$, and a closed embedding; o is not on it. Sublevels of arrival at most $c \geq v_0$ lie in the compact subset $K(c) = \big([v_0, c] \times [2m_{\min}(c), R(c)] \times [-\Phi_c, \Phi_c]\big) \cap \{r \geq 2m(v)\}$ of the closure, with $m_{\min}(c) = \min_{[v_0,c]} m$ and $\Phi_c = \sqrt{q}\,(c - v_0)/(2m_{\min}(c))$, by $\mathrm{d}r/\mathrm{d}v < \frac{1}{2}$ and $|\mathrm{d}\varphi/\mathrm{d}v| \leq \sqrt{w}/r$; for $c < v_0$ they are empty. Precompactness in the closure is what GMP require[6] (Definition 1.2). The space $\mathcal{L}^+$ is not empty: a radial null segment from o to $r = r_T$, outgoing with $\mathrm{d}r/\mathrm{d}v = s_+$ bounded below on $[r_0, r_T]$ since $m \leq m(v_0)$ there, or ingoing, followed by the null curve at $r = r_T$ with $|\mathrm{d}\varphi/\mathrm{d}v| = \sqrt{wf/q}/r_T$, bounded below because $r_T > 2\sup m$. Every future null geodesic from o to γ has swept angle exactly $|\Theta|$ (its angular momentum is conserved, and $L = 0$ cannot reach $\Theta \neq 0$), so by Lemma VII.7 it has index zero and a non-conjugate end point.

Count. Setting (P) holds with $t = v$, $b = 2m$, $s_\pm = -w \pm \sqrt{qw}$ and $a = \sqrt{wf/q}/r$, with the margin $\sup b = 2\sup m < r_T$, $a(v, r_T) \geq \sqrt{(q-1)f_*/q}/r_T$ with $f_* = 1 - 2\sup m/r_T > 0$, and $s_- \leq -(q-1) - \sqrt{q(q-1)} < 0$ on $r \geq r_T$. We reuse only the tail construction, its H^1 continuity and the contraction in the proof of Proposition III.11; these do not use global hyperbolicity. The outgoing tail from an event of a curve in $\tilde{\Lambda}$ stays above that curve by Lemma III.9, hence in $\tilde{\Lambda}$, and the ingoing and angular tails stay in $r \geq r_T > 2\sup m$; continuity is local on the compact images, not uniform up to the boundary. So $\mathcal{L}^+$ is contractible, and the algebraic count at the end of that proof gives exactly one null geodesic γ_Θ. Non-emptiness was proved above and compactness by $K(c)$; earliest arrival is obtained separately below.

Earliest arrival. Let T be the time function of[6] and $Q(z) = \int_0^1 (\mathrm{d}(T \circ z)/\mathrm{d}s)^2 \mathrm{d}s$, with T the time function chosen there. A curve of $\mathcal{L}^+$ arriving at $\gamma(a)$ satisfies $Q(z) \geq [T(\gamma(a)) - T(o)]^2$, with equality when $T \circ z$ is affine, and $T \circ \gamma$ is increasing. Reparametrize γ_Θ on $[0,1]$ so that $T \circ \gamma_\Theta$ is affine: it then represents the unique critical point of Q, with value $[T(\gamma(a_\Theta)) - T(o)]^2$, $\gamma(a_\Theta)$ being its arrival. If a causal curve in $\tilde{\Lambda}$ arrived before γ_Θ, Lemma VII.8 would give a null broken geodesic z with the same arrival $a < a_\Theta$, which after reparametrization with $T \circ z$ affine has $Q(z) < Q(\gamma_\Theta)$. For $Q(z) < b < Q(\gamma_\Theta)$, b is not a critical value, since the only critical value exceeds it, and the sublevel $\mathcal{L}^+ \cap Q^b$ is not empty, so its zeroth Betti number is at least one; in the Morse relations on sublevels[6] (Theorem 7.1), proved in the same setting with boundary, the constant term gives (number of critical points of index zero below b) $= b_0 + s_0 \geq 1$, while the left side is zero, a contradiction. Finally, a causal curve arriving at the same time as γ_Θ is a null pregeodesic, otherwise local push-up and the timelike receiver would give an earlier arrival, so it is γ_Θ. □

Lemma VII.10 (angular reduction)**.** *Let $z = (v, r, \Omega)$ be a future causal curve in Λ from o, absolutely continuous, and $\theta = d_{S^2}(\Omega, \Omega_0) \in [0, \pi]$. The planar curve $\bar{z} = (v, r, \theta)$, in a fixed plane through the center and Ω_0, is future causal, lies in Λ, arrives at angular distance θ at the same value of v, and its lifted angle stays in $[0, \pi]$.*

Proof. The distance to Ω_0 is 1-Lipschitz on the sphere, so θ is absolutely continuous with $|\theta'| \leq |\Omega'|$ almost everywhere, also where z passes through Ω_0 or its antipode. The cone $(r' + wv')^2/q + r^2|\Omega'|^2 \leq wv'^2$ depends on the angular motion only through $|\Omega'|$, and v, r are unchanged. □

Theorem 10 (the arrival cut for decreasing mass)**.** *Assume (H_{dec}) and $\hat{E} > 0$, and fix a target at radius $r_T > 2\sup m$. Let γ be a non-radial future null geodesic of G from o to that target, contained in Λ, with terminal swept angle δ.*

(1) If $\delta \in (0, \pi)$, γ is the unique global earliest arrival at the target, up to reparametrization, among all future causal competitors in Λ, in three dimensions.

(2) If $\delta = \pi$, γ is an earliest arrival at the target, not unique, and the target event is its first cut point.

Moreover, every segment of a non-radial null geodesic from o contained in Λ with swept angle greater than π is not minimizing, whatever the radius of its end point.

Proof. Use the plane through the center, Ω_0 and the target. By Lemma VII.10 every 3D competitor reduces to a planar causal curve in the class $\Theta = \delta$ with the same arrival, so by Proposition VII.9 no competitor arrives before γ_δ, the planar ray of that class, which is therefore a 3D earliest arrival; $\gamma = \gamma_\delta$ by uniqueness in the class. For $\delta < \pi$, let z be a 3D earliest arrival: then $\bar{z}$ is a planar one, hence $\bar{z} = \gamma_\delta$, whose angle is monotone; since $\bar{z}$ is null, the cone inequality for z forces $|\Omega'| = |\theta'|$ almost everywhere, so the projection of z to the sphere has length δ and is the unique minimizing arc, and $z = \gamma_\delta$. For $\delta = \pi$, the rotations about the axis through the center and Ω_0 give a family of distinct earliest arrivals. No earlier event of γ is a cut point: a chronological advantage at a prefix would propagate to the target by push-up, and a second maximal segment to an earlier event, concatenated with the tail of γ, would create a corner that push-up improves, unless the tangents coincide, which by uniqueness of geodesics would make the two segments equal. So the target event is the first cut point. Beyond π the out-of-plane Jacobi field generated by the rotations, proportional to $r\sin\varphi$ in the plane, has an interior zero; a variation inside a compact neighborhood of the segment contained in Λ gives a timelike connection to the end point, and the static line through the end point, timelike near it since $f > 0$, yields an earlier arrival. This last step is local and needs no margin on the radius of the end point. □

f. An admissible profile. Let $q \in \{3/2, 2\}$, $m(v) = 1 + \frac{1}{5}/\big(1 + e^{(v-5)/5}\big)$ (units with $m_\infty = 1$), $v_0 = 0$, $r_0 = 6$, $r_T > 12/5$. With $s = 1/(1+e^{(v-5)/5})$, $-m' = d = \frac{1}{25}s(1-s) \le \frac{1}{100}$ and $|m''| \le \frac{1}{5}d$, so $-q < m' < 0$, and the bracket in F is at most $-d(1-(4+5/q)d) \le -d/2$; hence $F \le (\frac{2}{5} - \frac{5}{12})d < 0$. For Sturm, $d \le e^{1-v/5}/25$ and $R = 6 + v/2$. Since $H_2(z) = z + z^2(24+40z)/(3+4z) \ge z$ and $Re^{-v/5}$ decreases for $v \ge 0$, $Rd \le 6e/25 < 1 \le m$, so $d < H_2(m/R)$. Since $H_{3/2}(z) = \frac{23}{3}z^2 + \frac{28}{3}z^3/(\frac{3}{2} + 4z) \ge \frac{23}{3}z^2$ and $R^2e^{-v/5}$ decreases, $R^2d \le 36e/25 < \frac{23}{3} \le \frac{23}{3}m^2$, so $d < H_{3/2}(m/R)$. Hypothesis ($\mathrm{H_{dec}}$) holds on the whole future, at the threshold $q = 3/2$ as well. For $q > 3/2$ the bound $H_q(z) \ge (2q-3)z$ gives more room; for a fixed profile the admissible launch radii are bounded, but a smaller amplitude admits any finite r_0.

The statement concerns null geodesics of G, the arrival extremals, not arbitrary rail curves, and it requires the margin $r_T > 2\sup m$ on the receiver. A mass that becomes exactly constant makes the boundary null in the future and is not covered by this argument; neither is a mass whose derivative changes sign, where the boundary is partly spacelike.

VIII. CONFORMAL KERR: A BARRIER, AND SMALL SPIN

For $a \neq 0$ the exterior $r > 2M$ is not globally hyperbolic for g_{opt}: $\Delta(2M) = a^2$, so $r = 2M$ lies at finite h_0-distance, the Randers condition degenerates there, $|\beta|_{h_0} \to 1$, and $N \to 1$. Causal curves reach $r = 2M$ in finite η, consistently with Paper II, where the compact-control problem ends at $r = 2M$. We show that fast competitors cannot get there.

a. Hypothesis (H2). $0 < |a| \le M$, $\hat{E} > 0$, $A \in C^\infty(I)$ positive with $\sup_I A < \hat{E}$ (no freezing on $r > 2M$).

A. The cost of an excursion

Let

$$\theta(r) = 1 - \frac{2M|a|}{r\sqrt{\Delta}}, \qquad c(r) = \theta(r)\sqrt{P_0(r)} = \theta(r)\frac{r}{\sqrt{f\Delta}}, \qquad C(r_1, r_2) = \int_{r_1}^{r_2} c(r)\,\mathrm{d}r, \quad (24)$$

with $f = 1 - 2M/r$ and $P_0 = h_{0,rr}$ at the equator. θ increases, since $r^2\Delta$ does, from 0 at $r = 2M$ to 1, and $C(r_1, r_2)$ stays finite as $r_1 \to 2M^+$ when $a \neq 0$.

Lemma VIII.1 (the equator is the worst latitude)**.** *For $r > 2M$ and $|a| \le M$, the function $F(r,\vartheta) = (1 - |\beta|_{h_0})\sqrt{h_{0,rr}}$ is, at fixed r, minimal at the equator, where it equals $c(r)$; it is non-increasing in $\sin\vartheta$, strictly if $a \neq 0$.*

Proof. With $x = \sin\vartheta$, $\Sigma = r^2 + a^2 - a^2x^2$ and $k = 2M|a|r/\sqrt{\Delta}$ one has $|\beta|_{h_0} = kx/\Sigma$ and $\sqrt{\Delta}\,F = G(x) = (\Sigma - kx)/\sqrt{\Sigma - 2Mr}$, with $(\Sigma - 2Mr)^{3/2}G'(x) = a^2x(4Mr - \Sigma) - k\Delta$. For $a = 0$ it vanishes and F is constant in ϑ. For $a \neq 0$, if $4Mr \le \Sigma$ it is negative. Otherwise $\Sigma \ge r^2$ and $x \le 1$ give $a^2x(4Mr - \Sigma) \le a^2r(4M - r)$, and it remains to show $|a|(4M - r) < 2M\sqrt{\Delta}$ for $2M < r < 4M$: with $r = 2M + s$, $4M^2\Delta - a^2(4M - r)^2 = s(8M^3 + 4Ma^2) + s^2(4M^2 - a^2) > 0$. □

Lemma VIII.2 (excursion cost)**.** *A causal curve that starts at $r \ge r_2$, reaches $r \le r_1$ and returns to $r \ge r_2$ uses a conformal time at least $2C(r_1, r_2)$, at any latitude and with any polar motion.*

Proof. For a future causal vector, $N \ge 1$ and Lemma II.2 give $v^\eta \ge (1 - |\beta|_{h_0})|v_x|_{h_0} \ge F(r,\vartheta)|v^r| \ge c(r)|v^r|$ by Lemma VIII.1. With $B' = c$, $\int c(r)|r'| \ge |B(r_{\mathrm{end}}) - B(r_{\mathrm{start}})|$; apply it to each of the two crossings of $[r_1, r_2]$. This is a total-variation bound: no monotone pieces are needed. □

B. First arrival within the margin

Theorem 11 (conformal Kerr within the margin)**.** *Assume (H2), $2M < r_1 < r_2$, a launch event o at $r_0 \ge r_2$ and a target x_f, different from the spatial point of o, with $r_f \ge r_2$, and a rail curve from o to x_f of duration $T_{\mathrm{up}} < 2C(r_1, r_2)$ with $[\eta_0, \eta_0 + T_{\mathrm{up}}] \subset I$. Then every rail curve from o reaching x_f within T_{up} stays in $r > r_1$; a first-arrival rail curve to x_f exists and is a null pregeodesic of g_{opt} with no interior conjugate point. Moreover, let γ be a rail extremal from o whose end point lies in $r \ge r_2$ and whose elapsed duration is at most T_{up}. For this segment and its prefixes, first arrival is characterized by the null cut point in the confined problem: the segment is globally minimal among all three-dimensional competitors precisely up to that point, the first conjugate point or the first meeting with another minimizing null geodesic from o, and the minimum is unique before it.*

Proof. Choose $\delta > 0$ with $T_{\mathrm{up}} < 2C(r_1 + \delta, r_2)$ and a smooth $\psi \ge 1$ on (r_1, ∞), equal to 1 for $r \ge r_1 + \delta$, with $\int_{r_1} \sqrt{\psi P_0}\,\mathrm{d}r = +\infty$, and replace h_0 by h_0^ψ (radial part multiplied by ψ). The cones of the modified metric $\tilde g$ lie inside those of g_{opt}, and the two metrics coincide on $r \ge r_1 + \delta$. With $b_1 = 2M|a|/(r_1\sqrt{\Delta(r_1)}) < 1$ the slice metric is $\ge (1 - b_1^2)h_0^\psi$, complete on $(r_1, \infty) \times S^2$; the lapse lies between 1 and $(1 - b_1^2)^{-1/2}$ and the shift is bounded by $b_1/(1 - b_1^2)^{1/2}$. On compact time windows lapse and shift bound the speed relative to h_0^ψ, so a causal curve with an interior time end point has a spatial limit and extends: $(\tilde D, \tilde g)$ is globally hyperbolic (equivalently, Fact III.2). By Lemma VIII.2 with $r_1 + \delta$, causal curves of either metric that reach x_f within T_{up} stay where the metrics coincide, so the first arrival is the same in both problems; it is null, hence a rail curve. Proposition III.6 in $\tilde g$ gives existence and the cut-point characterization. For a prefix of a segment ending in $r \ge r_2$, a competitor to the prefix concatenated with the tail is confined by the barrier, so the characterization holds for prefixes too. □

At $r_2 = 6M$ the largest margin $2C(2M^+, r_2)$ is, numerically, $16.7M$ for $a = 0.5M$, $12.9M$ for $a = 0.9M$ and $12.4M$ for $a = 0.99M$ (`kerr_barrier.py` and `kerr_3d_excursion.py` in the accompanying repository). Theorem 11 characterizes the minimum; it does not say which extremal is before its cut point. That is what persistence supplies for small spin.

C. Small spin

Theorem 12 (small spin)**.** *Let γ_0 be an arrival extremal for $a = 0$, from o to x_f, covered by Theorem 2 or Theorem 3, with $\Delta\varphi < \pi$. There is $\delta(\gamma_0) > 0$ such that for $0 < |a| < \delta(\gamma_0)$, with the same A, $\hat{E}$ and end points, the continuation of γ_0 is the unique global minimum among all three-dimensional rail curves.*

Proof. Reference. At $a = 0$, γ_0 is the unique global minimum and has no conjugate point along the reference segment, so its shooting map is invertible; the reference may depend on η.

Common competitor. The spatial arc of γ_0 lifts to a rail curve for small a; its clock depends continuously on a and arrives within some $T_{\mathrm{up}} > T_0$ inside I.

Confinement, inner side. Let $r_2 = \min(r_0, r_f)$ and choose $r_1 \in (2M, r_2)$ with $2C(r_1, r_2; 0) = 2\int_{r_1}^{r_2} \mathrm{d}r/f > T_{\mathrm{up}}$, possible because the integral diverges as $r_1 \to 2M$. At fixed $r_1 > 2M$, $C(r_1, r_2; a)$ is continuous in a, so the strict inequality persists for small a, and Lemma VIII.2 confines all competitors arriving within T_{up} to $r > r_1$.

Confinement, outer side. For $|a| \le a_*$ and $r \ge r_1$,

$$|\beta_a|_{h_{0,a}} \le b_* = \frac{2Ma_*}{r_1\sqrt{r_1^2 - 2Mr_1 + a_*^2}} < 1, \qquad h_{0,a,rr} \ge \Big(1 + \frac{a_*^2}{r_1^2}\Big)^{-1},$$

so a causal curve has $|\mathrm{d}r/\mathrm{d}\eta| \le v_* = \sqrt{1 + a_*^2/r_1^2}/(1 - b_*)$ and stays below $r_0 + v_* T_{\mathrm{up}}$.

Persistence. On the resulting compact shell $f_\vartheta > 0$ uniformly, and $F_a = \sqrt{N_a}|v|_{h_{0,a}} + \beta_a(v)$ is a strongly convex Randers norm with the origin uniformly inside its unit body; its support function is smooth for $p \neq 0$ and depends smoothly on a. Choose the shell with outer radius strictly larger than $r_0 + v_* T_{\mathrm{up}}$ and inner radius r_1 with the barrier margin, so that the spatial constraint is inactive. Lemma VI.1, with the spin as parameter and existence supplied by the barrier, gives a unique minimum in the shell, the continuation of γ_0; by confinement it is global. □

The threshold depends on γ_0, A, $\hat{E}$, the end points and the window, and may tend to zero near $\Delta\varphi = \pi$. We do not show that the cut point stays at π for finite spin. In three dimensions, global selection beyond the margin of Theorem 11, where competitors can approach the stationary limit, requires a separate analysis of the state-constrained problem and is left for future work. On the equator it is settled in section IX: at every spin for constant A, and for variable A under explicit conditions on its derivatives and on the spin.

IX. CONFORMAL KERR ON THE EQUATOR, AT EVERY SPIN

On the equatorial plane, which is totally geodesic for g_{opt} by the reflection symmetry, the arrival problem can be solved at every spin. The competitors are the future causal *equatorial* curves; in three dimensions, beyond Theorems 11 and 12, the problem at finite spin remains open (see the end of the section). We set $M = 1$, write $u = 1/q = A^2/\hat{E}^2$, $f = 1 - 2/r$, $\Delta = r^2 - 2r + a^2$, $r_\pm = 1 \pm \sqrt{1 - a^2}$, and assume $a \ge 0$, the case $a < 0$ following from $\varphi \mapsto -\varphi$. Throughout, $\hat{E}$ is a non-zero constant and A is smooth and positive on an open interval $I \supset [\eta_0, \infty)$ with $q \ge q_{\min} > 1$ there, so $u \le u_{\max} = 1/q_{\min} < 1$; the profile is needed on the whole future, since otherwise classes with many windings are not reachable. The launch $o = (\eta_0, r_0, \varphi_0)$ and the static receiver $\gamma(\eta) = (\eta, r_f, \varphi_f)$ have $r_0, r_f > 2$, and the target is distinct: $r_f \neq r_0$ or $\varphi_f \neq \varphi_0$ modulo 2π.

A. The extended auxiliary metric

On the equator fg_{opt} extends across $r = 2$:

$$\begin{aligned} G &= -f\,\mathrm{d}\eta^2 - \frac{4a}{r}\,\mathrm{d}\eta\,\mathrm{d}\varphi + \frac{Nr^2}{\Delta}\,\mathrm{d}r^2 + G_{\varphi\varphi}\,\mathrm{d}\varphi^2, \\ N &= \frac{1}{1-fu}, \qquad G_{\varphi\varphi} = r^2 + a^2\Big(1+\frac{2}{r}\Big) + \frac{u\Delta}{1-fu}, \end{aligned} \tag{25}$$

on $D = \{(\eta, r, \varphi) :\ \eta \in I,\ r > r_+\}$. For $r > 2$, $G = fg_{\mathrm{opt}}$ is a positive conformal multiple of g_{opt} and has the same null geodesics and conjugate points. For $r_+ < r < 2$, the ergoregion of the seed metric, ∂_η is spacelike and G is not conformal to g_{opt}; the results below are statements about G, and their physical reading inside the ergoregion, where the rail crosses a region in which the coordinates themselves are dragged, is left open.

Lemma IX.1 (structure of G)**.** *For $|a| \le 1$, $u \in [0, u_{\max}]$ and $r > r_+$: $1 - fu > 0$; $\det G = -N^2r^2$; the (η, φ) block has determinant $-N\Delta < 0$; $G^{rr} = \Delta/(Nr^2) > 0$; $G^{\eta\eta} = -(r^4 + a^2r^2 + 2a^2r + 4a^2u)/(r^2\Delta) < 0$, so η is a time function; and $G_{\varphi\varphi} \ge r^2$. Every future causal vector satisfies*

$$\Big|\frac{\mathrm{d}r}{\mathrm{d}\eta}\Big| \le \frac{\Delta}{r\sqrt{G_{\varphi\varphi}}} \le \frac{\Delta}{r^2} \le 1, \qquad \Big|\frac{\mathrm{d}\varphi}{\mathrm{d}\eta}\Big| \le \frac{2a}{r^3} + \frac{\sqrt{N\Delta}}{r^2} \le 2 + \sqrt{N_{\max}},$$

with $N_{\max} = 1/(1 - u_{\max})$, and a null vector tangent to $r = \mathrm{const}$ is a multiple of $X_\pm = \partial_\eta + \Omega_\pm\partial_\varphi$, $\Omega_\pm = (2a/r \pm \sqrt{N\Delta})/G_{\varphi\varphi}$.

Proof. The identities are direct computations (the second form of $G_{\varphi\varphi}$ uses $f(r^2 + a^2 + 2a^2/r) + 4a^2/r^2 = \Delta$). Completing the square in the tangential block,

$$G = -\frac{N\Delta}{G_{\varphi\varphi}}\mathrm{d}\eta^2 + G_{\varphi\varphi}\Big(\mathrm{d}\varphi - \frac{2a}{rG_{\varphi\varphi}}\mathrm{d}\eta\Big)^2 + \frac{Nr^2}{\Delta}\mathrm{d}r^2,$$

so a causal vector has $(Nr^2/\Delta)\dot r^2 \le (N\Delta/G_{\varphi\varphi})\dot\eta^2$, and its angular velocity lies between Ω_- and Ω_+. Finally $\Delta \le r^2$ since $a^2 \le 1 \le 2r$, $r > r_+ \ge 1$, and $N \le N_{\max}$ since $f \le 1$. The last two terms of $G_{\varphi\varphi}$ are non-negative, so $G_{\varphi\varphi} \ge r^2$; hence $|\Omega_\pm| \le (2a/r + \sqrt{N\Delta})/r^2$, which gives the bound on $\mathrm{d}\varphi/\mathrm{d}\eta$ with $a \le 1 < r$ and $\Delta \le r^2$. □

B. Compactness, existence and the count in each class

Let F be a primitive of r^2/Δ on (r_+, ∞): for $|a| < 1$, with $S = r_-$, $F = r + \frac{r_+^2}{r_+ - S}\log(r - r_+) - \frac{S^2}{r_+ - S}\log(r - S)$, and for $|a| = 1$, $F = r + 2\log(r-1) - 1/(r-1)$. In both cases F maps (r_+, ∞) onto $\mathbb{R}$, and by Lemma IX.1, $|\mathrm{d}F(r)/\mathrm{d}\eta| \le 1$ along every causal curve: the boundary $r = r_+$ is never reached in finite time, including at the extremal spin.

Proposition IX.2 (existence in every class)**.** *Let $|a| \le 1$. Then (D, G) and its azimuthal cover $\tilde D$ are globally hyperbolic. For every lifted angle $\Theta \in \Delta\varphi + 2\pi\mathbb{Z}$ the earliest arrival $m(\Theta)$ at the lifted receiver γ_Θ is attained by a null pregeodesic of G, except for $\Theta = 0$ when $r_f = r_0$, where the receiver contains o and $m(0) = \eta_0$; and*

$$m(\Theta) - \eta_0 \ge \frac{|\Theta|}{2 + \sqrt{N_{\max}}}.$$

Proof. η is a time function, so D and $\tilde D$ are causal. Causal curves from an event at η_1 that end before η_2 keep $|F(r) - F(r_1)| \le \eta_2 - \eta_1$, hence r in a compact interval of (r_+, ∞), and

the lifted angle within $(2+\sqrt{N_{\max}})(\eta_2-\eta_1)$ of its initial value. So the causal diamonds lie in compact sets; they are closed by the limit-curve argument of Theorem 5, and causality with compact diamonds is global hyperbolicity[31]. Every lifted receiver is reachable, inside $r>2$: go radially from r_0 to r_f at constant φ, along $\mathrm{d}\eta = \sqrt{Nr^2/(f\Delta)}\,|\mathrm{d}r|$, in finite time since f and Δ are positive and N is bounded between the two radii; then follow the circle $r=r_f$ with angular velocity $\Omega_+>0$ or $\Omega_-<0$, both bounded away from zero for $u\in[0,u_{\max}]$, until the lifted angle Θ; corners are allowed in a witness. Not every vertical line of D is timelike, since $G_{\eta\eta}=-f\geq 0$ in the ergoregion, but the proof of Proposition III.6(1)–(2) uses only the receiver line, which is timelike at $r_f>2$: the earliest arrival before the witness lies in a compact diamond, is attained on the horismos, since a chronological contact could be moved earlier, and is reached by a null pregeodesic. The last bound integrates the angular bound of Lemma IX.1. □

The count needs a time-like direction that does not degenerate in the ergoregion, where ∂_η is spacelike. Since $\partial_u G_{\varphi\varphi} = \Delta/(1-fu)^2 > 0$ and the other tangential coefficients do not depend on u, the interval (Ω_-,Ω_+) of timelike angular velocities shrinks as u grows: the one at $u_{\max}$ is contained in all the others. We therefore fix a field that does not depend on time. Let $S_*(r)=G_{\varphi\varphi}(r,u_{\max})$, choose $r_1\in(2,\min(r_0,r_f))$ and a cut-off $\chi(r)\in[0,1]$ equal to 1 for $r\leq 2$ and to 0 for $r\geq r_1$, and set

$$\omega_*(r)=\chi(r)\,\frac{2a}{rS_*(r)},\qquad T=\partial_\eta+\omega_*(r)\,\partial_\varphi.$$

$2a/(rS_*)$ is the angular velocity of the zero-angular-momentum observers at $u_{\max}$, inside the interval at $u_{\max}$; for $r>2$ so is 0, and the interval is convex. So T is timelike on D for every $u\leq u_{\max}$, and $-G(T,T)$ has a positive lower bound on every compact radial band, uniformly in time. This fixes a reference field, not the factor A. Launch and receiver lie where $\omega_*=0$.

Proposition IX.3 (one ray per class)**.** *Let $|a|\leq 1$ and $\Theta\in\Delta\varphi+2\pi\mathbb{Z}$, with $\Theta\neq 0$ if $r_f=r_0$. If no future null geodesic of $\tilde{D}$ from $\tilde{o}$ to γ_Θ has a point conjugate to $\tilde{o}$ in $(0,1]$, there is exactly one such geodesic, up to reparametrization, and it is the earliest arrival of its class.*

Proof. In the coordinates η, r, $\psi=\varphi-\omega_*(r)\eta$ one has $\partial_\eta=T$ exactly, because ω_* depends only on r. A spatial curve $(r(s),\psi(s))$ from the launch to the receiver, with velocity $V=(0,p,\psi'+\omega_*'\eta p)$, $p=r'$, has a unique future null lift,

$$\eta'=\frac{b+\sqrt{b^2+H_T c}}{H_T},\qquad H_T=-G(T,T)>0,\quad b=G(T,V),\quad c=G(V,V)\geq 0,$$

with $\eta(0)=\eta_0$. Writing $\Phi(x,\eta,x')$ for the right-hand side, with $x=(r,\psi)$, on a compact radial band and time interval

$$|\Phi(x,\eta_x,x')-\Phi(y,\eta_y,y')|\leq C|x'-y'|+C\big(|x-y|+|\eta_x-\eta_y|\big)\big(|x'|+|y'|\big),$$

also at zero velocity, since the square root is a norm on the spatial velocities; and $0\leq\eta'\leq C(1+|\eta|)(|p|+|\psi'|)$. Grönwall excludes blow-up, and with the embedding of H^1 in C^0 the same estimate in L^2 gives locally Lipschitz dependence of the lift in H^1. A null curve with a given spatial projection is this lift, rests included, because T is timelike. So the space $\mathcal{L}^+$ of[6] ((1.4)) is homeomorphic to the space of H^1 spatial curves between the end points; in the cover, whose spatial section is a strip, that space is contractible by interpolation toward a fixed path. The hypotheses of[6] (Theorem 1.7) hold with $\Lambda=\tilde{D}$: $\tilde{D}$ is globally hyperbolic, the receiver is timelike and a closed embedding, o is not on it, the sublevels are precompact by Proposition IX.2, and non-degeneracy is assumed. The algebraic count at the end of the proof of Proposition III.11 then gives one geodesic, and Proposition IX.2 identifies it with the earliest arrival of the class. Classes are preserved because $\omega_*=0$ near both end points. □

C. Only two classes compete

Let $L = G(k, \partial_\varphi)$, conserved along every null geodesic since ∂_φ is a Killing field for every A, and let Θ_0 be the lifted angle at which the null geodesic from o with $L = 0$ reaches $r = r_f$. When $r_f = r_0$ we set $\Theta_0 = 0$ and extend the arrival value by $m(0) = \eta_0$: this value represents the coincident end point and is not a non-constant null ray, and statements that count rays concern distinct targets; m is continuous at 0 by small causal arcs on the circle, and in this case Lemma IX.5 is applied only at $\Theta \neq 0$.

Lemma IX.4 (zero angular momentum)**.** *Along a null geodesic of G with $L = 0$, $\dot r$ never vanishes. Exactly two null geodesics from o have $L = 0$, one outgoing and one ingoing, and when $r_f \neq r_0$ the one directed toward r_f reaches it.*

Proof. If $\dot r = 0$, k lies in the Lorentzian (η, φ) block, where the orthogonal of the spacelike ∂_φ is timelike; $L = 0$ would make k timelike. Along it $\mathrm{d}r/\mathrm{d}\eta = \pm\Delta/(r\sqrt{G_{\varphi\varphi}})$, bounded away from zero between r_0 and r_f, uniformly in η. □

Lemma IX.5 (first variation in the receiver angle)**.** *If c is an earliest arrival at γ_Θ with tangent k and $E_f = -G(k, \partial_\eta) > 0$ at the arrival, then $m(\Theta + \epsilon) \leq m(\Theta) + (L/E_f)\,\epsilon + o(\epsilon)$ for ϵ of either sign.*

Proof. Take a point e' of c in a convex normal neighborhood of the arrival e. The nearby receivers are timelike and meet the null cone of e' once, transversally, at times $\tau(\epsilon)$ with $\tau(0) = m(\Theta)$. Along the variation by null geodesics from e', $G(k, J)$ is constant and vanishes at e'; at the arrival $J = \tau'(0)\partial_\eta + \partial_\varphi$, so $-E_f\tau'(0) + L = 0$. The prefix of c followed by the null segment from e' is causal and arrives at $\tau(\epsilon)$. □

Theorem 13 (two classes)**.** *Let $|a| \leq 1$. If $r_f \neq r_0$, m is strictly decreasing on $(-\infty, \Theta_0]$ and strictly increasing on $[\Theta_0, \infty)$, and the null geodesic with $L = 0$ is the earliest arrival on the circle $r = r_f$. If $r_f = r_0$, m is strictly decreasing on $(-\infty, 0)$ and strictly increasing on $(0, \infty)$. Hence, with $\Theta_- \leq \Theta_0 < \Theta_+ = \Theta_- + 2\pi$ the two lifted angles of the target adjacent to Θ_0, the earliest arrival lies in the class Θ_- or Θ_+, and every further winding in either sense arrives strictly later. On each circle $r = r_f$ there is exactly one physical angle at which the two adjacent classes tie.*

Proof. m is Lipschitz, by a circular arc at r_f, and coercive by Proposition IX.2. At a weak local minimum of m, with $\Theta \neq 0$ when $r_f = r_0$ (there $\Theta = 0$ is the coincident end point), Lemma IX.5 with both signs of ϵ forces $L = 0$. By Lemma IX.4, when $r_f \neq r_0$ the only such point is Θ_0, which is also the global minimum; when $r_f = r_0$ there is none with $\Theta \neq 0$, since $\dot r \neq 0$ along a null geodesic with $L = 0$, so it does not return to r_0. If $a < b$ in $[\Theta_0, \infty)$ had $m(a) \geq m(b)$, the minimum of m on $[a, \infty)$, which exists by coercivity, would be a weak local minimum different from Θ_0; the left side is the same. For the tie, $D(\theta) = m(\theta) - m(\theta - 2\pi)$ is continuous and strictly increasing on $[\Theta_0, \Theta_0 + 2\pi]$, negative at the left end and positive at the right end, so it has exactly one zero. □

Theorem 13 uses neither conjugate points nor the size of A'; it uses the Killing field ∂_φ, the first variation and existence in each class.

D. No conjugate points: three curvature certificates

In $2 + 1$ dimensions the screen is one-dimensional and a transverse Jacobi field obeys $y'' + \mathrm{Ric}(k, k)\, y = 0$ in an affine parameter. If $\mathrm{Ric}(k, k) \leq 0$ along a null geodesic, then $\int y'^2 = \int \mathrm{Ric}(k, k) y^2 \leq 0$ on any interval where y vanishes at both ends, so y has no second zero: there are no conjugate points. Conjugate points are conformally invariant and $\mathrm{Ric}(k, k)$ is not, so the representative can be chosen.

a. The representative $h = G/N$, constant A. In the frame of the zero-angular-momentum observers of h, with $k = e_0 + c\,e_1 + s\,e_2$, $c^2 + s^2 = 1$,

$$\mathrm{Ric}_h(k,k) = Uc^2 + Vs^2 + Js.$$

With $R = r_+ \in [1,2]$, $a^2 = R(2-R)$ and $x = R/r \in (0,1)$ the coefficients are rational in (x, R, u) up to a single square root in J:

$$U = \frac{x^3 \mathcal{D} P_U}{R^3 \tilde{w} H}, \qquad V = -\frac{x^3 P_V}{R^4 \tilde{w}^2 H}, \qquad J^2 = \frac{4x^8 (2-R) \mathcal{D} P_J^2}{R^7 \tilde{w}^3 H^2},$$

with $\tilde{w} = R(1-u)+2ux$, $\mathcal{D} = (1-x)(R(1+x)-2x)$ and $H = (r^4+a^2r^2+2a^2r+4a^2u)x^4/R$, all positive; P_U, P_V, P_J are polynomials. At large r, $U \sim -uz^3/(1-u)$ and $V \sim z^3(5u-3)/(1-u)$, $z = 1/r$: the natural threshold is $u \le 3/5$.

Proposition IX.6 (K_∞)**.** *For constant A with $q \ge 5/3$ and every $|a| \le 1$, $\mathrm{Ric}_h(k,k) \le 0$ for every null k and every $r > r_+$. Hence no equatorial null geodesic of G has conjugate points.*

Proof. On $s \in [-1,1]$ the maximum of $U(1-s^2) + Vs^2 + Js$ is non-positive if $V \le 0$, $V^2 \ge J^2$ and one of: the value at the vertex is non-positive; $V \ge U$; or $U \le 0$ and $J^2 \le U^2$. These are polynomial inequalities in $(x, R, u) \in [0,1] \times [1,2] \times [0, 3/5]$, proved by an exact Bernstein subdivision with 19 cells (section S7), which fails, as it must, if u is allowed up to $2/3$. □

b. Variable A. With $u = u(\eta)$ the curvature acquires jets, and in h they group exactly as

$$\mathrm{Ric}_h(k,k) = R_{\mathrm{stat}} + D_1 u' - C\,c^2\Big[u'' + \frac{f}{2w}u'^2\Big], \qquad C = \frac{2a^2}{r^2 \Delta w}, \quad w = 1 - fu,$$

with D_1 proportional to c. For a weight $\sigma(r)$, on the same null vector,

$$\mathrm{Ric}_{e^{2\sigma}h}(k,k) = \mathrm{Ric}_h(k,k) - \sigma' \,\mathrm{Hess}_h r(k,k) + (\sigma'^2 - \sigma'')(k^r)^2,$$

and $\mathrm{Hess}_h r$ contains no jet, so D_1 and C are the same in every such representative. At $a = 0$ all jets vanish and no condition on A' is needed. Since $-f/(2w) \le \frac{1}{2}$ on $r > r_+$ (there $f > -1$ and $0 < 1-u \le 1$, so $-f(1-u) < 1$, that is $-f < w$),

$$\mathrm{Ric}_{e^{2\sigma}h}(k,k) \le R_\sigma + |D_1||u'| + Cc^2\big[(-u'')_+ + \tfrac{1}{2}u'^2\big],$$

where R_σ is the static part in the weighted representative.

Proposition IX.7 (K_{var})**.** *Let $q \ge 5/2$ on I.*

(1) If $|a| \le 33/50$ and, at every time, $\frac{5}{4}|u'| + \frac{1}{3}(-u'')_+ < 1$, then in $\bar{g}_\infty = N^{-1} g_{\mathrm{opt}}$, $\mathrm{Ric}(k,k) < 0$ for every null k at $r > 2$.

(2) Let $\sigma = -\frac{1}{2}\log(1 - r_+/r)$. If $r_+ \in [R_{\mathrm{lo}}, R_{\mathrm{hi}}]$ is in one of the bands

band	$[R_{\mathrm{lo}}, R_{\mathrm{hi}}]$	*a, approximately*	K_1	K_2
B1	$[8/5, 44/25]$	$[0.650, 0.800]$	3/2	7/2
B2	$[143/100, 8/5]$	$[0.800, 0.903]$	3	9
B3	$[131/100, 143/100]$	$[0.903, 0.951]$	5	19
B4	$[57/50, 131/100]$	$[0.951, 0.9902]$	14	110

and, at every time, $\delta = K_1|u'| + K_2[(-u'')_+ + \frac{1}{2}u'^2] < 1$, then $\mathrm{Ric}_{e^{2\sigma}h}(k,k) \le 0$ for every null k at $r \ge r_+ + (r_+ - 1)/10$.

The band conditions are implied by the uniform one $14|u'| + 110[(-u'')_+ + \frac{1}{2}u'^2] < 1$*; B1 starts at* $a = \sqrt{264}/25$ *and B4 ends at* $a = \sqrt{2451}/50 \approx 0.9902$*.*

Proof. (1) In $\bar{g}_\infty$, with its own static part R_{stat} and jet coefficients D_1, D_2, D_3 (not those of h), the coefficients of u'' and u'^2 are non-positive for $r > 2$, so $\mathrm{Ric} \le R_{\mathrm{stat}} + |D_1||u'| + |D_2|(-u'')_+$; exact Bernstein certificates give $R_{\mathrm{stat}} < 0$ for $q \ge 2$, $|a| \le 33/50$, and $|D_1| \le \frac{5}{4}(-R_{\mathrm{stat}})$, $|D_2| \le \frac{1}{3}(-R_{\mathrm{stat}})$ for $q \ge 5/2$. (2) In the weighted representative R_σ contains the single radical $a\sqrt{\Delta w}$, and so does D_1^2. Exact Bernstein certificates in (x, t, u, R), $t = \tan(\alpha/2)$, prove on each band and for $x \le 10R_{\mathrm{lo}}/(11R_{\mathrm{lo}} - 1)$ the three inequalities $-R_\sigma \ge 0$, $Cc^2 \le K_2(-R_\sigma)$ and $D_1^2 \le K_1^2 R_\sigma^2$; then $\mathrm{Ric}_{e^{2\sigma}h} \le R_\sigma(1 - \delta) \le 0$. □

The rays of case (2) stay where the certificate holds. At a radial minimum of a null geodesic with affine tangent $k = k^\eta X_\pm$, $\mathrm{d}^2r/\mathrm{d}\lambda^2 = \frac{1}{2}G^{rr}(k^\eta)^2\partial_r G(X_\pm, X_\pm)$, with no derivative of u, since G has no mixed terms with $\mathrm{d}r$ and $k^r = 0$ there; an exact certificate shows $\partial_r G(X_\pm, X_\pm) < 0$ on $(r_+, r_+ + (r_+ - 1)/10]$ for $r_+ \in [57/50, 44/25]$ and $u \in [0, 2/5]$, so a null geodesic with both end points beyond 2 has no radial minimum there, and stays in $r > r_+ + (r_+ - 1)/10$. At the boundary itself, $\partial_r G(X_H, X_H) = -(r_+ - 1)/(2(1 - f(r_+)u))$. In case (1) the rays of interest stay in $r > 2$ by a light-convex collar: for $|a| < a^*(q_{\min})$, with

$$a^*(q) = \sqrt{\frac{4q}{\sqrt{(4q+1)^2 + 3} + 4q + 1}} \qquad \text{(increasing, from 0.6234 to } 1/\sqrt{2}\text{)},$$

the prograde tangential cost has negative radial derivative at $r = 2^+$, its numerator $3a^4u^2 + 8a^2u + 32a^2 - 16$ being increasing in u and negative at $u_{\max}$, and the retrograde one is smaller still; so tangent null geodesics have $\mathrm{d}^2r/\mathrm{d}\lambda^2 < 0$ on some $(2, r_*]$, uniformly in u, and $\{r > r_m\}$, $2 < r_m < \min(r_0, r_f, r_*)$, is light-convex. Since $33/50 < a^*(5/2) \approx 0.672$, case (1) applies there.

The certificates are exact, in rational arithmetic, and are distributed as portable files with an independent verifier that recomputes every polynomial, every Bernstein enclosure and the subdivision tree without the code that produced them; the identification of the polynomials with the geometric quantities is a separate symbolic computation. Section S7 of the supplementary material lists the 33 certificates with their variables, domains and sizes, and the steps that the verifier does not check.

E. Global minimality on the equator

Theorem 14 (equatorial global minimality)**.** *Assume one of:*

(i) A constant, $q \ge 5/3$, $|a| \le 1$*; competitors: all future causal equatorial curves in* D*;*

(ii) $q \ge 5/2$, $|a| \le 33/50$*, and the condition of Proposition IX.7(1) at every future time; competitors: all future causal equatorial curves in* $r > 2$*;*

(iii) $q \ge 5/2$*,* r_+ *in one of the bands, and the condition of Proposition IX.7(2) at every future time; competitors: all future causal equatorial curves in* D*.*

Then each winding class, except the class $\Theta = 0$ *of the coincident end point when* $r_f = r_0$*, contains exactly one null geodesic from* o *to the receiver, and it is the earliest arrival of its class; the earliest arrival overall lies in one of the two classes adjacent to* Θ_0*, and is unique unless these two tie; on each circle* $r = r_f > 2$ *they tie at exactly one angle, where the two minimizers are distinct; and the equatorial null cut locus of* o*, restricted to events with* $r > 2$*, is the set of these tie events.*

Proof. (i) Proposition IX.6 gives non-degeneracy, Proposition IX.3 one ray per class, and Theorem 13 the two classes and the tie. (iii) The same, with Proposition IX.7(2) on the

region where the rays stay, which contains them by the radial certificate, which confines every null geodesic between the end points, not only the minimizers.

(ii) The competitors live in $\{r > 2\}$, where $G = f g_{\mathrm{opt}}$; for $a = 0$ the statement is contained in Theorem 2. Choose $2 < r_m < \min(r_0, r_f, r_*)$ with the collar above. The projection $r \mapsto \max(r, r_m)$ keeps η and φ, hence arrival time and class; in the collar the tangential costs of both orientations decrease toward r_m, so removing the radial velocity keeps the curve causal with the same clock. A projected minimizing sequence stays in $r \geq r_m$ with bounded time, radius and lifted angle, a compact subset of $\{r > 2\}$, and its causal limit attains the infimum of the original problem. A minimizer is a null pregeodesic by local push-up, and it cannot touch r_m, where it would have a radial minimum with negative acceleration. The same holds for every null geodesic between the end points: if it went down to r_m or below, it would have a radial minimum in the collar. So all of them lie in $\Lambda = \{r > r_m\}$, whose boundary is timelike ($G^{rr} > 0$) and light-convex in the sense of[6]: a null geodesic of the closure tangent to it from inside would have a radial minimum there with negative acceleration. On Λ the lift holds with $T = \partial_\eta$, the sublevels are precompact in the closure, and[6] (Theorem 1.7) with Proposition IX.7(1) gives one geodesic per class; it is therefore the only one in $\{r > 2\}$. Theorem 13 holds with the same proof: coercivity holds for competitors in $r > 2$, and Lemma IX.5 is used at $r_f > 2$.

Cut locus. Before the tie the winner is unique and has no conjugate point. After the tie event e, consider an event of the losing ray c_1 at $r > 2$: concatenate the prefix of the other minimizer c_2 up to e with the tail of c_1. The tangents of c_1 and c_2 at e are distinct, otherwise uniqueness for the geodesic equation would make the two rays coincide, so the concatenation has a corner, and push-up gives a timelike connection; since the receiver at $r > 2$ is timelike, the chronological advantage becomes a strictly earlier arrival, so c_1 is no longer minimizing beyond e. □

Corollary IX.8 (physical rails)**.** *Assume $\hat{E} > 0$. In cases (i) and (iii) of Theorem 14, suppose that at least one of the G-minimizing curves remains entirely in $r > 2$. Then the minimum arrival time among equatorial rail curves of charge $\hat{E}$ entirely in $r > 2$ is attained and equals the minimum for G. Its minimizers are exactly the G-minimizers that remain entirely in $r > 2$, up to reparametrization. In particular, there are two physical minimizers precisely when the two adjacent classes tie and both tied minimizers remain entirely in $r > 2$; otherwise the physical minimizer is unique.*

Proof. On $r > 2$, $G = f g_{\mathrm{opt}}$ with $f > 0$, and for $\hat{E} > 0$ Proposition II.3 identifies the rail curves of charge $\hat{E}$ with the future null curves of g_{opt}; so the equatorial rails entirely in $r > 2$ are causal curves of G in D, a subset of the competitors of Theorem 14. Let T_G be the minimum for G. Every rail arrives no earlier than T_G, and a G-minimizer entirely in $r > 2$ is null, hence a rail, arriving at T_G: the restricted minimum is T_G and is attained. Conversely, a rail arriving at T_G is a G-minimizer, and it lies in $r > 2$. So the restricted minimizers are exactly the G-minimizers in $r > 2$, and the count follows from Theorem 14. □

If no G-minimizer remains entirely in $r > 2$, the corollary gives no conclusion about attainment in the restricted rail problem. The physical interpretation of G inside the ergoregion, which would decide the rail reading of such curves, remains open; the corollary makes the statement about physical rails independent of that question.

The cut locus is described only for receivers at $r > 2$, where the static receiver is timelike; inside the ergoregion one would need receivers adapted to T. In case (i) the theorem covers the extremal spin $|a| = M$, where no collar exists: at $a = 1$ both $\Omega_\pm$ decrease at r_+, and compactness rests on F. The thresholds are those of the method: below $q = 5/3$ the pointwise criterion fails at large radii for constant A, which does not by itself produce conjugate points, and the weighted asymptotics $\sigma = b/r$, $b = (1 - 2u)/(1 - u)$, suggest, as an indication of the method and not as a theorem, that a threshold at $q = 3/2$ may be reachable also with spin. Open on the equator, at non-zero spin and for variable A beyond the stated conditions: spins above $\sqrt{2451}/50 \approx 0.990152\, M$, and $2 \leq q < 5/2$; at $a = 0$ Theorem 2 already covers every factor with $q \geq 3/2$. In three dimensions only Theorems 11

and 12 are proved here, and Theorem 13 does not transfer: Lemma IX.4 fails when the ray can move in ϑ, and the section $r > 2M$ is simply connected, so windings do not classify homotopy; uniqueness in three dimensions needs the conjugate points of the polar mode, which are not controlled here.

X. THE PROPER-TIME BRANCH

The proper-time branch minimizes a different functional, and the methods above do not apply to it. Its structure is in fact different near $r = 2M$.

Lemma X.1 (proper-time density)**.** *Along a rail curve, for any a,*

$$\mathrm{d}\tau = \frac{A(\eta)^2 f}{\hat{E}}\big(\mathrm{d}\eta - \beta(\mathrm{d}x)\big) = \frac{A(\eta)^2 f\sqrt{N}}{\hat{E}}\,|\mathrm{d}x|_{h_0}. \tag{26}$$

For constant A and $a = 0$ this is the length of $h_\tau = (A^4 N/\hat{E}^2)(\mathrm{d}r^2 + f\,r^2\mathrm{d}\Omega^2)$, whose radial coefficient tends to $A^4/\hat{E}^2$ and whose angular coefficient tends to zero at $r = 2M$.

Proof. With $w = u^\eta - \beta(u) = \hat{E}/(A^2 f)$ (proof of Proposition II.3), $w\,\mathrm{d}\tau = \mathrm{d}\eta - \beta(\mathrm{d}x)$ along the curve, and on the null cone of g_{opt} $\mathrm{d}\eta - \beta(\mathrm{d}x) = \sqrt{N}|\mathrm{d}x|_{h_0}$. The limits follow from $N \to 1$ as $f \to 0$. □

For non-constant A, (26) is a density depending on the clock η, not a static Riemannian metric. For $a = 0$, in the uniform estimate with $A_{\max}$, the cost of angular legs near $r = 2M$ tends to zero while radial motion there costs a finite amount; for constant A this means that the stationary limit is at finite proper-time distance and has zero proper-time diameter.

Theorem 15 (the dive)**.** *Let $a = 0$, let the time domain contain the whole future of the launch, $I \supseteq [\eta_0, \infty)$, and let $A_{\max} = \sup_{\eta \ge \eta_0} A < \hat{E}$, $N_{\max} = (1 - A_{\max}^2/\hat{E}^2)^{-1}$. For a launch at r_0 and a target at r_f, at any angular separation,*

$$\inf \tau \;\le\; \frac{A_{\max}^2\sqrt{N_{\max}}}{\hat{E}}\big[(r_0 - 2M) + (r_f - 2M)\big].$$

An interior proper-time extremal of larger proper time is not a global minimum.

Proof. Go radially to $2M + \epsilon$, along a great circle there, and radially to r_f. By Lemma X.1 the radial legs cost at most $A_{\max}^2\sqrt{N_{\max}}\,\mathrm{d}r/\hat{E}$ and the arc at most $A_{\max}^2\sqrt{N_{\max}}\sqrt{f(2M+\epsilon)}(2M+\epsilon)\pi/\hat{E} \to 0$. The arc is Lipschitz, away from the boundary, and its clock arrives in finite time for each ϵ; corners can be smoothed at arbitrarily small extra cost. Let $\epsilon \to 0$. □

The conformal time of the angular leg diverges like $\epsilon^{-1/2}$, and that of the radial legs logarithmically, so a fixed finite window does not allow the limit. For a constant conformal factor, the circular leg of this explicit competitor requires proper acceleration $|\mathfrak{a}| \sim \hat{E}^2 f^{-3/2}/(4A^3 M)$, whereas its radial legs are geodesic; corners require regularization. This particular approximating family has no uniform thrust bound; a problem with a prescribed thrust bound requires a separate analysis.

Proposition X.2 (completion, constant factor)**.** *Let $a = 0$ and A constant, $c = A^2/\hat{E}$, $q = \hat{E}^2/A^2 > 1$, $R(r) = c\int_{2M}^{r}\sqrt{N}\,\mathrm{d}s$ and $\rho = c\,r\sqrt{Nf}$, so that $h_\tau = \mathrm{d}R^2 + \rho(R)^2\mathrm{d}\Omega^2$. The metric completion adds a single point p_*, with $d(x, p_*) = R(r)$; it is a proper length space. For the identification with the infimum over rail curves, assume that the time domain contains the whole future of the launch and that the arrival time is otherwise unrestricted.*

With ℓ_{int} the infimum of the lengths of interior geodesics between the end points ($+\infty$ if there are none),

$$\inf_{\text{rail curves in } r>2M} \tau = \min\{\ell_{\mathrm{int}},\, R(r_0) + R(r_f)\}.$$

If $R(r_0) + R(r_f) < \ell_{\mathrm{int}}$, the infimum is not attained by any rail curve.

Proof. Near $R = 0$, $\rho^2 \sim 2McR \to 0$, so all Cauchy sequences toward $r = 2M$ have the same limit, and none escapes to infinity because $R \to \infty$. The closed ball of radius L about p_* is the compact quotient of $[0, L] \times S^2$ with $\{0\} \times S^2$ collapsed, so the completion is proper; Hopf–Rinow for proper length spaces gives a minimizer. One through p_* has parts of lengths $R(r_0)$ and $R(r_f)$, hence is two radial segments; one avoiding p_* stays away from the boundary and is a smooth interior geodesic. Every rail curve in $r > 2M$ has proper time equal to the h_τ-length of its spatial path (Lemma X.1); conversely every rectifiable interior spatial path on a compact interval is the path of a rail curve with finite clock, its image being compact in $r > 2M$, and the time domain contains the whole future. So the infimum over rail curves is the distance in the completion, computed by interior curves. A path through p_* is approximated by two radial segments to $r = 2M + \epsilon$ joined by an arc at that radius, of length $O(\rho(2M + \epsilon)) \to 0$. Hence the infimum is $\min\{\ell_{\mathrm{int}}, R(r_0) + R(r_f)\}$, and if $R(r_0) + R(r_f) < \ell_{\mathrm{int}}$ every minimizer of the completion passes through p_*, so no rail curve attains it. □

In case of equality one must check whether an interior geodesic of the same length exists. The completion describes a limit of the cost; curves through p_* are not admissible.

Theorem 16 (antipodal targets)**.** *Under the hypotheses of Proposition X.2, including the unrestricted future, for antipodal end points the proper-time infimum is exactly $R(r_0)+R(r_f)$ and is attained by no rail curve in $r > 2M$.*

Proof. $\mathrm{d}\rho/\mathrm{d}R = (r - M)/(r\sqrt{f}) + M\sqrt{f}/(r(q - f)) > 1$, because $(r - M)^2 - r(r - 2M) = M^2 > 0$; with $\rho(0) = 0$ this gives $\rho(R) > R$ for $R > 0$. The metric $\mathrm{d}R^2 + \rho^2\mathrm{d}\Omega^2$ thus dominates the Euclidean metric in polar form $\mathrm{d}R^2 + R^2\mathrm{d}\Omega^2$, in which antipodal points are at distance $R(r_0) + R(r_f)$, realized by the dive. A curve in $r > 2M$ changes its angular direction at some $R > 0$, where $\rho > R$, and cannot attain equality. □

A. Equal radii: the critical separation

For constant A, $a = 0$, unrestricted arrival time and end points at the same radius $r_0 = r_f$, write $\rho_0 = \rho(R_0)$ and, on the plane of a geodesic, $w(\rho) = \mathrm{d}R/\mathrm{d}\rho \in (0, 1)$ (by the proof of Theorem 16).

Lemma X.3 (symmetric dips)**.** *Every non-radial interior geodesic joining the two end points is a symmetric dip with a single periastron $\rho = L \in (0, \rho_0)$ (its Clairaut constant), sweeping*

$$\Delta(L) = 2\int_L^{\rho_0} \frac{L\, w(\rho)}{\rho\sqrt{\rho^2 - L^2}}\,\mathrm{d}\rho, \qquad 0 < \Delta(L) < 2\arccos\frac{L}{\rho_0} < \pi,$$

with length $\ell(L)$. Δ extends continuously by zero to $L = 0$ and $L = \rho_0$, so $\Delta_{\max} = \max\Delta < \pi$, attained in the interior.

Proof. In arclength, $\dot{R}^2 + L^2/\rho^2 = 1$ and $\ddot{R} = L^2\rho'/\rho^3 > 0$: R is strictly convex, there is no apoastron, and a geodesic returning to the same radius has one periastron and two symmetric halves. The bound follows from $w < 1$ and $2\int_L^{\rho_0} L\,\mathrm{d}\rho/(\rho\sqrt{\rho^2 - L^2}) = 2\arccos(L/\rho_0)$. As $L \to \rho_0$ the bound gives $\Delta \to 0$. As $L \to 0$, $\rho^2 \sim 2McR$ gives $w(\rho) \sim \rho/(Mc)$; for fixed small ρ_1 the part over $[L, \rho_1]$ is $O(L\,\mathrm{arcosh}(\rho_1/L))$ and the rest is $O(L)$, so $\Delta \to 0$. The substitution $\rho = L\sec\theta$ removes the singularity and gives continuity inside. □

Lemma X.4 (first variation)**.** *With $\mathcal{S}(L) = \ell(L) - L\Delta(L) = 2\int_{R_t}^{R_0} \sqrt{1 - L^2/\rho^2}\,\mathrm{d}R$ one has $\mathcal{S}'(L) = -\Delta(L)$, hence*

$$\ell'(L) = L\,\Delta'(L), \qquad \ell(L) - 2R_0 = L\Delta(L) - \int_0^L \Delta(s)\,\mathrm{d}s,$$

with $\ell(0^+) = 2R_0$ and $\ell(\rho_0^-) = 0$.

Proof. The boundary term at the moving periastron vanishes because the integrand does, and differentiation under the integral gives $\mathcal{S}' = -\Delta$. The limits follow from the same regularized integrals and $w = O(\rho)$ at the inner end. □

Proposition X.5 (critical separation)**.** *Let $d(\delta)$ be the distance in the completion between the two end points at angular separation $\delta \in [0, \pi]$. There is δ_c with $0 < \delta_c < \Delta_{\max} < \pi$ such that $d(\delta) < 2R_0$ for $\delta < \delta_c$ and $d(\delta) = 2R_0$ for $\delta \geq \delta_c$. Below δ_c the proper-time minimum is attained by at least one interior geodesic; at δ_c an interior geodesic ties with the dive; above δ_c the infimum is the dive and is not attained by any rail curve in $r > 2M$.*

Proof. The dive gives $d \leq 2R_0$, and d is continuous in δ. If the minimizer at δ_2 is interior, rescaling φ by $\delta_1/\delta_2 < 1$ along its radial profile gives a strictly shorter curve at δ_1; if it passes through p_*, $d(\delta_2) = 2R_0 \geq d(\delta_1)$. So d is non-decreasing, strictly so at every value with an interior minimizer. For small δ the arc at R_0 has length $\rho_0\delta < 2R_0$, and for $\delta > \Delta_{\max}$ no interior geodesic exists, so $d = 2R_0$: a unique threshold δ_c exists. At any maximizer L_m of Δ, Lemma X.4 gives $\ell(L_m) - 2R_0 = L_m\Delta_{\max} - \int_0^{L_m} \Delta > 0$; the set of maximizers is compact and away from the ends, so all dips with angle close to $\Delta_{\max}$ are longer than the dive and $\delta_c < \Delta_{\max}$. At δ_c, interior minimizers for $\delta_n \uparrow \delta_c$ have L_n bounded away from 0 and ρ_0 (where $\Delta \to 0$), and a limit gives an interior geodesic of length $2R_0$. Above δ_c an interior minimizer of length $2R_0$ would, rescaled to an angle in (δ_c, δ), give a curve shorter than the distance there. □

Uniqueness of the interior minimizer below δ_c follows once Δ has a single maximum. This is not true for every q, but it is a theorem for $q \geq 3/2$. The reason is that the dip angle has already been studied in Paper II. Write $b = q - 1$ and $V(r) = r^2(r - 2M)/(br + 2M)$, the potential of Paper II. For constant $A > 0$ the proper-time radius satisfies $\rho^2 = A^2V$, while $c\sqrt{N}/\rho = 1/\sqrt{r(r - 2M)}$; the factors of A cancel in the angle, which is

$$\tfrac{1}{2}\Delta = \sqrt{V_t}\int_t^{r_0} \frac{\mathrm{d}r}{\sqrt{r(r - 2M)}\,\sqrt{V(r) - V_t}}, \qquad V_t = V(t),$$

which is the proper-time half-angle Φ_τ of Paper II at frozen energy $E^2 = q$ and $a = 0$ ([2], proposition 'where the τ half-angle decreases' and its appendix). Moreover $L = A\sqrt{V(t)}$ and $w(A\sqrt{V}) = 2VW$, with $W = K/V'$ as in Paper II, so our edge formula for $\Delta'(L)$ is its equation (app-g) after this rescaling. The number of maxima, the critical periastron and the angles do not depend on A; R, ρ, L and ℓ scale with it.

Theorem 17 (a single maximum for $q \geq 3/2$)**.** *Let $q \geq 3/2$ and $r_0 > 2M$. Then Δ has exactly one critical point on $(2M, r_0)$, a non-degenerate maximum. Hence:*

(1) every $0 < \delta < \Delta_{\max}$ is the angle of exactly two dips, a deep one and a shallow one, and $\delta = \Delta_{\max}$ of exactly one;

(2) the deep dip is longer than the dive, $\ell > 2R_0$;

(3) along the shallow branch ℓ decreases strictly and equals $2R_0$ at exactly one periastron t_*, with $\delta_c = \Delta(t_*)$;

(4) for $0 < \delta < \delta_c$ the proper-time minimizer is unique up to reparametrization, the shallow dip of angle δ. At δ_c the minimizers in the metric completion are exactly that dip and the radial path through p_*; in the physical domain $r > 2M$ only the dip is attained, the radial path being the limit of a sequence of dives.

Proof. The single non-degenerate maximum is part (v) of Paper II's proposition, applied at $a = 0$, $E^2 = q$. Its proof sets $g = VW$, $W = K/V'$, $K = 1/\sqrt{r(r-2M)}$, and shows $g' > 0 > g''$ for $E^2 - 1 \geq 1/2$ by polynomials with positive coefficients; the concavity of g then gives a single zero of Δ'. Since $L = \rho(t)$ is increasing in t, the same holds in L. Items (1)–(3) then follow from $\ell'(L) = L\Delta'(L)$ and $\ell(L) - 2R_0 = L\Delta(L) - \int_0^L \Delta$ (lemma X.4). On the deep branch Δ increases, so $\int_0^L \Delta < L\Delta(L)$. On the shallow branch ℓ decreases from a value above $2R_0$ (at the maximum) to 0. Item (4) combines (1)–(3) with proposition X.5 and proposition X.2. □

Below $q = 3/2$ the single maximum can fail. Paper II certifies, by interval arithmetic, that at $E^2 = q = 1.01$ and $r_0 = 10^4 M$ the derivative of Φ_τ has signs $+, -, +, -$ at periastra $3M, 6M, 30M, 9000M$. Hence Δ has at least two maxima, and there are more than two dips for some angles. Proposition X.5 does not use the single maximum and still holds. Items (1)–(4) need it, and below 3/2 we establish it on a grid.

Proposition X.6 (certified grid)**.** *For $q = 6/5$ and $r_0 \in \{4M, 6M, 10M, 20M\}$, Δ has exactly one critical point, a maximum, so items (1)–(4) of theorem 17 hold. For the twenty pairs $q \in \{6/5, 3/2, 2, 4, 10\}$, $r_0 \in \{4M, 6M, 10M, 20M\}$, the enclosures of δ_c/π are those of table II; in particular $\delta_c < \pi/2$ at $r_0 = 4M$ for $q = 6/5, 3/2, 2$.*

TABLE II. Certified enclosures of δ_c/π (end points rounded outwards).

$q \backslash r_0$	$4M$	$6M$	$10M$	$20M$
6/5	[0.478032, 0.478041]	[0.560637, 0.560647]	[0.628871, 0.628880]	[0.691007, 0.691015]
3/2	[0.486257, 0.486267]	[0.575456, 0.575466]	[0.652406, 0.652416]	[0.725796, 0.725805]
2	[0.493846, 0.493856]	[0.588406, 0.588417]	[0.671513, 0.671523]	[0.751130, 0.751141]
4	[0.504191, 0.504201]	[0.605064, 0.605075]	[0.694412, 0.694424]	[0.778986, 0.778998]
10	[0.509874, 0.509885]	[0.613780, 0.613792]	[0.705767, 0.705779]	[0.792024, 0.792036]

Proof (computer-assisted). Since angles and the number of maxima do not depend on A, we set $M = A = 1$. With the regular kernel $r = t + (r_0 - t)u^2$, $u \in [0, 1]$, and $\rho^2(r) - \rho^2(t) = (r - t)W/D$, where $W = (q-1)rt(r + t - 2) + 2(r^2 + rt + t^2 - 2r - 2t)$ and $D = ((q-1)r + 2)((q-1)t + 2)$, the integrands of Δ and ℓ are smooth on $[0, 1]$ and analytic in the periastron t. The claim then reduces to finitely many sign conditions: $\Delta' > 0$ near $2M$ and $\Delta' < 0$ near r_0, from explicit monotone bounds in the variable ρ; a tiling of the remaining interval of periastra by pieces on which $\Delta' > 0$, $\Delta'' < 0$ or $\Delta' < 0$, with no piece of the last kind before one of the first, which leaves exactly one non-degenerate zero of Δ'; and a sign change of $\ell - 2R_0$ across a short interval on which $\Delta' < 0$, which brackets δ_c. Each condition is verified in outward-rounded interval arithmetic, with q an interval containing the rational value; the end-point bounds, the pieces and the remainders are in section S8 of the supplementary material. Items (1)–(4) then follow as in the proof of theorem 17. □

The certificate was also run for the sixteen grid cases with $q \geq 3/2$, as an independent check of theorem 17; all pass. The enclosures agree with thirty-digit quadrature. Over the grid, δ_c increases with r_0 and with q; this is an observation, not a proved monotonicity.

B. Selection of the minimizer: the Maxwell rule

Theorem 17 uses the single maximum only to select the minimizer. The selection itself reduces to a scalar problem for every $q > 1$. We keep the setting of this section and use the function $\mathcal{S}$ of lemma X.4, extended by $\mathcal{S}(0) = 2R_0$ and $\mathcal{S}(\rho_0) = 0$, and

$$F_\delta(L) = \mathcal{S}(L) + \delta L, \qquad L \in [0, \rho_0], \qquad \frac{\mathrm{d}F_\delta}{\mathrm{d}L} = \delta - \Delta(L).$$

Derivatives of Δ are taken in L in this subsection.

Lemma X.7 (analyticity)**.** *For $q > 1$, Δ and $\mathcal{S}$ are real analytic on $(0, \rho_0)$, and Δ is constant on no interval.*

Proof. In the regular kernel of the proof of proposition X.6 the integrand depends analytically on the periastron t on compact subsets of $(2M, r_0)$, with a positive denominator, and integration over $u \in [0, 1]$ preserves analyticity. $L = \rho(t)$ is analytic with $\mathrm{d}L/\mathrm{d}t > 0$. Δ is positive inside and tends to zero at both ends (lemma X.3), so it is not constant, and a non-constant analytic function is constant on no interval. □

Theorem 18 (Maxwell rule, equal radii)**.** *Let $q > 1$, $r_0 > 2M$, and let $\hat{\mathcal{S}}$ be the lower convex envelope of $\mathcal{S}$ on $[0, \rho_0]$. For $\delta \in (0, \pi]$:*

(1) $d(\delta) = \min_{[0,\rho_0]} F_\delta$. The minimizers in the completion are, up to reparametrization, the radial path through p_* when $L = 0$ minimizes and the dips of Clairaut constant L for the interior minimizers L of F_δ; these have $\Delta(L) = \delta$. $L = \rho_0$ never minimizes.

(2) The minimizers are the contact points of $\mathcal{S}$ with its supporting line of slope $-\delta$. The minimizer is unique except at angles δ for which $-\delta$ is the slope of a maximal affine face of $\hat{\mathcal{S}}$ with at least two contact points; these angles form an at most countable set. Between two minimizers $L_1 < L_2$ at such an angle,

$$\int_{L_1}^{L_2} \big(\Delta(s) - \delta\big)\, \mathrm{d}s = 0.$$

A face may have further contact points in its interior.

(3) The critical separation of proposition X.5 is

$$\delta_c = \max_{0<L\le\rho_0} \frac{1}{L} \int_0^L \Delta(s)\, \mathrm{d}s.$$

The maximum is attained inside, where $\Delta(L_c) = \delta_c$ and $\ell(L_c) = 2R_0$.

(4) d is concave and non-decreasing. Every choice of minimizer $L_*(\delta)$ is non-increasing, the one-sided derivatives of d are $d'_+(\delta) = \min \operatorname{argmin} F_\delta$ and $d'_-(\delta) = \max \operatorname{argmin} F_\delta$, and $d' = L_*$ where the minimizer is unique.

(5) Let L_c^+ be the largest maximizer in item (3). The minimizer is unique for every $0 < \delta < \delta_c$ if and only if Δ is non-increasing on $[L_c^+, \rho_0]$; it is then strictly decreasing there and $\mathcal{S}$ is strictly convex. Otherwise there is $\delta_i \in (0, \delta_c)$ at which two distinct dips in $r > 2M$ are both global minimizers.

Proof. (1) By proposition X.2 a minimizer exists in the completion and is the radial path through p_*, of length $2R_0 = F_\delta(0)$, or an interior geodesic. By lemma X.3 an interior geodesic is a dip of some $L \in (0, \rho_0)$ with $\Delta(L) < \pi$, so for $\delta \le \pi$ neither the other orientation nor windings occur, and its length is $\ell(L) = \mathcal{S}(L) + L\Delta(L) = F_\delta(L)$. Hence $d(\delta)$ is the minimum of F_δ over $\{0\} \cup \{\Delta = \delta\}$. F_δ is continuous on $[0, \rho_0]$ (lemma X.4); an interior minimum has $\Delta = \delta$, and ρ_0 is not a minimum because $\Delta \to 0$ there, so $\mathrm{d}F_\delta/\mathrm{d}L > 0$ nearby. Thus the global minimum of F_δ lies in that set and equals $d(\delta)$. Note that $F_\delta(L)$ for $\Delta(L) \neq \delta$ is not the length of a curve to the target; only its minimizers are.

(2) L minimizes F_δ if and only if the line of slope $-\delta$ through $(L, \mathcal{S}(L))$ supports the graph. If it touches at two points, $\hat{\mathcal{S}}$ is affine between them, so both lie on a maximal affine face of that slope. Distinct maximal faces of positive length have disjoint interiors, hence countably many slopes. For two contacts $\mathcal{S}(L_2) - \mathcal{S}(L_1) = -\delta(L_2 - L_1)$, which is the equal-area identity by $\mathcal{S}' = -\Delta$.

(3) $L = 0$ minimizes if and only if $\delta L \ge 2R_0 - \mathcal{S}(L) = \int_0^L \Delta$ for all L, that is $\delta \ge m(L) = L^{-1} \int_0^L \Delta$. The mean m is continuous on $(0, \rho_0]$, tends to zero at 0, and $m' = (\Delta - m)/L < 0$

near ρ_0, so its supremum is an interior maximum L_c with $\Delta(L_c) = m(L_c)$, and lemma X.4 gives $\ell(L_c) = 2R_0$.

(4) d is a minimum of affine functions of δ with slopes $L \geq 0$. For $\delta_1 < \delta_2$ with minimizers L_1, L_2, adding the two minimality inequalities gives $(\delta_2 - \delta_1)(L_2 - L_1) \leq 0$. The one-sided derivatives follow from the compactness of the argmin (Danskin's theorem[34]).

(5) By (4) and comparison with L_c^+ at δ_c, every minimizer for $\delta < \delta_c$ lies in $(L_c^+, \rho_0]$, and L_c^+ itself does not minimize there since $\mathrm{d}F_\delta/\mathrm{d}L = \delta - \delta_c < 0$ at it. If Δ is non-increasing on $[L_c^+, \rho_0]$, lemma X.7 makes it strictly decreasing, so $\mathcal{S}$ is strictly convex there and every supporting line touches once. If not, $\mathcal{S}$ is not convex on $[L_c^+, \rho_0]$, so $\hat{\mathcal{S}} < \mathcal{S}$ somewhere in (L_c^+, ρ_0). On a component (a, b) of the open set $\{\hat{\mathcal{S}} < \mathcal{S}\}$ in (L_c^+, ρ_0), $\hat{\mathcal{S}}$ is affine and touches $\mathcal{S}$ at a and b, which are therefore two minimizers of F_{δ_i}, with $-\delta_i$ the slope of that face; $a \geq L_c^+$ because L_c^+ is a contact point. Since $\mathcal{S}$ decreases strictly, $\delta_i > 0$. The slopes of $\hat{\mathcal{S}}$ do not decrease and the face of slope $-\delta_c$ through $(0, 2R_0)$ has L_c^+ as its last contact, so $\delta_i < \delta_c$. Then $a \neq L_c^+$, where $\mathrm{d}F_{\delta_i}/\mathrm{d}L = \delta_i - \delta_c < 0$, and $b \neq \rho_0$ by (1); both contacts are interior, so they are dips. □

The construction is the classical one of conjugate functions and convex envelopes[35]; for Clairaut geodesics on surfaces of revolution see[36,37], and for Maxwell strata and wave fronts[38,39]. What is specific here is the vertex p_* of the completion, which makes $L = 0$ a competitor and gives item (3). For $q \geq 3/2$, Δ has a single maximum and theorem 17 is the special case in which the only face is the one from $L = 0$. Item (5) is the exact uniqueness criterion, and the second branch of Δ beyond L_c^+ is what matters, not the number of maxima: at $q = 1.01$, $r_0 = 10^4 M$, where Δ has two maxima, a numerical evaluation finds both maxima under the face from $L = 0$ and Δ decreasing after its contact point, so the minimizer is unique below δ_c (not certified). Uniqueness can nevertheless fail.

Proposition X.8 (an interior Maxwell angle)**.** *Let $q = 1001/1000$ and $r_0 = 1500M$. There is a unique angle δ_* with*

$$\delta_*/\pi \in [0.677106398,\ 0.677112765]$$

at which exactly two interior dips, with periastra in $[21.33M, 30M]$ and $[105M, 140M]$, are global minimizers in the completion, up to reparametrization. Both are strictly shorter than the radial path through p_*, so $\delta_* < \delta_c$ and the minimizer is not unique below δ_c.

Proof (computer-assisted). Three steps, each verified in the interval arithmetic of proposition X.6 with q an interval containing $1001/1000$. (A) On two intervals of periastra Δ is strictly decreasing and crosses every angle of a short window $[\delta_a, \delta_b]$, so each interval contributes exactly one dip, which minimizes F_δ there. (B) The difference of the two lengths changes sign on the window and is strictly increasing in δ by lemma X.4, so there is exactly one tie δ_*. (C) Every other candidate, the radial path through p_* included, is longer, by enclosures of F_δ on cells of periastra. The intervals, the excluded cells and the margins are in section S8 of the supplementary material. □

The value of δ_* may depend on q within the interval; the certificate does not classify the Maxwell angles outside the stated interval, nor give δ_c for this case. In the theorem, $\mathrm{d}^2F_\delta/\mathrm{d}L^2 = -\mathrm{d}\Delta/\mathrm{d}L$ is the convexity of the reduced scalar problem, not the second variation of length.

C. Unequal radii

Let $2M < r_f < r_0$, with $\rho_i = \rho(R_i)$, and write the half-angle from periastron L to radius r_i as $A_i(L) = \int_L^{\rho_i} L\, w\, \mathrm{d}\rho/(\rho\sqrt{\rho^2 - L^2})$. By the convexity of R (proof of lemma X.3) an interior geodesic from P_f to P_0 is either *monotone*, with angle $\Phi_- = A_0 - A_f$, or a *dip* with periastron below r_f and angle $\Phi_+ = A_0 + A_f$, with $L \in (0, \rho_f]$. Both families meet at the geodesic tangent to $r = r_f$, of angle $\delta_{\tan} = A_0(\rho_f)$. Since $w < 1$, $A_i(L) < \arccos(L/\rho_i) < \pi/2$, hence

$\Phi_\pm < \pi$; for $\delta \le \pi$ the other orientation and windings, which need a swept angle of at least $2\pi - \delta \ge \pi$, do not occur. The radial path through p_* has length $R_0 + R_f$. Parametrized by the launch angle $\theta \in (0, \pi)$ at P_f, with $L = \rho_f \sin\theta$, the front $\theta \mapsto (\Phi, \ell)$ is smooth, also at $\theta = \pi/2$, and $\mathrm{d}\ell = L\,\mathrm{d}\Phi$ along it.

Theorem 19 (unequal radii)**.** *Let $A > 0$ be constant, $a = 0$, $q \ge 3/2$ and $r_0 > r_f > 2M$. Then Φ_+ has a single non-degenerate maximum $\Phi_{\max}$, and there is a unique critical separation $0 < \delta_c < \Phi_{\max}$. Below it the unique minimizer, up to reparametrization, is the monotone geodesic when $\delta \le \delta_{\tan}$ and the shallow dip when $\delta_{\tan} < \delta < \delta_c$. Above it only the radial path through p_* minimizes in the completion and the physical infimum is not attained. At δ_c the interior geodesic and the radial path tie in the completion.*

Proof. Single maximum. With $x = (L/A)^2$, $V_i = V(r_i)$, $y_i = V_i - x$ and $g = VW$ as in theorem 17, identity (app-g) of Paper II gives for each half-angle

$$J_i(x) = \sqrt{x}\,\frac{\mathrm{d}A_i}{\mathrm{d}x} = 2\sqrt{y_i}\int_0^1 g'(x + y_i u^2)\,\mathrm{d}u - \frac{g(V_i)}{\sqrt{y_i}},$$

$$J_i'(x) = -\frac{1}{\sqrt{y_i}}\int_0^1 g'\,\mathrm{d}u + 2\sqrt{y_i}\int_0^1 g''(x + y_i u^2)(1 - u^2)\,\mathrm{d}u - \frac{g(V_i)}{2y_i^{3/2}}.$$

For $q \ge 3/2$, $g' > 0 > g''$ (Paper II, part (v)), so $J = J_0 + J_f$ is strictly decreasing on $(0, V_f)$. It tends to $+\infty$ as $x \to 0$ and to $-\infty$ as $x \to V_f$, through the term $-g(V_f)/\sqrt{y_f}$. Hence $\mathrm{d}\Phi_+/\mathrm{d}x$ has one zero x_{pk}, with $\mathrm{d}^2\Phi_+/\mathrm{d}x^2 = J'(x_{\mathrm{pk}})/\sqrt{x_{\mathrm{pk}}} < 0$. Applied to a single term, the same computation is a shorter variant of Paper II's proof of part (v), under the same signs of g.

Ordering. Φ_- increases strictly, since its domain $[\rho_f, \rho_0]$ is fixed and the integrand increases with L, and $\Phi_+ > \Phi_-$. So a dip and a monotone geodesic of equal angle have $L_{\mathrm{dip}} < L_{\mathrm{mono}}$, and two dips of equal angle are ordered by $\sin\theta$. Hence, of two points of the front with equal angle, the one with larger θ has smaller L.

Cusp. By the single maximum, each $\delta \in (0, \Phi_{\max})$ is the angle of exactly two interior geodesics, $\theta_1(\delta) < \theta_2(\delta)$, which meet at the cusp $\delta = \Phi_{\max}$. With $\mathrm{d}\ell_k/\mathrm{d}\delta = L_k$,

$$\ell_2(\delta) - \ell_1(\delta) = \int_\delta^{\Phi_{\max}} \big(L_1(s) - L_2(s)\big)\,\mathrm{d}s > 0.$$

ℓ_1 increases from $R_0 - R_f$ (the radial geodesic), and ℓ_2 increases from $R_0 + R_f$ (the dip tending to p_*), so $\ell_1(\Phi_{\max}) = \ell_2(\Phi_{\max}) > R_0 + R_f$ and there is a unique δ_c with $\ell_1(\delta_c) = R_0 + R_f$. Proposition X.2, with $R_0 + R_f$ for $2R_0$, completes the proof. If $\delta_c \le \delta_{\tan}$, no dip ever minimizes. □

For $\delta > \delta_{\tan}$ the monotone geodesics are no longer candidates (at $\delta = \delta_{\tan}$ the tangent geodesic is the common end of both families), and the scalar minimization and the supporting-line construction of theorem 18 apply on the dips, with $\mathcal{S}_+ = \ell_+ - L\Phi_+$ and $L \in [0, \rho_f]$; if $\delta_c \ge \delta_{\tan}$, then $\delta_c = \max_{0<L\le\rho_f} L^{-1}\int_0^L \Phi_+$. The concavity of d holds only in that regime: along the monotone branch $\mathrm{d}^2\ell/\mathrm{d}\delta^2 = \mathrm{d}L/\mathrm{d}\delta > 0$. Open: the case $\delta < \delta_{\tan}$ when Φ_+ has several maxima, which requires comparing the monotone branch with the dips; the number of maxima of Δ for $1 < q < 3/2$ off the grid; and the number of interior Maxwell angles.

For ingoing Vaidya with frozen mass the same rail and proper time give the same results with $A = 1$, $M = m_0$.

Remark X.9 (spin)**.** For $a \ne 0$ the equatorial coefficients of h_τ behave differently: at the stationary limit the radial one tends to zero but the azimuthal one tends to $A^4a^2/\hat{E}^2$. In the full three-dimensional domain $f_\vartheta > 0$ the boundary nevertheless collapses to one point of the completion (constant A, $0 < |a| < M$): along $r = r_s(\vartheta) + \epsilon$, $r_s = M + \sqrt{M^2 - a^2\cos^2\vartheta}$,

one has $\Sigma - 2Mr = \epsilon(2r_s - 2M + \epsilon)$ and $r_s'^2/\Delta \le a^2/(M^2 - a^2)$, so meridians cost $O(\sqrt{\epsilon})$ and connect any two boundary points through the pole. In the strictly equatorial problem the boundary is instead a curve of positive length; that problem is treated in section XII. The distance from the collapsed boundary in three dimensions is not treated here.

XI. THE PROPER-TIME BRANCH WITH A TIME-DEPENDENT FACTOR

In this section $a = 0$, $M > 0$, $\hat{E} > 0$, and $A \in C^1(I)$ is positive with $\sup_I A < \hat{E}$; thus only C^1 regularity is needed in place of the smoothness assumed in (H1). We assume $I \supseteq [\eta_0, \infty)$ and an unrestricted arrival time. With $f = 1 - 2M/r$, N as in (6) and

$$\omega(\eta, r) = \frac{A(\eta)^2 f(r)}{\hat{E}}, \qquad v = \frac{f}{\sqrt{N}}, \qquad w_0 = \frac{\sqrt{f}}{r\sqrt{N}},$$

lemma X.1 gives $\mathrm{d}\tau = \omega\,\mathrm{d}\eta$ along a rail curve, and a rail curve is a graph $x(\eta)$ with $|\dot{x}|_{h_0} = N^{-1/2}$. For constant A the reduction to the Riemannian metric h_τ is the stationary proper-time reduction of Perlick[9] (Proposition 3.1 there); for $A' \neq 0$ the selector ∂_η is not a Killing field and $\mathrm{d}\tau$ depends on the clock.

A. Relaxation, radial reduction and existence

A *relaxed curve* is a Lipschitz graph $x(\eta)$, $\eta \in [\eta_0, T]$, with $|\dot{x}|_{h_0} \le N^{-1/2}$, i.e. a causal curve of g_{opt}, with cost $C = \int_{\eta_0}^{T} \omega(\eta, x(\eta))\,\mathrm{d}\eta$. This set of velocities is the convex hull of the rail cone, and C equals the proper time only on rails. The cost contains no derivative: the proper time is the integral in conformal time of a positive potential under a speed limit.

Lemma XI.1 (radial reduction)**.** *For every relaxed curve from x_0 to x_f there is a rail curve with the same radial profile $r(\eta)$, the same duration and the same cost.*

Proof. Write $|\dot{x}|_{h_0}^2 = \dot{r}^2/f^2 + (r^2/f)|\dot{\Omega}|^2$. On a rail with radial profile r the angular speed is $w = (\sqrt{f}/r)\sqrt{N^{-1} - \dot{r}^2/f^2}$, and the relaxed curve has $|\dot{\Omega}| \le w$, so $B = \int w\,\mathrm{d}\eta$ is at least its angular length, hence at least the angular distance δ of the end points. Let $\alpha : [0, B] \to S^2$ be a minimal arc of length δ followed by an out-and-back arc of total length $B - \delta$, parametrized by arclength, and put $\Omega(\eta) = \alpha(\int_{\eta_0}^{\eta} w)$. This is Lipschitz with speed w almost everywhere. The cost depends only on (η, r). □

Hence the infima over rails and over relaxed curves coincide for every A, and the problem is equivalent to choosing a radial profile with $|\dot{r}| \le v$, $r(\eta_0) = r_0$, $r(T) = r_f$, subject to the angular budget $\int w_0\sqrt{1 - u^2}\,\mathrm{d}\eta \ge \delta$, where $\dot{r} = vu$.

Proposition XI.2 (minimizers are rails)**.** *Assume $x_f \neq x_0$. Every relaxed minimizer attained at a finite time inside I is a rail curve almost everywhere. No assumption on the sign of A' is needed.*

Proof. Write $\dot{x} = v_{\mathrm{s}} \sum u_i E_i$ in an h_0-orthonormal frame, $|u| \le 1$, $v_{\mathrm{s}} = N^{-1/2}$, and adjoin the clock to the state, with costate p_η. The maximized Hamiltonian is $\mathcal{H} = p_\eta + v_{\mathrm{s}}|p|_{h_0^*} + p_0\omega$. The free clock at the target gives $p_\eta(T) = 0$, and the free final time of the autonomous system gives $\mathcal{H} \equiv 0$. If $p_0 = 0$ then $|p(T)| = 0$ and all multipliers vanish, so $p_0 = -1$. Where $p \neq 0$, $|u| = 1$. On $\{p = 0\}$ the adjoint equation gives $\dot{p} = \mathrm{d}_x\omega$, whereas an absolutely continuous function has zero derivative almost everywhere on its zero set; since $\partial_r\omega = 2MA^2/(\hat{E}r^2) > 0$, that set is null. □

Proposition XI.3 (the angular budget of a minimizer)**.** *Let $0 < \delta < \pi$ be the angle between the end directions. For a radial profile with $\dot{r} = vu$, $|u| \le 1$, call $\int_{\eta_0}^{T} w_0\sqrt{1 - u^2}\,\mathrm{d}\eta$*

its angular budget. Every global minimizer attained at a finite time, relaxed or rail, saturates it:

$$\int_{\eta_0}^{T} \frac{\sqrt{f}}{r\sqrt{N}} \sqrt{1 - \frac{N\dot{r}^2}{f^2}}\, \mathrm{d}\eta = \delta,$$

and its direction runs along the unique short great-circle arc between the end directions, without returns. In particular every minimizer is planar, in the plane through the center and the two end points, and its angular momentum $L = p_\varphi$ is a non-zero constant.

Proof. A relaxed curve with a given radial profile has angular speed at most $w = w_0\sqrt{1-u^2}$, so its spherical length, which is at least δ, is at most the budget $B_{\mathrm{ang}} = \int w\,\mathrm{d}\eta$. Let a minimizer have $B_{\mathrm{ang}} > \delta$. Replacing its direction by the short arc run at the speed $(\delta/B_{\mathrm{ang}})w$ gives a relaxed curve with the same radial profile, duration and cost, hence a minimizer, whose velocity is strictly inside the rail cone wherever $w > 0$, a set of positive measure since $B_{\mathrm{ang}} > 0$. This contradicts Proposition XI.2. Hence $B_{\mathrm{ang}} = \delta$. The minimizer itself is a rail almost everywhere, so its spherical length is $B_{\mathrm{ang}} = \delta$, the spherical distance of its end directions; a curve of that length is the short great-circle arc, which is unique for $\delta < \pi$. Finally $L = 0$ would give $\dot{\varphi} = 0$ almost everywhere, against $\delta > 0$. □

For constant A the radial barrier of proposition X.2 has the non-stationary analogue

$$\begin{aligned} C \;\geq\; \int \frac{A^2\sqrt{N}}{\hat{E}}\,|\mathrm{d}r| \;\geq\; \big|R_a(r_2) - R_a(r_1)\big| \quad \text{when } A \geq a, \\ R_a(r) = \frac{a^2}{\hat{E}} \int_{2M}^{r} \Big(1 - \frac{a^2 f(s)}{\hat{E}^2}\Big)^{-1/2} \mathrm{d}s, \end{aligned} \tag{27}$$

because N increases with A and $f|\mathrm{d}x|_{h_0} \geq |\mathrm{d}r|$.

Theorem 20 (existence and persistence)**.** *Let $A \geq a_* > 0$ on $[\eta_0, \infty)$ and $x_f \neq x_0$.*

(1) If some rail curve costs less than $R_{a_*}(r_0) + R_{a_*}(r_f)$, a global minimizer exists, and every minimizer, relaxed or not, is a rail curve.

(2) Let $A_\epsilon \geq a_*$, $\sup A_\epsilon \leq A_{\max} < \hat{E}$, $A_\epsilon \to A_0$ uniformly on compact sets, with A_0 constant, and let the frozen problem have a unique minimizer γ_0 of cost $C_0 < R_{a_*}(r_0) + R_{a_*}(r_f)$. Then for small ϵ minimizers γ_ϵ exist, are rails, and every choice converges to γ_0 uniformly as a graph over η, with arrival times and costs converging.

Proof. (1) By (27) a minimizing sequence of relaxed curves stays in a shell $r_1 \leq r \leq r_2$; there $\omega \geq a_*^2 f(r_1)/\hat{E}$, which bounds the durations. The curves are 1-Lipschitz for h_0; extended as constants after arrival, Arzelà–Ascoli gives a uniform limit. The speed constraint passes to the limit through $d_{h_0}(x(\eta), x(\eta')) \leq \int_\eta^{\eta'} N^{-1/2}$, and the cost converges because it contains no derivative. Lemma XI.1 turns the relaxed minimizer into a rail of equal cost, and proposition XI.2 shows that every minimizer is a rail. (2) The frozen minimizer, slowed by a factor $\lambda < 1$, is admissible for A_ϵ for small ϵ, with cost tending to C_0/λ; hence $\limsup m_\epsilon \leq C_0$. The barrier and the duration bound are uniform, and every limit of minimizers is a relaxed frozen minimizer, which for constant A is the rail γ_0. □

B. Necessary conditions

By the spherical symmetry minimizers are planar (the vector angular momentum is conserved; Proposition XI.3 gives the plane and the arc). In the plane, with heading ψ against the outward radial direction, $\dot{r} = v\cos\psi$, $\dot{\varphi} = w_0 \sin\psi$.

Proposition XI.4 (extremals)**.** *Along a normal extremal with $p \neq 0$, with $|p|^2 = f^2 p_r^2 + fL^2/r^2$:*

$$\cos\psi = \frac{f p_r}{|p|}, \qquad \sin\psi = \frac{\sqrt{f}L}{r|p|}, \qquad |p| = \sqrt{N}(\omega - p_\eta),$$
$$p_\eta(T) = 0, \qquad \dot{p}_\eta = \frac{AA' f}{\hat{E}}\Big[\frac{N(\omega - p_\eta)}{\hat{E}} + 2\Big],$$

with $L = p_\varphi$ constant. Hence the generalized Clairaut relation

$$\rho_{\text{eff}} \sin\psi = L, \qquad \rho_{\text{eff}} = r\sqrt{N/f}\,(\omega - p_\eta).$$

For constant A, $p_\eta \equiv 0$ and $\rho_{\text{eff}} = (A^2/\hat{E})\, r\sqrt{Nf}$ is the ρ of proposition X.2. If A is monotone on $[\eta, T]$, $p_\eta(\eta)$ has the sign opposite to that of A'. Where the minimal remaining cost $V(\sigma, x)$ from the event (σ, x) of a minimizer is differentiable in σ, with $p_\eta = -\partial_\sigma V$, waiting at x in the relaxed problem during $[\sigma, \sigma + \epsilon]$ and then continuing optimally changes the cost by $\epsilon\,(\omega - p_\eta(\sigma)) + o(\epsilon) = \epsilon\,|p|/\sqrt{N} + o(\epsilon)$, which is positive for small ϵ.

The identities follow from the maximization of the Hamiltonian; the last statement is the first variation of the remaining cost with respect to its starting clock, the wait itself costing ω per unit clock.

C. The escape price

Write $A_{\max} = \sup_{\eta \geq \eta_0} A$ and $N_{\max} = (1 - A_{\max}^2/\hat{E}^2)^{-1}$. Let $D_\downarrow(\eta_0, r_0)$ be the limit, as $r \downarrow 2M$, of the cost of the full-speed radial descent $\dot{\rho} = -v(\eta, \rho)$, $\rho(\eta_0) = r_0$. It is finite, and the descent reaches $2M$ only as $\eta \to \infty$, since $|\dot{r}| \leq f$ for every relaxed curve.

Lemma XI.5 (optimal descent)**.** *For every A as above and every $r_\epsilon \in (2M, r_0)$, among relaxed curves from (η_0, r_0) to their first arrival at r_ϵ, with free final direction, the full-speed radial descent minimizes the cost.*

Proof. A competitor has $\dot{r} \geq -v(\eta, r)$, so by scalar comparison $r(\eta) \geq \rho(\eta)$ before its arrival, and it arrives no earlier than the descent, at T_*. Since ω is strictly increasing in r, $C \geq \int_{\eta_0}^{T_*} \omega(\eta, r(\eta))\,\mathrm{d}\eta \geq \int_{\eta_0}^{T_*} \omega(\eta, \rho(\eta))\,\mathrm{d}\eta$. □

Theorem 21 (escape price)**.** *Let $A(\eta) \to A_\infty > 0$ as $\eta \to \infty$, and let*

$$B = \liminf_{\epsilon \downarrow 0} \{C(\gamma) : \min_\gamma r \leq 2M + \epsilon\}$$

over relaxed curves joining the end points in finite time. Then

$$B = D_\downarrow(\eta_0, r_0) + R_{A_\infty}(r_f),$$

and if the infimum m of the problem satisfies $m < B$, a global minimizer exists and is a rail curve. Equivalently, if no minimizer exists then $m = B$.

Proof. ($\leq$) Descend radially to $2M + \epsilon$, which happens after a conformal time of order $2M \log(1/\epsilon)$; there turn to the final direction along an arc of angle at most π, of duration at most $\pi r_\epsilon \sqrt{N_{\max}}/\sqrt{f_\epsilon}$ and cost at most $\pi r_\epsilon A_{\max}^2 \sqrt{N_{\max}}\sqrt{f_\epsilon}/\hat{E}$, with $r_\epsilon = 2M + \epsilon$, $f_\epsilon = f(r_\epsilon)$; then ascend radially, at a time when $|A - A_\infty| < \kappa$, at cost at most $R_{A_\infty + \kappa}(r_f)$. Let $\epsilon \to 0$, then $\kappa \to 0$. Corners are admissible. ($\geq$) For a competitor reaching $r_\epsilon = 2M + \epsilon$ first at σ_ϵ, lemma XI.5 bounds the part before σ_ϵ below by the descent cost to r_ϵ. From $|\dot{r}| \leq f$, $\sigma_\epsilon \geq \eta_0 + r_0 - r_\epsilon + 2M \log((r_0 - 2M)/\epsilon)$ uniformly over competitors, so for small ϵ one has $A \geq A_\infty - \kappa$ after σ_ϵ, and (27) bounds the rest below by $R_{A_\infty - \kappa}(r_f) - R_{A_\infty - \kappa}(r_\epsilon)$. If $m < B$, a minimizing sequence cannot approach $2M$; the outer barrier and the duration bound confine it, and the proof of theorem 20(1) applies. □

This refines theorem 15, which uses $A_{\max}$ on both legs: the descent pays A along its own history, the ascent pays A_∞. For constant A it is the dive $R(r_0)+R(r_f)$ of proposition X.2. Equality $m = B$ does not by itself decide whether a minimizer exists; an interior minimizer can tie with the escape price.

D. Global selection near a constant factor

Write $q_0 = \hat{E}^2/A_0^2$ for the frozen energy parameter, $p_{\eta,\mathrm{in}} = p_\eta(\eta_0)$ for the initial clock costate, $\|g\|_{C^1_b(J)} = \sup_J |g| + \sup_J |g'|$, and $\Phi_A(\psi_{\mathrm{in}}, p_{\eta,\mathrm{in}}, T) = (r(T) - r_f,\ \varphi(T) - \delta,\ p_\eta(T))$ for the planar shooting map.

Theorem 22 (perturbative global selection)**.** *Let $0 < A_0 < \hat{E}$ and $0 < \delta < \pi$. Suppose that the frozen problem has a unique interior global minimizer γ_0, with data $(\psi^*_{\mathrm{in}}, 0, T_0)$ and $C_0 < R_{A_0}(r_0) + R_{A_0}(r_f)$, and that its planar shooting map is non-degenerate. There is $\varrho > 0$ such that, if $\|A - A_0\|_{C^1_b([\eta_0,\infty))} < \varrho$:*

(1) the proper-time problem has a unique global minimizer, up to reparametrization, among all admissible three-dimensional rail curves. It is planar and is the extremal whose data $z(A)$ form the unique zero of Φ_A near $(\psi^*_{\mathrm{in}}, 0, T_0)$;

(2) $A \mapsto z(A)$ is continuous, and $|p_{\eta,\mathrm{in}}| \le C\,\|A'\|_{L^\infty([\eta_0,\eta_0+H])}$ with C, H uniform;

(3) for $A_\epsilon = A_0 + \epsilon\alpha$, $\alpha \in C^1_b$,

$$m_\epsilon = C_0 + \epsilon \int_{\eta_0}^{T_0} \frac{A_0 f(N_0+1)}{\hat{E}}\,\alpha(\eta)\,\mathrm{d}\eta + o(\epsilon),$$

the integral being taken along γ_0, with $N_0 = (1 - A_0^2 f/\hat{E}^2)^{-1}$.

Proof. Choose $a_* < A_0$ with $C_0 < R_{a_*}(r_0) + R_{a_*}(r_f)$. For ϱ small, theorem 20 gives global minimizers, a uniform shell and a uniform duration H, and convergence to γ_0. By propositions XI.2 and XI.3 every minimizer is a normal extremal, planar and on the short arc because $0 < \delta < \pi$, with $L \neq 0$ since $\delta > 0$; uniform convergence to γ_0 fixes its angular lift. The clock costate solves a linear equation with coefficients of order $d = \|A'\|_{L^\infty([\eta_0,\eta_0+H])}$ and $p_\eta(T) = 0$, so Grönwall gives $\|p_\eta\|_\infty \le C_H d$. Then $|p| = \sqrt{N}(\omega - p_\eta)$ is bounded below on the shell, the extremal equations are C^1 in the state, and the shooting data of every minimizer lie in a compact set; continuity of the flow identifies every limit with the data of γ_0. For constant A, $p_\eta(T) = p_{\eta,\mathrm{in}}$, so the last row of $D\Phi_{A_0}$ is $(0,1,0)$ and $\det D\Phi_{A_0} = -\det \partial(r,\varphi)/\partial(\psi_{\mathrm{in}}, T) \neq 0$. The right-hand sides involve A and A' but not A'', so Φ_A is C^1-close to Φ_{A_0} and has a single zero in a fixed neighborhood U of the reference data. The data of every global minimizer lie in U and a global minimizer exists, so it is unique and is that zero. For $A_\epsilon = A_0 + \epsilon\alpha$ with $\alpha \in C^1_b$ the right-hand sides depend on ϵ only through the values A_ϵ and A'_ϵ, which are affine in ϵ, and smoothly on them on the interior tube, so Φ is C^1 in (z, ϵ) without using α'', and $z(A_\epsilon)$ is C^1 in ϵ by the implicit function theorem. For item (3), with the adjoint equation and the stationarity of the control, the variation of the cost is $-\int \partial_\epsilon \mathcal{H}\,\mathrm{d}\eta$ plus the boundary term $p_\eta(T_0)T'_0 = 0$; at $\epsilon = 0$, $p_\eta = 0$, $|p| = \sqrt{N_0}\,\omega_0$ and $N_0 A_0^2 f/\hat{E}^2 = N_0 - 1$ give the integrand. □

The theorem is global among all competitors: *perturbative* refers only to the distance of A from a constant, and the proof brings every possible minimizer into the neighborhood of the reference. A merely convergent family $A_\epsilon \to A_0$ gives $z(A_\epsilon) \to z(A_0)$, but the rate $p_{\eta,\mathrm{in}} = O(\epsilon)$ needs $\|A_\epsilon - A_0\|_{C^1} = O(\epsilon)$: the family $A_0 + \sqrt{|\epsilon|}\,\alpha$ can have $p_{\eta,\mathrm{in}}$ of order $\sqrt{|\epsilon|}$. Uniform C^1 closeness on the whole future is sufficient, not necessary: it suffices to have global bounds $0 < a_* \le A \le A_{\max} < \hat{E}$ with $C_0 < R_{a_*}(r_0) + R_{a_*}(r_f)$, and C^1 closeness to A_0 on a common window $[\eta_0, \eta_0 + H]$ containing every competitive curve; the barrier

supplies such a uniform H. The weight $A_0 f(N_0+1)/\hat{E}$ is positive, so raising A along the way raises the cost to first order.

For $a = 0$, Theorem 2 certifies the exact arrival extremals underlying the adiabatic expressions of Papers I–II whenever $q \geq 3/2$ throughout and the swept angle is less than π. This concerns minimality of the exact extremal, not the accuracy or exact optimality of its first-order approximation; the expansion of the minimizer itself is in section XI F. For the proper-time branch, Theorem 22 certifies the continuation to a fixed target of a unique, non-degenerate frozen minimizer below the escape barrier. A slow family may be treated on a fixed time window provided global bounds supply a uniform strict barrier margin; slowness alone, or a bounded excursion of the factor, does not supply this margin.

Corollary XI.6. *The hypotheses of theorem 22 hold for $q_0 \geq 3/2$, $r_0 \geq r_f > 2M$ and $0 < \delta < \delta_c$, with δ_c from theorem 17 or 19, including the tangent separation $\delta = \delta_{\tan}$ whenever $\delta_{\tan} < \delta_c$.*

Proof. Uniqueness and $C_0 < R(r_0) + R(r_f)$ are those theorems. For non-degeneracy with transversal arrival, $\psi_{\text{in}} \mapsto L$ and $T \mapsto \ell$ (with $\ell_T = \omega > 0$) are local coordinates. For $F(L,\ell) = (r,\varphi)$, dependent columns would give, choosing $\ell(L)$ so as to stay on $r = r_f$, an arrival angle Θ with $\Theta'(L) = 0$; but $\Theta' \neq 0$ on the minimizing branch: $\Delta' < 0$ on the shallow branch, $\Phi'_- > 0$ on the monotone geodesics, and the argument with J in the proof of theorem 19 gives a single zero of Φ'_+ with strict signs on each side. At the tangent the arrival at r_f is not transversal. Launch the frozen geodesic backwards from the inner end point with angle $\theta = \pi/2$, a regular coordinate: writing $\Phi = A_0(L) + \operatorname{sgn}(\theta - \pi/2) A_f(L)$ with $L = \rho_f \sin\theta$, one has $A_f(L) = w(\rho_f)|\theta - \pi/2| + o(|\theta - \pi/2|)$ while $\frac{\mathrm{d}}{\mathrm{d}\theta} A_0(\rho_f \sin\theta)\big|_{\theta=\pi/2} = 0$, so $\mathrm{d}\Phi/\mathrm{d}\theta = w(\rho_f) > 0$; and the arrival at r_0 is transversal. Full rank of the endpoint map is equivalent to the absence of a non-zero Jacobi field vanishing at both ends, which is symmetric under reversal of the frozen geodesic. □

Remark XI.7 (numerical observations)**.** The following are floating-point diagnostics, not certified. For $q_0 = 2$, $r_0 = r_f = 6M$, $\delta = \pi/2$ and A dropping smoothly from 1 to 1/2 around η_{d}:

- shooting on $(\psi_{\text{in}}, p_{\eta,\text{in}}, T)$ and a direct optimization of the reduced problem agree to 10^{-6}–10^{-5};
- the continued interior extremal crosses the escape price of theorem 21 near $\eta_{\mathrm{d}} \approx 8.03$; its family folds near $\eta_{\mathrm{d}} \approx 10.35$;
- for $\eta_{\mathrm{d}} = 15$ no interior extremal was found, and the direct optimization drifts toward $r = 2M$ with cost approaching $B \approx 3.938$.

A semi-Lagrangian dynamic program for the value function in (η, r, b), with b the remaining angular budget, on domains that contain every competitor of cost below the candidate, gives values compatible with the minimality of the continued extremal at $\eta_{\mathrm{d}} = 4$ and for an increasing factor. Theorem 22 does not quantify ϱ; finite variations with explicit data are treated in section XI E.

E. Finite variations of the factor: explicit examples

Theorem 22 does not quantify ϱ. For explicit data the selection can be proved for finite variations of A, including a pulse of amplitude 1/2 and a drop from 1 to 1/2. In this subsection $M = 1$, $\hat{E} = \sqrt{2}$ and $r_0 = r_f = 6$. By Proposition XI.3 every minimizer lies in the plane of the end points; in it, with $J = p_\varphi$ constant,

$$\omega = \frac{A^2 f}{\sqrt{2}}, \quad a_H = \frac{f^2}{N}, \quad b_H = \frac{f}{N r^2}, \quad S = \sqrt{a_H p_r^2 + b_H J^2}, \quad \mathcal{H} = -\omega + S,$$

$\dot r = a_H p_r/S$, $\dot\varphi = b_H J/S$, and the clock costate is $p_\eta = -\mathcal{H}$, so $S = \omega - p_\eta$; S is $|p|/\sqrt{N}$ of Proposition XI.4. Derivatives in r are taken at fixed A, p_r, J, and $\alpha = \frac{1}{2}(\log a_H)_r$, $\gamma = \frac{1}{2}(\log b_H)_r$. For constant A write $\rho_A = \omega/\sqrt{b_H} = (A^2/\sqrt{2})\, r\sqrt{Nf}$ and R_A as in (27), so that $R_A' = (A^2/\sqrt{2})\sqrt{N}$ and

$$\frac{\mathrm{d}\rho_A}{\mathrm{d}R_A} = \frac{r-1}{r\sqrt{f}} + \frac{A^2\sqrt{f}}{r\,(2-A^2 f)} > 1 \qquad (r>2,\ 0<A\le 1). \tag{28}$$

In the coordinates (R_A, φ) the frozen metric h_τ is $\mathrm{d}R_A^2 + \rho_A^2 \mathrm{d}\varphi^2$; by (28) it dominates the Euclidean metric $\mathrm{d}R_A^2 + R_A^2 \mathrm{d}\varphi^2$.

A pulse at a small separation

Theorem 23 (global selection for a spherical pulse)**.** *Let $\delta = 1/50$ and $A_\tau = 1 - \frac{1}{2}\sin^2(\pi\eta/\tau)$ on $[0,\tau]$, $A_\tau = 1$ afterwards. For every $\tau > 0$ the proper-time global minimizer is unique among all three-dimensional rail curves, as a curve parametrized by the clock. Its arrival time $T(\tau)$ is in $C^1((0,\infty))$, and there is exactly one τ^* with $T(\tau^*) = \tau^*$, with*

$$\frac{10}{61} < \tau^* < \frac{4000}{24253}, \qquad T'(\tau^*) - 1 < -\frac{7}{10}.$$

The minimizer crosses the end of the pulse for $\tau < \tau^*$ and arrives before it for $\tau > \tau^*$. The two reflected orientations of the target have the same arrival time and threshold.

Structure of the proof. The argument is that of Theorems 30 and 31, without the magnetic term and with new constants. (1) *Localization*: every minimizer stays in a thin shell around $r = 6$, arrives at a clock in $(4/27, 1/5)$, and ends within $1/5$ of the tangential heading; only $1/2 \le A \le 1$ is used. (2) *Uniqueness*: on the rectangle of arrival clocks and terminal headings, explicit bounds on the coefficients of the variational equations give the hypotheses of the triangular criterion (Lemma XII.10), so the backward shooting map has at most one zero. (3) *Sector change*: at every zero of $T(\tau)-\tau$ the dilation argument gives $T'-1 < -7/10$, so the zero is unique, and closed-form bounds on the mean angular speed enclose τ^*. The constants are in section S9 of the supplementary material. □

Corollary XI.8 (an explicit radius of selection)**.** *Let $\delta = 1/50$ and $A_0 = 3/4$, so $q_0 = 32/9$. Every A with $\|A - 3/4\|_{C_b^1([0,\infty))} < 1/4$ has a unique proper-time global minimizer among all three-dimensional rail curves, and its clock costate obeys $\|p_\eta\|_{L^\infty([0,T])} \le \frac{25}{97}\|A'\|_\infty \le \frac{1}{3}\|A'\|_\infty$; the last inequality is strict if $\|A'\|_\infty > 0$, and $p_\eta \equiv 0$ if A is constant. Thus $\varrho = 1/4$ is admissible in item (1) of Theorem 22 for this reference, with the backward shooting map on Q in place of Φ_A.*

Proof. Steps 1–2 of the previous proof use only $1/2 \le A \le 1$ and $T\,|(2/A + A/(3 - A^2))A'(T)| \le 46/7$ for $T \in [3/25, 21/100]$; the factor in brackets is at most $46/11$, so $|A'| \le 1100/147$ suffices, and here $|A'| < 1/4$. On the shell $f \le 17/25$, $N \le 50/33$ and $T < 1/5$; writing the equation of Proposition XI.4 as $p_\eta' = b_0 + b_1 p_\eta$, $p_\eta(T) = 0$, and putting $d = \|A'\|_\infty < 1/4$, one has $|b_0| \le \frac{5}{4}d$ and $|b_1| \le \frac{3}{5}d$. Grönwall gives $\|p_\eta\|_{L^\infty([0,T])} \le \frac{d}{4}e^{3d/25} \le \frac{25}{97}d \le \frac{d}{3}$, using $e^{3/100} \le 100/97$. □

Structure in the separation

Proposition XI.9 (the escape threshold in the separation)**.** *Let $r_0 = r_f = r_* > 2M$ and $0 < a_* \le A \le A_{\max} < \hat E$ with $A \to A_\infty > 0$; let $m_A(\delta)$ be the infimum at separation $\delta \in [0,\pi]$ and B the escape price of Theorem 21, which does not depend on δ. Then m_A*

is non-decreasing and Lipschitz, $m_A(0) = 0$ and $m_A \le B$. If a minimizer is attained at $\delta_2 > 0$, then $m_A(\delta_1) < m_A(\delta_2)$ for every $\delta_1 < \delta_2$. If $\{\delta : m_A(\delta) = B\}$ is not empty, it is an interval $[\delta_c, \pi]$ with $\delta_c > 0$: minimizers exist for $\delta < \delta_c$ and do not exist for $\delta > \delta_c$. No monotonicity of A is assumed.

Proof. Lowering the angular budget on the same radial profile keeps the cost and the relaxed admissibility, so m_A is non-decreasing; a circular arc of angle $|\delta_2 - \delta_1|$ at r_* added at the target, of cost at most $\rho_{A_{\max}}(r_*)|\delta_2 - \delta_1|$, gives the Lipschitz bound. If a minimizer at δ_2 had the value $m_A(\delta_1)$ with $0 < \delta_1 < \delta_2$, running the short arc of angle δ_1 at reduced speed on its radial profile would give a relaxed minimizer at δ_1 that is not a rail, against Proposition XI.2; for $\delta_1 = 0$ the value is 0. By continuity and monotonicity the set where $m_A = B$ is a closed terminal interval; below it Theorem 21 gives existence, and above it an attained minimizer would contradict the strict increase toward an intermediate angle. □

At δ_c an interior minimizer may tie with the escape price. Uniqueness does not follow from monotonicity of A: Proposition X.8 gives two minimizers for a constant factor, and keeping A constant until after both arrivals and then raising it smoothly gives a non-decreasing, non-constant factor with the same two minimizers, since every competitor costs at least the frozen length.

A drop at a large separation

Let $\delta = \pi/2$ and, for $d \ge 0$ and $\epsilon > 0$,

$$A_d(\eta) = \begin{cases} 1, & \eta \le d, \\ 1 - \frac{1}{2}(3z^2 - 2z^3),\ z = (\eta - d)/\epsilon, & d \le \eta \le d + \epsilon, \\ \frac{1}{2}, & \eta \ge d + \epsilon, \end{cases}$$

a non-increasing C^1 factor. Write m_d for the infimum and $B_d = D_\downarrow(0, 6) + R_{1/2}(6)$ for the escape price. Since the family is ordered by translation in d, m_d and B_d are non-decreasing in d.

Lemma XI.10 (a single periastron)**.** *For a non-increasing factor, every minimizer between two points at angular separation $0 < \delta \le \pi$ has $p_\eta \ge 0$ and p_r vanishes at most once, crossing zero upwards (here $J \ne 0$, by the budget argument, with a chosen half great circle at $\delta = \pi$); if the two points are at the same radius, p_r vanishes exactly once, and the radius decreases to a single periastron and then increases.*

Proof. By Proposition XI.4, $p'_\eta = (AA'f/\sqrt{2})[N(\omega - p_\eta)/\sqrt{2} + 2] \le 0$ with $p_\eta(T) = 0$, so $0 \le p_\eta < \omega$ ($S > 0$ because $J \ne 0$). At a zero of p_r, $p'_r = \omega_r - \gamma S$. By (28) $\rho_A = \omega/\sqrt{b_H}$ increases, i.e. $\omega_r - \gamma\omega > 0$; if $\gamma \ge 0$ then $p'_r = (\omega_r - \gamma\omega) + \gamma p_\eta > 0$, and if $\gamma < 0$ then $p'_r = \omega_r - \gamma S > 0$. Every zero of p_r is thus crossed upwards, so there is at most one; as $\dot r$ has the sign of p_r and the radial end points coincide, there is exactly one. □

Theorem 24 (finite drops at a large separation)**.** *Let $\epsilon = 1/20000$.*

(1) For $0 \le d \le 1/20000$ there is a unique proper-time global minimizer among all three-dimensional rail curves, as a curve parametrized by the clock. More generally this holds for every $A \in C^1$ with $1/2 \le A \le 1$ and $A = 1/2$ after a clock h, $0 \le h \le 10^{-4}$.

(2) For $d \ge 10^7$, $m_d = B_d$ and no rail curve attains it; this holds for every $0 < \epsilon \le 1/100$.

(3) The first parameter d_* with $m_{d_*} = B_{d_*}$ satisfies $1/20000 < d_* < 10^7$. For $d < d_*$ the value is below B_d and a minimizer exists; at d_* an interior minimizer exists and ties with the escape price.

Structure of the proof. (1) After the layer the problem is frozen at $A = 1/2$; its value is smooth, with a controlled Hessian, on a small rectangle around the launch point, and every minimizer leaves the layer on the graph of minus its gradient. Through a layer of duration $h \le 10^{-4}$ the backward canonical flow is so close to the identity that the map from states on that graph to launch states is injective; so all minimizers share their state and covector at the clock h, and hence coincide. (2) Waiting costs, so a minimizer of cost at most B_d is close to $r = 2M$ when the factor drops, with a small angular momentum, and cannot sweep $\pi/2$; this is the mechanism of Theorem 25, here with sharper constants. (3) m_d and B_d are continuous in d, and the minimizers for bounded d form a compact family, which gives the first contact. The constants and the details are in section S9 of the supplementary material. □

Not proved: uniqueness for $1/20000 < d < d_*$, the uniqueness of the contact with B_d, monotonicity of $m_d - B_d$, and the fold of the continued interior extremals seen in Remark XI.7. The threshold 10^7 is far from the numerical values: it closes non-attainment analytically for all sufficiently late drops. The numerical family of Remark XI.7 is not identified with the drop above, so its numbers do not transfer.

a. Late changes: the comparison that decides. For a late drop of the conformal factor, compare the frozen value C_0 with $\bar B = R_{A_0}(r_0) + R_{A_\infty}(r_f)$, an upper bound for the escape price: descend with the initial factor and climb out with the final one. If $\bar B < C_0$ and the transition cannot sweep the prescribed angle, sufficiently late drops destroy the minimum (Theorem 25). For Vaidya the escape price includes the return frontier; late accretion is instead controlled by the radial lower bound of Theorem 36, and leaves the frozen minimizer in place. Both proofs rest on two elementary bounds: waiting above a radius s for a clock Δ costs at least Δ times the potential at s, and every radial passage costs at least a fixed weight per unit radius.

Theorem 25 (late drops of the factor)**.** *Let $a = 0$, $M > 0$, $\hat E > 0$, $r_0, r_f > 2M$ and $0 < \delta \le \pi$. Let $0 < A_\infty < A_0 < \hat E$, $\epsilon > 0$, and for $d \ge 0$ let $A_d \in C^1$ be non-increasing with $A_d = A_0$ on $[0, d]$ and $A_d = A_\infty$ on $[d + \epsilon, \infty)$. Assume*

$$m_{A_0}(\delta) > R_{A_0}(r_0) + R_{A_\infty}(r_f) \qquad \text{and} \qquad \delta > \frac{\epsilon}{3\sqrt{3}\,M}, \tag{29}$$

where $m_{A_0}(\delta)$ is the frozen value at the constant factor A_0. Then there is D such that for every $d \ge D$ the infimum equals the escape price B_d of Theorem 21 and is attained by no rail curve.

The first condition holds, for instance, if $\sqrt{R_0^2 + R_f^2 - 2R_0R_f\cos\delta} > R_0 + R_{A_\infty}(r_f)$ with $R_0 = R_{A_0}(r_0)$, $R_f = R_{A_0}(r_f)$, since the frozen metric dominates the Euclidean one in the coordinates (R_{A_0}, φ): $\mathrm{d}\rho_A/\mathrm{d}R_A > 1$ for every M, $\hat E$ and $0 < A < \hat E$, which generalizes (28). The second only excludes the angle that the transition alone can sweep, $|\dot\varphi| \le \sqrt{f}/r \le 1/(3\sqrt{3}M)$.

Proof. Idea. Waiting above a radius s during $[0, d]$ costs at least $d\,\omega_{A_0}(s)$, while a minimizer costs at most the bounded price B_d; so for large d it must be close to $r = 2M$ at the clock d, with small angular momentum. A curve with small angular momentum sweeps a small total angle in each frozen part, and the transition alone cannot sweep δ.

Constants. Put $c = A_0^2/\hat E$, $n = (1 - A_0^2/\hat E^2)^{-1/2}$, $k = (A_\infty/A_0)^2 < 1$, $\bar C = R_{A_0}(r_0) + R_{A_\infty}(r_f)$, $L = \bar C/c$ and $g = \delta - \epsilon/(3\sqrt{3}M) > 0$; then

$$\alpha = \frac{g}{4\pi + 8}, \quad r_s = 2M + \frac{M\alpha^2}{4}, \quad j_* = \min\Big\{\frac{\rho_{A_0}(r_s)}{2}, \rho_{A_\infty}(r_s)\Big\}\sin\alpha,$$

$$z = \min\Big\{M,\ Mf(r_s),\ \frac{j_*^2}{3Mc^2n^2}\Big\}, \qquad K = \max\Big\{1, \frac{2n}{1-k}\Big\}, \qquad D = L\Big(1 + \frac{4MK}{z}\Big).$$

Structure of a minimizer. Suppose a minimizer exists for some $d \geq D$. It is a normal rail extremal with saturated budget and $J > 0$ (Proposition XI.3; for $\delta = \pi$ a half great circle). As in Lemma XI.10, whose proof uses only $A' \leq 0$, $J \neq 0$ and the increase of ρ_A with r, which holds for every M and $\hat{E}$ (see the paragraph after the statement), $0 \leq p_\eta < \omega_A$ and every zero of p_r is crossed upwards, so p_r has at most one zero, there is no apoastron and $r \leq \max(r_0, r_f)$; no equality of the radial end points is needed.

Compression. Since $A \leq A_0$, $B_d \leq \bar{C}$, so the minimizer costs $C \leq \bar{C}$. A curve arriving by d is a frozen curve at A_0 and costs at least $m_{A_0}(\delta) > \bar{C}$, so $T > d$. With $s = \min_{[0,d]} r$ and $r_d = r(d)$, the plateau gives $dc\, f(s) \leq \bar{C}$, hence $s - 2M \leq 2ML/(d - L)$. Splitting the prefix where $r = s$ and bounding every radial passage by its weight, with $R'_{A_\infty} \leq kR'_{A_0}$,

$$C \geq R_{A_0}(r_0) + R_{A_0}(r_d) - 2R_{A_0}(s) + R_{A_\infty}(r_f) - R_{A_\infty}(r_d),$$

so $(1 - k)R_{A_0}(r_d) \leq 2R_{A_0}(s)$; with $c \leq R'_{A_0} \leq cn$ this gives $r_d - 2M \leq K(s - 2M) \leq 2MLK/(d - L) \leq z/2$. This step does not use the single periastron.

Small costates. On the plateau $p_\eta = p_{\eta,0}$ is constant and $S = \omega_{A_0} - p_{\eta,0} \geq \sqrt{b_H} J$ at the clock d, so $p_{\eta,0} < \omega_{A_0}(r_d) \leq \omega_{A_0}(r_s)/2$ and $J \leq \rho_{A_0}(r_d)$, with $\rho_{A_0}(r_d)^2 = c^2 N_{A_0}(r_d)\, r_d(r_d - 2M) < 3Mc^2n^2z \leq j_*^2$.

Small angle. On the plateau the orbit is that of the frozen metric with $\rho_{\mathrm{pl}} = \rho_{A_0}(1 - p_{\eta,0}/\omega_{A_0})$ and $R'_{\mathrm{pl}} = R'_{A_0}(1 - p_{\eta,0}/\omega_{A_0})$; $p_{\eta,0} \geq 0$ gives $\mathrm{d}\rho_{\mathrm{pl}}/\mathrm{d}R_{\mathrm{pl}} \geq \mathrm{d}\rho_{A_0}/\mathrm{d}R_{A_0}$, hence $w_{\mathrm{pl}} \leq w_{A_0}$, and for every A

$$0 < w_A(r) < b(r) = \frac{r\sqrt{f}}{r - M}, \qquad b(r)^2 = 1 - \frac{M^2}{(r - M)^2},$$

so $w < b(r_s) < \alpha$ on $2M < r \leq r_s$. After the drop $p_\eta = 0$. Each frozen part has at most two monotone radial legs, and on each $|\mathrm{d}\varphi| = Jw\,\mathrm{d}\rho/(\rho\sqrt{\rho^2 - J^2})$: the part with $r \leq r_s$ sweeps at most $\alpha\pi/2$, the part with $r \geq r_s$ at most $\arcsin(J/\rho(r_s)) < \alpha$, since $\rho_{\mathrm{pl}}(r_s) \geq \rho_{A_0}(r_s)/2$. Four legs sweep less than $4(\alpha\pi/2 + \alpha) = g/2$, and the transition at most $\epsilon/(3\sqrt{3}M)$ because $|\dot{\varphi}| \leq \sqrt{f}/r$. Hence $\delta < g/2 + \epsilon/(3\sqrt{3}M) < \delta$, a contradiction; an arrival during the transition only removes a frozen part. The argument uses $C \leq B_d$, so it also excludes a minimizer tying with B_d, and Theorem 21 gives $m_d = B_d$. □

No constant depends on the shape of the drop or on a bound for A'; ϵ enters only through g. For $M = 1$, $\hat{E} = \sqrt{2}$, $r_0 = r_f = 6$, $\delta = \pi/2$, $A_0 = 1$, $A_\infty = 1/2$ and $\epsilon \leq 1/100$ the sharper estimates of part (2) of Theorem 24 give $D = 10^7$.

F. Adiabatic expansions of the minimizers

Papers I and II give the first-order response of an extremal to a slowly varying background at fixed launch data and fixed charges (Theorem I.5 of[1];[2], section 'Complete first-order response and true-flow validation', where the on-shell part alone is shown not to be the first-order response), and Paper II converts it to fixed spatial end points through the floating charge ([2], section 'Fixed-charge and fixed-endpoint perturbations'). These are statements about extremals. The selection theorems of this paper say when the extremal with fixed end points is the global minimizer; smooth dependence on the slow parameter then gives an expansion of the minimizer itself, uniform in time.

Let χ be the clock of the drift (η for the conformal factor, v for the Vaidya mass) and consider a family $p_\varepsilon(\chi) = \bar{p}(\varepsilon(\chi - \chi_0))$ of factors or masses, with $\bar{p} \in C^2$, on a fixed window $[\chi_0, \chi_0 + H]$; the value at the launch does not depend on ε, and ε ranges over an interval containing 0, possibly one-sided. Write the normal extremals as $Y(\chi; b, \varepsilon)$, solutions of the extremal system $\dot{Y} = X_\varepsilon(Y)$ with independent initial data b, at absolute time χ, and let $\Phi(b, T; \varepsilon)$ collect the end-point conditions at $\chi_0 + T$ together with, on the proper-time branch, the transversality $p_\eta(T) = 0$. For the planar arrival b is the launch direction; for the proper-time branch with a varying factor $b = (\psi_{\mathrm{in}}, p_{\eta,\mathrm{in}})$.

Proposition XI.11 (expansion of the minimizer)**.** *Let $z = (b, T)$ and assume:*

(R) on a compact neighborhood of the reference extremal, away from the boundary of the domain and from a vanishing spatial costate, the extremal flow and its launch parametrization are well posed, the shooting map Φ is jointly C^1 in (z, ε) and the frozen launch map is C^2 in b; $X_\varepsilon = X_0 + \varepsilon X_1 + O(\varepsilon^2)$ uniformly, with X_0 of class C^2 and X_1 locally Lipschitz, and the launch data have the same expansion, uniformly for b near b_0, with a locally Lipschitz first-order coefficient;

(N) $\Phi(z_0; 0) = 0$, $0 < T_0 < H$, and $D_z\Phi(z_0; 0)$ is invertible;

(S) for every small ε in the range, a global minimizer to the fixed target exists, and the shooting data of every global minimizer lie in a fixed neighborhood U of z_0 contained in the neighborhood of local uniqueness of the zeros of Φ given by (N).

Then the global minimizer is unique up to reparametrization, its data are the only zero $z(\varepsilon)$ of $\Phi(\cdot\,; \varepsilon)$ in U, $z(\varepsilon) = z_0 + \varepsilon z_1 + O(\varepsilon^2)$ with $z_1 = (b_1, T_1) = -D_z\Phi(z_0; 0)^{-1}\partial_\varepsilon\Phi(z_0; 0)$, and, with $\chi_s = \chi_0 + sT_0$,

$$Y\big(\chi_0 + sT(\varepsilon); b(\varepsilon), \varepsilon\big) = Y_0(\chi_s) + \varepsilon\Big[V(\chi_s) + D_bY(\chi_s)\, b_1 + sT_1\, X_0\big(Y_0(\chi_s)\big)\Big] + O(\varepsilon^2)$$

uniformly in $s \in [0, 1]$, turning points included. Here $V = \partial_\varepsilon Y$ at fixed b and fixed absolute time is the fixed-launch response, and $D_bY\, b_1$ is a solution of the homogeneous variational equation of the frozen extremal system.

Proof. For $\bar p \in C^2$ and χ in the window, $p_\varepsilon = \bar p(0) + \varepsilon(\chi - \chi_0)\bar p'(0) + O(\varepsilon^2)$ and $p'_\varepsilon = \varepsilon\bar p'(0) + O(\varepsilon^2)$, and both are jointly C^1 in (χ, ε). The arrival geodesic equations and the proper-time systems of Pontryagin's principle used in this paper (Proposition XI.4 for the conformal factor) involve the profile and its first derivative only, which gives (R) in the cases considered here. Let V solve the linearized equation $\dot V = DX_0(Y_0)V + X_1(Y_0)$ with the linearized launch. For b near b_0 the difference $Y - Y_0 - \varepsilon V$ satisfies an equation whose source is $O(\varepsilon^2)$ on the compact neighborhood, and Grönwall gives $O(\varepsilon^2)$ uniformly on the window and in b; since V is locally Lipschitz in b, replacing b_0 by $b_0 + O(\varepsilon)$ in εV costs $O(\varepsilon^2)$; the same bound holds for Φ. With (N) and the C^1 implicit function theorem, Φ has a unique zero $z(\varepsilon)$ in U and $z(\varepsilon) - z_0 = O(\varepsilon)$; inserting this in $0 = D_z\Phi(z_0; 0)[z(\varepsilon) - z_0] + \varepsilon\partial_\varepsilon\Phi(z_0; 0) + O(\varepsilon^2)$ gives z_1 with an $O(\varepsilon^2)$ remainder, without a second-order implicit function theorem. A global minimizer is an extremal with data in U, by (S), hence is this zero. The chain rule at fixed absolute time splits the first-order term into the fixed-launch response, the variation of the initial data and the shift of the arrival time. The remainder is uniform because the clock χ does not degenerate at a turning point, where only r does. □

Remark XI.12 (relation with Papers I and II)**.** On a common sub-arc on which $|\mathrm{d}r_0/\mathrm{d}\chi| \geq c > 0$, the shape variation at fixed radius is $\delta\varphi|_r = \delta\varphi|_\chi - (\dot\varphi_0/\dot r_0)\,\delta r|_\chi$, and the time-shift term drops out because it is tangent to the path. With the same fixed-charge launch convention, the part coming from V is the fixed-charge response of Papers I and II, with its off-shell source. On the arrival branch, with the frozen costate normalization fixed and the angular charge J a regular launch coordinate, the part coming from $D_bY\, b_1$ is the floating-charge term of Paper II. On the proper-time branch it also contains the variation $p_{\eta,\mathrm{in},1}$ of the initial clock costate, which is in general non-zero for a varying factor; all independent shooting variables must be retained. In the spherical case with fixed normalization, $J = C(r_0)\sin\alpha$, so a tangential launch is not parametrized regularly by J and the launch angle must be kept; in general the criterion is the regularity of the charge map. In the radial representation the coefficients may lose uniformity at a turning point; in time they do not. Reconstructing the two spectral formulas in r across a periastron still requires their matching, and the proposition does not by itself place the first-order term in a given class of transcendental functions. In a numerical example (arrival branch, $a = 0$, $A = A_0[1 + \varepsilon(\eta - \eta_0)/M]$, $q_0 = 2$, $r_0 = r_f = 6M$, $\delta = 0.8\pi$, periastron at $4.17M$) the residual of the

expansion decreases as $\varepsilon^{2.03}$ uniformly along the path, including the periastron, and the fixed-launch response alone misses the target at first order; on the ingoing arc away from the periastron the largest fixed-end-point shape coefficient is about fifteen times the largest fixed-launch one. These are floating-point observations, not certified.

Remark XI.13 (where hypothesis (S) holds)**.** The global statements below are made for smooth profiles $\bar{p}$, the setting of the theorems they invoke; the expansion itself needs only $\bar{p} \in C^2$. Choose the window with $T_0 < H$; then $T(\varepsilon) < H$ for small ε, and for an earliest arrival a competitor that could beat it reaches the target before $\chi_1 = \chi_0 + H$ and sees the profile only on the window. A profile can therefore be extended beyond χ_1 to meet the conditions that a theorem asks at every future time; the extension may change later rays and arrivals, but not the earliest arrival before χ_1. This argument does not apply to the proper time, where a late competitor may cost less.

On the arrival branch with $a = 0$, (S) holds by Corollary V.7 when $q \geq q_{\min} > 1$ on the perturbed duration and the radii are below $R(q_{\min})$; a linearly growing factor is admissible on a window, but not under Theorem 2 on the whole future. For Vaidya with a slowly accreting mass, $\varepsilon \geq 0$ and $\hat{E}^2 \geq 3/2$, it holds by Theorem 8, after a smooth non-decreasing extension of the mass (m_{ext} defined as u_{ext} below), provided the shooting segment stays in the domain and arrives within the window. For a decreasing mass it holds by Theorem 10 for profiles that satisfy (H_{dec}), together with its domain, Sturm and target conditions, on the whole future. In these spherical cases the swept angle is below π, longer classes are excluded, and the uniqueness of the ray in its class identifies the zero with the first arrival. On the proper-time branch it holds by Theorems 22 and 33, under their global bounds; slowness alone does not supply the barrier margin.

On the equator with spin, (S) holds by Theorem 14(ii) or (iii), applied in full, when $q \geq 5/2 + \kappa$ on the window for some $\kappa > 0$, the spin is in the stated range, the end points are at admissible radii with $r_f > 2M$, and the target is not at a tie. Let $u = A^2/\hat{E}^2$, so that $u \leq 1/(5/2 + \kappa)$ on the window, and fix a collar length $L > 0$ in units of the clock, independent of ε, and $c \in C^\infty([0,1])$ with $0 \leq c \leq 1$, $c = 1$ near 0 and $c = 0$ near 1. Set $u_{\text{ext}} = u$ up to χ_1,

$$u_{\text{ext}}(\chi_1 + t) = u(\chi_1) + \int_0^t c(s/L)\, u'(\chi_1 + s)\, \mathrm{d}s, \qquad 0 \leq t \leq L,$$

and u_{ext} constant for $t \geq L$. It coincides with u near χ_1 and is as regular as u; $u'_{\text{ext}} = cu' = O(\varepsilon)$, $u''_{\text{ext}} = L^{-1}c'u' + cu'' = O(\varepsilon)$, and the total increment is $O(\varepsilon L)$. For small ε the extensions keep $q > 5/2$ with a common bound $u_* < 2/5$, which serves for the timelike field in the proof of Proposition IX.3, and the jet conditions of Proposition IX.7 hold at every time; a small negative part of u''_{ext} is allowed. In the extended problem the two adjacent classes have shooting branches given by the implicit function theorem, with arrival times $T_\varepsilon^\pm \to T_0^\pm$, and the theorem identifies them with the earliest arrivals of their classes and restricts the global candidates to them; since $T_0^+ \neq T_0^-$ the order is kept for small ε. The winner arrives before χ_1, so it is the earliest arrival of the original problem. In case (iii) the selection refers to the auxiliary first-arrival problem for G, whose competitors may enter the ergoregion; its identification with the minimum among the physical rails requires an exterior minimizer as in Corollary IX.8, a condition that persists for small ε when the frozen winning segment has a positive margin from $r = 2M$. Case (ii) has no such difference.

Not covered here: $2 \leq q < 5/2$ and spins beyond the bands on the equator; three-dimensional rotation with a varying factor; windows of length $O(\varepsilon^{-1})$; and targets at a tie.

XII. THE PROPER-TIME BRANCH ON THE EQUATOR, WITH SPIN

In this section $0 < a \leq M$ (the case $a < 0$ follows by $\varphi \mapsto -\varphi$), the competitors are the rail curves *in the equatorial plane* $\vartheta = \pi/2$, the domain is $r > 2M$, $A \in C^1([\eta_0, \infty))$ with

$0 < a_* \le A \le A_{\max} < \hat{E}$, and the arrival time is free. Fix $x_0 = (r_0, 0)$ and a physical target $x_f \neq x_0$, with a representative $\varphi_f \in (-\pi, \pi]$. Write $\delta = |\varphi_f|$ and $\Lambda_j = \varphi_f + 2\pi j$, $j \in \mathbb{Z}$, for the angular lifts. Unless a lift is specified, m and d_α denote the infimum and the frozen distance to the physical target, minimized over all lifts. Prograde means increasing φ. With the data (8), $\beta = \beta_\varphi \,\mathrm{d}\varphi$, $\beta_\varphi = -2Ma/(r-2M)$, $\omega = A^2 f/\hat{E}$, N as in (6), and $k = f^2 h_0 = (fr^2/\Delta)\,\mathrm{d}r^2 + \Delta\,\mathrm{d}\varphi^2$. Unlike the case $a = 0$, the boundary $r = 2M$ does not collapse in the equatorial completion: it is a ring of positive length. The relaxed cost contains a magnetic term, and approaching the ring in finite conformal time is kinematically possible. For $0 < a < M$, the boundary of the frozen three-dimensional completion collapses through the poles, as discussed after Theorem 19; no comparison with three-dimensional competitors is asserted here. Covector norms are taken with h_0^{-1} unless explicitly stated otherwise.

A. Relaxation, rails and extremals

A *relaxed curve* is a Lipschitz graph $x(\eta)$ with $\dot{x} \in K(\eta, x) = \{v : \beta(v) + \sqrt{N}|v|_{h_0} \le 1\}$, the causal cone of g_{opt}; K is a translated ellipse containing 0 in its interior, since $|\beta|_{h_0} < 1 \le \sqrt{N}$ (Lemma II.2). Its cost is

$$C = \int \theta, \qquad \theta = \omega(\mathrm{d}\eta - \beta) = -\frac{g(W,\cdot)}{\hat{E}}, \qquad C = \int \omega\,\mathrm{d}\eta + \int \frac{2MaA^2}{\hat{E}r}\,\dot{\varphi}\,\mathrm{d}\eta, \tag{30}$$

which equals the proper time on rails (Lemma X.1); we write $c_A(v) = \omega_A(1 - \beta(v))$ for the cost rate of a velocity v. The last term is magnetic, so the radial reduction of Lemma XI.1 is not available.

Lemma XII.1. *For relaxed curves the following hold.*

(1) $C \ge \int (A^2/\hat{E})\sqrt{N}|\dot{x}|_k\,\mathrm{d}\eta \ge L_{a_*}$, *the length in the frozen metric* $h_\tau^{(a_*)} = a_*^4 k/(\hat{E}^2 - a_*^2 f)$; *and* $\omega(1-\beta(\dot{x})) \ge \omega/(1+|\beta|_{h_0})$.

(2) On a common finite clock interval and in a common compact subset of $r > 2M$, *the cost is continuous under uniform convergence of curves and weak-∗ convergence of velocities, and the constraint passes to the limit.*

(3) The infima over rails and over relaxed curves coincide.

(4) If $A_1 \le A_2$ *then* $K_{A_2} \subset K_{A_1}$ *and the cost for* A_1 *is not larger, so the infimum, and every escape price below, is non-decreasing in* A.

Proof. (1) One has

$$1 - \beta(\dot{x}) \ge \sqrt{N}|\dot{x}|_{h_0}, \qquad \omega\sqrt{N}|\dot{x}|_{h_0} = \frac{A^2}{\hat{E}}\sqrt{N}|\dot{x}|_k,$$

and $A^2\sqrt{N_A} = A^2\hat{E}/\sqrt{\hat{E}^2 - A^2 f}$ increases with A; for the rate, $y = 1 - \beta(\dot{x})$ satisfies $1 - y \le |\beta||\dot{x}| \le |\beta| y/\sqrt{N}$. (2) The cost is affine in $\dot{x}$ with continuous coefficients. (3) The cost is affine, so the convex hull of the rail velocities with their costs is the relaxed set; the relaxation theorem[40] (Theorem 2.7.2), applied on a compact neighborhood of a relaxed curve of finite duration, approximates it uniformly with its cost by rail curves. The end point is corrected by a short spatial curve $z(l)$ near the target with the clock solved from $\mathrm{d}\eta/\mathrm{d}l = \beta(z_l) + \sqrt{N}|z_l|_{h_0}$, at cost and duration proportional to the error. The correction is made in the target chart of the same angular lift; uniform approximation and this local correction preserve that lift. (4) N increases with A, and $1 - \beta(v) > 0$ on K. □

Proposition XII.2. *Every relaxed minimizer in* $r > 2M$ *attained at finite time is a rail curve almost everywhere, for every* $A \in C^1$, *with no condition on the sign of* A'.

Proof. Write $K = \{c + Eu : |u| \le 1\}$ with c, E of class C^1 in (η, x). Abnormal multipliers vanish: for $p_0 = 0$, $H(T) = 0$ and $0 \in \operatorname{int} K$ give $p(T) = 0$, and the adjoint equation is linear in p. For $p_0 = -1$, $H = -\omega + \tilde p \cdot \dot x$ with $\tilde p = p + \omega\beta$, and where $\tilde p \neq 0$ the maximizer lies on ∂K. On $\{\tilde p = 0\}$, almost everywhere $\dot{\tilde p} = 0$, and the adjoint equation at fixed control gives $\partial_r\omega(1 - \beta_\varphi\dot\varphi) = \omega\beta'_\varphi\dot\varphi$ and $\beta_\varphi\partial_\eta\omega + \dot r\,\partial_r(\omega\beta_\varphi) = 0$, whose only solution is $\dot\varphi = 1/a$, $\dot r = 2rA'/A$. This velocity is not in K: already $\beta_\varphi/a + \sqrt{\Delta}/(fa) = 1 + r(\sqrt{\Delta} - a)/((r-2M)a) > 1$, since $\Delta - a^2 = r(r-2M)$. For $a = 0$ the first equation reads $\partial_r\omega = 0$, which is impossible, as in Proposition XI.2. □

Here ψ is the heading in an h_0-orthonormal frame, measured from the outward radial direction, and $\dot x^\flat = h_0(\dot x, \cdot)$. Along a normal extremal, where $\tilde p \neq 0$, put $\lambda = \tilde p \cdot \dot x > 0$ and $p_\eta = -H = \omega - \lambda$. Then $p = (\omega - p_\eta)\sqrt{N}\,\dot x^\flat/|\dot x|_{h_0} - p_\eta\beta$; the constant $L = p_\varphi$ gives $\rho_{\mathrm{eff}} \sin\psi = L - 2Ma\,p_\eta/(r-2M)$ with $\rho_{\mathrm{eff}} = (\omega - p_\eta)\sqrt{N\Delta}/f$, and

$$p_\eta(T) = 0, \qquad \dot p_\eta = \big(1 - \beta(\dot x)\big)\frac{AA'f}{\hat E}\Big[2 + \frac{N(\omega - p_\eta)}{\hat E}\Big].$$

For constant A, $p_\eta \equiv 0$ and L is the Clairaut constant of h_τ; for $a = 0$ these are the formulas of Proposition XI.4.

B. Constant factor: the ring and the escape price

For constant A, $q = \hat E^2/A^2$, the cost is the length of Perlick's metric $h_\tau = A^4 k/(\hat E^2 - A^2 f) = \mathrm{d}R^2 + \rho(R)^2\mathrm{d}\varphi^2$, with $\rho = (A^2/\hat E)\sqrt{N\Delta}$ and $R(r) = (A^2/\hat E)\int_{2M}^r \sqrt{Nf}\,s\,\Delta^{-1/2}\mathrm{d}s$.

Lemma XII.3. *ρ increases strictly with r, from $\rho_h := \rho(2M) = A^2 a/\hat E$; near $r = 2M$, $\rho - \rho_h \asymp R^{2/3}$. The metric completion is the closed half-cylinder $[0,\infty)\times S^1$ with the continuous metric $\mathrm{d}R^2 + \rho^2\mathrm{d}\varphi^2$; the boundary is a ring of length $2\pi\rho_h$. Moreover $R(r) \to \infty$ as $r \to \infty$, the completion is a proper length space, as is its angular universal cover, and*

$$\frac{\mathrm{d}\rho}{\mathrm{d}R} = \frac{r - M + M\Delta/(r^2(q-f))}{r\sqrt{f}} > 1.$$

Proof. N and Δ increase with r. With $s = r - 2M$, $\sqrt{h_{\tau,rr}} = (A^2/\hat E)\sqrt{2Ms}/a + O(s^{3/2})$ and $\rho = \rho_h + (A^2/\hat E)(M/a + a/(4qM))s + O(s^2)$. Near $R = 0$ the metric is comparable to $\mathrm{d}R^2 + \rho_h^2\mathrm{d}\varphi^2$, and distinct ring points are at distance at least ρ_h times their angular separation. The radial density tends to $A^2/\sqrt{\hat E^2 - A^2} > 0$ at infinity, and $\mathrm{d}s \ge |\mathrm{d}R|$, $\mathrm{d}s \ge \rho_h|\mathrm{d}\varphi|$ make closed balls compact. The last inequality follows from $(r-M)^2 - r^2 f = M^2$, as in Theorem 16. □

For $0 \le L \le \rho_h$ put

$$\mathcal{S}_L(r) = \int_0^{R(r)} \sqrt{1 - L^2/\rho^2}\,\mathrm{d}R, \qquad \Phi_L(r) = \int_0^{R(r)} \frac{L}{\rho\sqrt{\rho^2 - L^2}}\,\mathrm{d}R,$$

the reduced action and the angle swept by the geodesic of Clairaut constant L from radius r to the ring; Φ_{ρ_h} is finite, the integrand being $O(R^{-1/3})$. Let $\delta_{\mathrm{ring}} = \Phi_{\rho_h}(r_0) + \Phi_{\rho_h}(r_f)$ and, for a class of angular lift Λ,

$$W(\Lambda) = \mathcal{S}_{\rho_h}(r_0) + \mathcal{S}_{\rho_h}(r_f) + \rho_h|\Lambda|, \qquad \mathcal{B}_A(\delta) = \max_{0\le L\le\rho_h}\big[\mathcal{S}_L(r_0) + \mathcal{S}_L(r_f) + L\delta\big]. \tag{31}$$

Proposition XII.4 (ring calibration and escape price)**.** *Let A be constant.*

(1) For every Lipschitz curve in the completion and every $L \in [0, \rho_h]$, length $\ge \int\sqrt{1 - L^2/\rho^2}\,|\mathrm{d}R| + L\,|\int \mathrm{d}\varphi|$.

(2) The infimum of the lengths of the curves of the class $|\Lambda| = \delta$ that come within ϵ of the ring tends to $\mathcal{B}_A(\delta)$ as $\epsilon \to 0$. For $\delta \geq \delta_{\text{ring}}$ the maximum is at $L = \rho_h$ and $\mathcal{B}_A = W$, attained in the completion only by the curve that arrives tangentially at the ring, follows it monotonically and leaves tangentially; for $\delta < \delta_{\text{ring}}$ it is at the unique L with $\Phi_L(r_0) + \Phi_L(r_f) = \delta$. In particular $\mathcal{B}_A(0) = R(r_0) + R(r_f)$.

(3) A minimizer in the completion that meets the ring is the tangent–arc–tangent curve; hence $|\Lambda| \geq \delta_{\text{ring}}$, and for $|\Lambda| < \delta_{\text{ring}}$ the minimum of the class is attained by an interior geodesic, i.e. by a rail curve.

(4) Every interior geodesic sweeps less than π: a monotone geodesic less than $\pi/2$, a symmetric dip less than $2\arccos(L/\rho(r_0))$, and a dip between different radii less than $\arccos(L/\rho(r_0)) + \arccos(L/\rho(r_f))$. The classes with $|\Lambda| \geq \pi$ are therefore decided by the ring.

Proof. (1) Cauchy–Schwarz with the unit vector $(\sqrt{\rho^2 - L^2}, L)/\rho$. (2) For a curve within ϵ of the ring, (1) gives at least $\mathcal{S}_L(r_0)+\mathcal{S}_L(r_f)+L\delta-2\mathcal{S}_L(r_\epsilon)$ for every L, and $\mathcal{S}_L(r_\epsilon) \leq R(r_\epsilon)$; only then take the maximum in L. The function maximized has derivative $\delta - \Phi_L(r_0) - \Phi_L(r_f)$ and is concave. Upper bounds: for $\delta < \delta_{\text{ring}}$ the two Clairaut geodesics of the maximizing L, with the same orientation, joined at the ring and truncated at $R(r_\epsilon)$, with L adjusted to keep the angle; for $\delta \geq \delta_{\text{ring}}$ the two tangent geodesics truncated at $R(r_\epsilon)$ and joined by an arc of that radius. Equality in Cauchy–Schwarz forces the Clairaut relation with $L = \rho_h$, R monotone on each side and φ monotone. (3) Between the first and last contact the curve is a shortest path between ring points, hence an arc, since $\mathrm{d}s \geq \rho_h|\mathrm{d}\varphi|$. Before the first contact it is an interior geodesic, whose one-sided tangent at the ring exists ($\varphi' = L/\rho^2$, $R' = \pm\sqrt{1 - L^2/\rho^2}$ have limits). If it made an angle with the arc, the flat shortcut in the coordinates $(R, \rho_h\varphi)$, which stays in $R \geq 0$, would save $c\,t$ at scale t, while the metric differs from the flat one by $O(R^{2/3})$, costing $O(t^{5/3})$. If the first and last contacts coincide, apply the same shortcut to the incoming and outgoing geodesic pieces: their one-sided unit tangents have radial components of opposite signs, and unless both are tangent to the ring with the same angular orientation the shortcut saves a positive multiple of t. Thus both Clairaut constants have magnitude ρ_h, with the same orientation; this covers the zero-length arc and forces $|\Lambda| = \delta_{\text{ring}}$ in that case. (4) By Lemma XII.3, $\mathrm{d}R/\mathrm{d}\rho < 1$ in the angle integrals, and $\int_L^{\rho_1} L\,\mathrm{d}\rho/(\rho\sqrt{\rho^2 - L^2}) = \arccos(L/\rho_1)$. □

Theorem 26 (the Maxwell rule with a ring)**.** *Let A be constant with $q = \hat{E}^2/A^2 \geq 3/2$, $0 < a \leq M$, and $r_0 \geq r_f > 2M$. There is a unique $\delta_c \in [\delta_{\text{ring}}, \Phi_{\max})$, where $\Phi_{\max} < \pi$ is the largest angle of an interior dip, such that:*

(1) for $0 < \delta < \delta_c$ the equatorial proper-time minimizer exists and is unique; it is a monotone geodesic or a shallow dip, and it is non-degenerate;

(2) at δ_c it is attained and ties with the ring curve, which is not admissible;

(3) for $\delta_c < \delta \leq \pi$ the infimum is $W(\delta)$ and is not attained by any rail curve in $r > 2M$.

Deep dips are never minimal.

Proof. With $V_a = r\Delta/((q-1)r + 2M)$ and $K_a = \sqrt{r(r-2M)}/\Delta$ one has $\rho^2 = A^2 V_a$ and $R'/\rho = K_a$, so the half-angle of a dip with periastron t is $\sqrt{V_t}\int_t^{r_0} K_a\,\mathrm{d}r/\sqrt{V_a - V_t}$, the proper-time half-angle of Paper II[2] at $E^2 = q$, with spin. Part (v) of its proposition *where the τ half-angle decreases* (proof in its appendix: $g = VW$, $W = K/V'$, $g' > 0 > g''$ for $E^2 \geq 3/2$ and every $a^2 \leq M^2$, with $g \asymp \sqrt{V - V(2M)}$ at the ring) gives a single non-degenerate maximum for equal radii. For unequal radii, with $x = V_t$, $y_i = V_i - x$ and A_i the two half-angles,

$$\sqrt{x}A_i' = 2\sqrt{y_i}\int_0^1 g'(x + y_i u^2)\,\mathrm{d}u - \frac{g(V_i)}{\sqrt{y_i}},$$

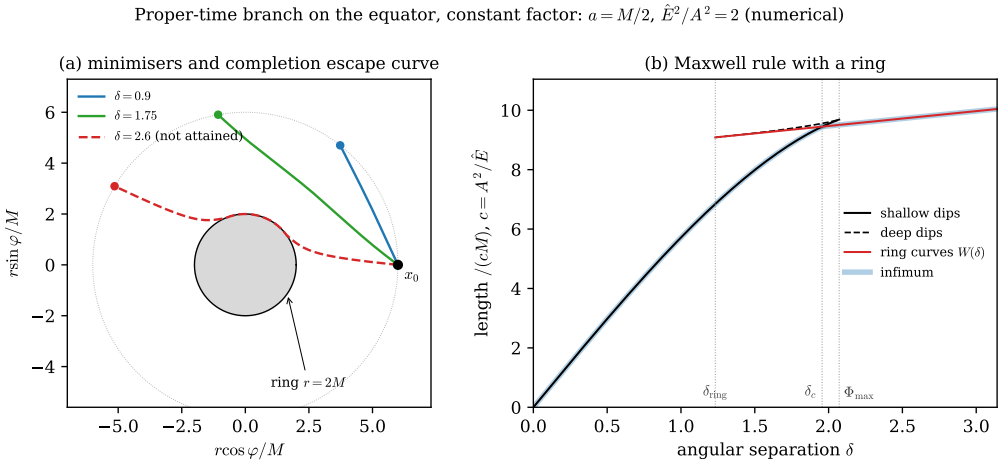


FIG. 2. Proper-time branch on the equator for a constant factor, $a = M/2$, $\hat{E}^2/A^2 = 2$, $r_0 = r_f = 6M$ (numerical, not certified). (a) Minimizers in the equatorial plane, drawn in the coordinates $(r\cos\varphi, r\sin\varphi)$ (not an isometric embedding of h_τ; the ring has coordinate radius $2M$ and length $2\pi\rho_h$): shallow dips for $\delta < \delta_c$, and for $\delta > \delta_c$ the tangent–arc–tangent curve of the completion, which no rail attains. (b) Lengths against the separation: shallow and deep dips, the ring curves $W(\delta) = \mathcal{S}(r_0) + \mathcal{S}(r_f) + \rho_h\delta$, and the infimum (Theorem 26); here $\delta_{\mathrm{ring}} \approx 1.232$, $\delta_c \approx 1.956$, $\Phi_{\max} \approx 2.072$.

whose derivative in x is $-y_i^{-1/2}\int_0^1 g' + 2\sqrt{y_i}\int_0^1 g''(1-u^2) - g(V_i)/(2y_i^{3/2}) < 0$; the sum tends to $+\infty$ at the ring and to $-\infty$ as $x \uparrow V_f$, so the dip angle has a single non-degenerate maximum $\Phi_{\max}$, which exceeds δ_{ring} (its limit at the ring) and the tangent separation $\delta_{\tan}$. Along each regular branch $\mathrm{d}\ell/\mathrm{d}\delta = L$ (as in Lemma X.4). The dip angle and length converge to their tangent values as the periastron $r_p \downarrow 2M$: on a fixed neighborhood of the ring their integrands are bounded by $C\sqrt{r-2M}/\sqrt{r-r_p}$, since $(\rho^2)'$ has a positive lower bound, so for $r_p \le 2M + \epsilon$ the contribution of $[r_p, 2M+\epsilon]$ is at most $C[\epsilon + 2\sqrt{(r_p - 2M)\epsilon}] \le 3C\epsilon$; one passes to the limit away from the ring and then lets $\epsilon \downarrow 0$. The deep branch starts at $W(\delta_{\mathrm{ring}})$ with $L > \rho_h$, so it stays above W. The principal branch (monotone up to $\delta_{\tan}$, then shallow dips) has $L > \rho_h$ for $\delta \ge \delta_{\mathrm{ring}}$, because the monotone angle at $L = \rho_h$ is $\Phi_{\rho_h}(r_0) - \Phi_{\rho_h}(r_f) < \delta_{\mathrm{ring}}$; hence $\ell_{\mathrm{main}} - W$ increases strictly on $[\delta_{\mathrm{ring}}, \Phi_{\max}]$ and is positive at $\Phi_{\max}$. For $\delta < \delta_{\mathrm{ring}}$ the minimum is interior (Proposition XII.4(3)), hence principal, and continuity of the distance gives $\ell_{\mathrm{main}}(\delta_{\mathrm{ring}}) \le W(\delta_{\mathrm{ring}})$. Other lifts have $|\Lambda_j| \ge \pi$, no interior geodesics, and completed distances $W(|\Lambda_j|)$. For $\delta \ge \delta_{\mathrm{ring}}$ these are no smaller than $W(\delta)$; for $\delta < \delta_{\mathrm{ring}}$ they are at least $W(\delta_{\mathrm{ring}}) \ge \ell_{\mathrm{main}}(\delta_{\mathrm{ring}}) > \ell_{\mathrm{main}}(\delta)$, because the principal length increases with the angle. The inequalities are strict for $\delta < \pi$; at $\delta = \pi$ the two ring orientations tie. Non-degeneracy follows as in Corollary XI.6, with the launch angle at the tangent separation. □

Below $q = 3/2$ the single-maximum conclusion is not valid throughout the parameter family (already at $a = 0$, Paper II certifies two maxima for $E^2 = 1.01$, $r_0 = 10^4 M$), and the selection must compare all branches with $\mathcal{B}_A$. As $a \to 0$, $\delta_{\mathrm{ring}} \to 0$ and $W \to R(r_0) + R(r_f)$, the dive of Theorem 15. Figure 2 illustrates Theorem 26.

C. Time-dependent factor

Lemma XII.5 (kinematics at the ring)**.** *For $a > 0$:*

(1) the largest radial speed in K is $\sqrt{f\Delta}/(r\sqrt{N - |\beta|^2})$, with the positive limit

$a\hat{E}/(2M\sqrt{A^2+2\hat{E}^2(1+2M^2/a^2)})$ *at* $r=2M$*; the ring is reached in finite conformal time;*

(2) along circular rails the conformal time per radian is $T_\pm=(\sqrt{N\Delta}\mp 2Ma/r)/f$*, with* $T_+(2M)=a+aA^2/(2\hat{E}^2)+2M^2/a$ *finite and* $T_-\sim 4aM/(r-2M)$*;*

(3) waiting at radius r *costs* ω *per unit time, which vanishes at the ring;*

(4) for constant A*, the clock along the frozen geodesic tangent to the ring is finite in the prograde sense and diverges logarithmically in the retrograde sense.*

Proof. (1) In the variables $X=r\dot{r}/\sqrt{f\Delta}$, $Y=\sqrt{\Delta}\,\dot{\varphi}/f$ the constraint is $-|\beta|Y+\sqrt{N}\sqrt{X^2+Y^2}\le 1$; the maximum of $(1+|\beta|Y)^2/N-Y^2$ is $1/(N-|\beta|^2)$, and $1-|\beta|^2=fg_{\varphi\varphi}/\Delta$. (2)–(3) are direct. (4) Along a curve with $\mathrm{d}r/\mathrm{d}\varphi=u$ the clock per radian exceeds $T_\pm$, in both senses, by $(\sqrt{N\Delta}/f)\big(\sqrt{1+fr^2u^2/\Delta^2}-1\big)\in[0,\sqrt{N}r^2u^2/(2\Delta^{3/2})]$. Along the frozen geodesic tangent to the ring, $L=\rho_h$ and $|\mathrm{d}\varphi/\mathrm{d}r|=\rho_hR'/(\rho\sqrt{\rho^2-\rho_h^2})$; by the expansions in the proof of Lemma XII.3, $R'\asymp\sqrt{r-2M}$ and $\rho^2-\rho_h^2\asymp r-2M$, so $\mathrm{d}\varphi/\mathrm{d}r$ has a positive finite limit and u is bounded. The prograde clock therefore has a bounded density in r, and the retrograde one has density $T_-|\mathrm{d}\varphi/\mathrm{d}r|+O(1)$, asymptotic to a positive multiple of $1/(r-2M)$ since $T_-\sim 4aM/(r-2M)$. □

For a class Λ define the escape price $B(\Lambda)$ as the limit, as $\epsilon\to 0$, of the infimum of the cost over relaxed curves of the class, of finite duration, that come within ϵ of $r=2M$.

Theorem 27 (existence for finite variations)**.** *Let* A *be as at the beginning of this section.*

(1) $B(\Lambda)\ge\mathcal{B}_{a_*}(|\Lambda|)$*, and every curve of the class costs at least* $\rho_h(a_*)|\Lambda|$*.*

(2) If the infimum m *satisfies* $m<\min_\Lambda B(\Lambda)$*, a global minimizer exists and every minimizer is a rail curve.*

(3) In particular, with d_α *the distance in the frozen completion for the factor* α *and* $\Gamma=(A_{\max}/a_*)^2\sqrt{(\hat{E}^2-a_*^2)/(\hat{E}^2-A_{\max}^2)}$*,*

$$d_{a_*}\le m\le d_{A_{\max}}\le\Gamma\, d_{a_*},$$

and $\Gamma d_{a_*}(x_0,x_f)<\mathcal{B}_{a_*}(\delta)$ *implies the existence of a minimizer for every* $A\in C^1$ *with values in* $[a_*,A_{\max}]$*, with no condition on* A'*.*

Proof. (1) Lemma XII.1(1) and Proposition XII.4. (2) Choose a bounded-cost minimizing sequence. The angular lower bound in (1) leaves only finitely many lifts; pass to a subsequence of fixed lift Λ. If its minimum radius approached $2M$, the definition of $B(\Lambda)$ would give $m\ge B(\Lambda)$, a contradiction; thus it stays in $r\ge 2M+\epsilon$. The bound $C\ge L_{a_*}$ and $\mathrm{d}s\ge|\mathrm{d}R_{a_*}|$ give an outer radial barrier. On this shell the rate is at least $a_*^2f(2M+\epsilon)/(2\hat{E})$, so durations and coordinate velocities are bounded. Compactness, Lemma XII.1(2) and Proposition XII.2 give an attained rail minimizer. (3) Near-minimizing frozen rails for $A_{\max}$ are admissible for A with no larger cost (Lemma XII.1(4)); since $|\Lambda_j|\ge\delta$ and $\mathcal{B}_{a_*}$ is non-decreasing, the criterion excludes every escape. The ratio of the frozen densities, $(A_{\max}^2/a_*^2)\sqrt{(\hat{E}^2-a_*^2f)/(\hat{E}^2-A_{\max}^2f)}$, is non-decreasing in $f\le 1$. □

Theorem 28 (asymmetry of the escape price)**.** *Let* $a>0$*,* $A\ge A_0$ *on* $[\eta_0,\infty)$ *and* $A\equiv A_0$ *on* $[\eta_0,\eta_0+T_*]$*. For a fixed lifted class with* $\Lambda\ge\delta_{\mathrm{ring}}(A_0)$*, suppose that* T_* *exceeds the limiting clock duration of its prograde tangent–arc–tangent curve, and that* $A\ge A_0+\kappa$ *on* $[\eta_2,\infty)$ *for some* η_2 *and* $\kappa>0$*. Then* $B(+\Lambda)=W_{A_0}(\Lambda)<B(-\Lambda)$*. This compares the two lifted classes, not the prices minimized over all lifts of a physical target.*

Proof. Both prices are at least W_{A_0} by Theorem 27(1). Interior approximations of the prograde ring curve have clocks converging to its limiting duration (Lemma XII.5(4)), which is less than T_*, so they run inside the window. For the retrograde class put $T^{\mathrm{base}} = (\sqrt{\Delta} + 2Ma/r)/f \le T_-$ and $T_n = \min(T^{\mathrm{base}}, n)$; every relaxed curve satisfies $\mathrm{d}\eta \ge T_n(r)(\mathrm{d}\varphi)^- \ge -T_n\,\mathrm{d}\varphi$. Use the frozen radial coordinate $R = R_{A_0}(r)$ and parametrize the curves by length on a common interval. For each fixed n the form $\theta_n = -T_n(R)\,\mathrm{d}\varphi$ is constant in R near the ring and Lipschitz on the compact set in question; for uniformly convergent curves of bounded length, Stokes' formula on the straight homotopy in (R, φ) bounds the difference of its integrals by $C_n\|\gamma_k - w_-\|_\infty(\operatorname{len}\gamma_k + \operatorname{len} w_-) + o(1)$, so the integrals converge for fixed n. Only afterwards let $n \to \infty$: on the retrograde ring curve w_- they diverge, since $T^{\mathrm{base}} \asymp (r-2M)^{-1}$ and $|\mathrm{d}\varphi/\mathrm{d}r|$ has a positive limit on a retrograde tangent. If $B(-\Lambda) = W_{A_0}$, take curves of costs $C_k \to W_{A_0}$ and split them at $\min(T_k, \eta_2)$. The calibration gives $L_{A_0}(\gamma_k) \ge W_{A_0} - o(1)$, and the late overprice, with factor $((A_0+\kappa)/A_0)^2 - 1 > 0$, forces the frozen length of the late parts to zero; they end at the target, so the near contact with the ring belongs to the early parts. These have end points converging to x_f and lengths tending to W_{A_0}; a subsequence parametrized by length converges to the unique equality case w_-, while their clocks stay below $\eta_2 - \eta_0$, contradicting the divergence above. □

Where the value function $V(\eta, x)$, the infimum of the cost from (η, x) to x_f, is differentiable off the target, it satisfies the Hamilton–Jacobi–Bellman equation

$$V_\eta + \omega - \sigma_K(\omega\beta - D_x V) = 0, \qquad \sigma_K(\xi) = \frac{\sqrt{(\xi\cdot\beta^\sharp)^2 + (N - |\beta|^2)|\xi|^2} - \xi\cdot\beta^\sharp}{N - |\beta|^2}, \tag{32}$$

with $V(\eta, x_f) = 0$, or equivalently $g_{\mathrm{opt}}^{-1}(\theta + \mathrm{d}V, \theta + \mathrm{d}V) = 0$ with $\omega + V_\eta \ge 0$. With $\mathcal{H}_\alpha(x,p) = -\omega_\alpha + \sigma_{K_\alpha}(p + \omega_\alpha\beta)$, regular characteristics satisfy $\dot X = \partial_p \mathcal{H}_{A(\eta)}$, $\dot P = -\partial_x \mathcal{H}_{A(\eta)}$, $P = -D_x V$, and $p_\eta = -V_\eta$. More generally, let U be locally Lipschitz on a domain containing all competitors under consideration, and suppose that every covector $\xi = (\xi_\eta, \xi_x)$ of its Clarke differential[40] (section 4.7) satisfies $|\xi_x + \xi_\eta\beta| \le \sqrt{N}(\omega + \xi_\eta)$ and $\omega + \xi_\eta \ge 0$. If $U(T, x_f) \le 0$ for every admissible arrival clock (and every target lift, when using the cover), then $C(\gamma) \ge U(\eta_0, x_0) - U(T, x_f) \ge U(\eta_0, x_0)$. A candidate γ_* is a global minimizer if, in addition, $(U\circ\gamma_*)' + \theta(\gamma_*') = 0$ almost everywhere and $U(T_*, x_f) = 0$.

Theorem 29 (global selection for finite variations)**.** *Let $A \equiv A_*$ on $[\eta_*, \infty)$, with $q_* = \hat E^2/A_*^2 \ge 3/2$, and let $d_* = d_{A_*}(\cdot, x_f)$.*

(1) Change after arrival. *If the frozen problem for A_0 has a unique minimizer γ_0, arriving at T_0, and $A \equiv A_0$ on $[\eta_0, T_0]$, $A \ge A_0$ afterwards, then γ_0 is the unique global minimizer.*

(2) Short initial window. *Let A be arbitrary in $[a_*, A_{\max}]$ on $[\eta_0, \eta_*]$, of length τ. Assume that d_* is C^2 on a neighborhood of a closed coordinate ball $\bar B_R(x_0) \subset \{r > 2M\}$, disjoint from every lift of the target, with unique and attained physical frozen tails; that on $\mathcal{D} = \{x \in \bar B_R,\ |p + Dd_*(x)| \le s\}$ the quantity $|p + \omega_\alpha\beta|$ is bounded below by $\chi > 0$ for $\alpha \in [a_*, A_{\max}]$; and let V_{loc} bound the coordinate speeds of K_α on $\bar B_R$, M_x, M_p, L bound the two components of $(\partial_p\mathcal{H}_\alpha, -\partial_x\mathcal{H}_\alpha)$ and its derivative on $\mathcal{D}$, and $S = \sup|D^2 d_*|$, $B_0 = 1 + S$ (here, exceptionally, Euclidean norms in (r,φ) and in (x,p)). If*

$$\tau V_{\mathrm{loc}} < R/4, \qquad \tau(M_p + SM_x) < s/2, \qquad B_0(e^{L\tau} - 1) < 1/2,$$

then the global minimizer exists and is unique; it crosses η_* and its tail is the frozen minimizer. No assumption is made on A'.

Proof. (1) Every competitor is admissible for A_0 with a cost not larger (Lemma XII.1(4)), hence costs at least the frozen minimum, which γ_0 attains inside the window; equality forces a frozen relaxed minimizer, i.e. γ_0.

(2) Before any first exit from $\bar{B}_R$, every relaxed curve from x_0 has displacement at most $\tau V_{\mathrm{loc}} < R/4$. Thus no first exit from $B_{R/4}$ is possible before η_*, and no target lift is reachable by that clock. Integrate the characteristics backwards from $X(\eta_*, z) = z$, $P(\eta_*, z) = -Dd_*(z)$, for $z \in \Omega := \bar{B}_{R/2}(x_0)$. Up to a possible first exit from $\mathcal{D}$, one has $\partial_p \mathcal{H} \in \partial K$ and hence

$$|X - z| \le \tau V_{\mathrm{loc}} < R/4, \qquad |P + Dd_*(X)| \le \tau(M_p + SM_x) < s/2.$$

These bounds keep X in $B_{3R/4}$ and exclude a first exit from the tube.

The variational matrix $Y = D_z(X, P)$ starts at $(I, -D^2 d_*)^{\mathsf{T}}$, with norm at most B_0. Since the canonical flow and its state derivative depend on A, not on A', Grönwall gives $|D_z X - I| \le k := B_0(e^{L\tau} - 1) < 1/2$. Convexity of Ω yields $|X(\eta, z) - X(\eta, w)| \ge (1 - k)|z - w|$. Writing $X = z + e_\eta(z)$, the map $z \mapsto x - e_\eta(z)$ is a contraction of Ω into itself for every $x \in B_{R/4}$, because $\|e_\eta\|_\infty < R/4$. Hence the projected field covers $B_{R/4}$ injectively at every clock.

Define on the covered region $V(\eta, X(\eta, z)) = d_*(z) + \int_\eta^{\eta_*} \omega(1 - \beta(\dot{X}))\,\mathrm{d}s$. Variation of the canonical action, with terminal covector $-Dd_*$, gives $D_x V = -P$ and $V_\eta = \mathcal{H}$. For $v \in K$,

$$c_A(v) + V_\eta + D_x V \cdot v = \sigma_K(P + \omega\beta) - (P + \omega\beta) \cdot v \ge 0,$$

with equality at a unique velocity since $\chi > 0$. For an arbitrary competitor with junction $y = x(\eta_*)$, the prefix costs at least $V(\eta_0, x_0) - d_*(y)$ and the tail at least $d_*(y)$, including non-attained tails approximated by escape curves. The characteristic through x_0 followed by its attained frozen tail has total cost $V(\eta_0, x_0)$. In the equality case the prefix solves the unique spatially C^1 field and the tail is the unique frozen one. □

No smallness of the amplitude or of A' is required in (2); the duration bound depends on the frozen field and on the allowed interval $[a_*, A_{\max}]$. For example, a pulse $A = A_* - b\sin^2(\pi(\eta - \eta_0)/\tau)$ of fixed allowed depth b has $\|A'\|_\infty = \pi b/\tau \to \infty$.

Corollary XII.6 (perturbative equatorial selection)**.** *Let $0 < A_0 < \hat{E}$, $q_0 = \hat{E}^2/A_0^2$, and $0 < \delta < \pi$. Suppose that the frozen equatorial problem has a unique interior global minimizer γ_0, with non-degenerate free-arrival shooting map and cost $C_0 < \mathcal{B}_{A_0}(\delta)$. There exists $\varrho > 0$ such that $\|A - A_0\|_{C_b^1([\eta_0, \infty))} < \varrho$ gives a unique global equatorial minimizer, with shooting data close to those of γ_0. For $A_\epsilon = A_0 + \epsilon\alpha$, $\alpha \in C_b^1$,*

$$m_\epsilon = C_0 + \epsilon \int_{\eta_0}^{T_0} \frac{A_0 f(N_0 + 1)}{\hat{E}} \big(1 - \beta(\dot{x}_0)\big)\alpha(\eta)\,\mathrm{d}\eta + o(\epsilon).$$

The frozen hypotheses hold for $q_0 \ge 3/2$ and $0 < \delta < \delta_c$, including unequal radii and, whenever $\delta_{\tan} < \delta_c$, the tangent separation.

Proof. Choose $a_* < A_0$ close enough that $C_0 < \mathcal{B}_{a_*}(\delta)$; this is possible because the ratio of the frozen densities tends to one uniformly. Theorem 27, with a slowed frozen competitor, supplies minimizers in a common shell and a common finite clock window, and compactness brings every minimizer to γ_0 as $A \to A_0$. The clock equation gives $p_\eta = O(\|A'\|_\infty)$, so $\omega - p_\eta$ stays positive on the shell; the normal shooting data are therefore compact and approach the reference data. Non-degeneracy and the implicit-function argument of Theorem 22, restricted to equatorial competitors, identify a unique zero containing every global minimizer. At the frozen reference $p_\eta = 0$, and $z(A_\epsilon)$ is C^1 in ϵ as in Theorem 22. At fixed p the velocity maximizes $p \cdot v - c_A(v)$, with $c_A(v) = \omega_A(1 - \beta(v))$, on K, and the constraint $\beta(v) + \sqrt{N}|v|_{h_0} = 1$ is active; stationarity gives $p + \omega\beta = \mu\, D_v\big(\beta(v) + \sqrt{N}|v|_{h_0}\big)$, and contracting with v, by homogeneity and $\mathcal{H} = 0$ at the frozen reference, $\mu = p \cdot v + \omega\beta(v) = \omega$. The derivative of the reduced cost is $-\partial_A \mathcal{H}$ at fixed p: the direct term $2A_0 f(1 - \beta)/\hat{E}$ plus the constraint term $\omega\, \partial_A(\sqrt{N})\, |v|_{h_0} = A_0 f(N_0 - 1)(1 - \beta)/\hat{E}$, which gives the weight. The strict frozen barrier follows from the corner shortcut below δ_{ring} and from the Maxwell rule above it. □

As in Theorem 22, global C^1 closeness may be replaced by closeness on a common competitive window, provided global bounds and a uniform strict escape margin already supply that window.

Remark XII.7 (an example)**.** Let $M = 1$, $a = 1/2$, $\hat{E} = \sqrt{2}$, $r_0 = 6$, $\eta_0 = 0$, a target at radius 6 with $\varphi_f = \pm 1/50$, so $\delta = 1/50$, and $A = 1 - \frac{1}{2}\sin^2(\pi\eta/\tau)$ on $[0, \tau]$, $A = 1$ afterwards. Write $m_\pm$ for the two targets. The following are proved.

- *Early arrival is impossible for* $\tau \le 7/50$. Since $|\dot{r}|^2 \le \Delta^2/(r^2 g_{\varphi\varphi}) < 1$ and the largest angular speed is that of the circle, $T_+ \ge 4394/625$ on $r \ge 293/50$, so no curve sweeps $1/50$ by then.
- *All minimizers arrive early for* $1/5 \le \tau \le 1$. For $1/5 \le \tau \le 2/5$, every curve with $T \ge \tau$ costs at least $(5985/23296)\tau > \tau/4$, while the prograde and retrograde circles at $r = 6$, run with the actual clock, arrive before τ and cost less than $\tau/4$. For $2/5 \le \tau \le 1$, the respective bounds are $57\tau/256 \ge 57/640$ and $\sqrt{291}/200$, with strict gap $57/640 - \sqrt{291}/200 > 0$. Uniqueness holds for every τ (Theorem 30).
- *Continuity and asymmetry.* $m_\pm$ is continuous in $\tau > 0$; $m_+ < m_-$ for small τ and $m_- < m_+$ for large τ.

For $\tau \le 2/5$, the radial speed bound gives $r \ge 28/5$ on every crossing prefix; hence $f \ge 9/14$, $|\beta| < 1/25$ and $c_A \ge (315/728)A^2$, while $\int_0^\tau A^2 \,\mathrm{d}\eta = 19\tau/32$. The two circles last less than $9/50$ and $93/500$, and their cost rates divided by A^2 are below $1/2$ and $10/21$. With $F(\nu) = \int_0^\nu (1 - \frac{1}{2}\sin^2 \pi u)^2 \,\mathrm{d}u$ one has $F(1) = 19/32$ and $F(1) - F(1-h) > h - 10h^3/3$; taking $h = 1/10$ and $7/100$ gives $F(9/10) < 1/2$ and $F(93/100) < 21/40$, hence both circle costs are below $\tau/4$ for $\tau \ge 1/5$. For $\tau \le 1$, instead, $r \ge 5$, $f \ge 3/5$ and $|\beta| < 1/15$ give $c_A \ge (3/8)A^2$; the frozen circle length bounds both actual circle costs by $\sqrt{291}/200$, and both durations are below $2/5$.

For existence and uniform compactness, the circle gives $m \le U := \sqrt{291}/200$, whereas $\mathcal{B}_{1/2}(1/50) \ge 2R_{1/2}(6) \ge 4/\sqrt{97} > U$, by $r(r-2)/\Delta \ge 32/97$ on $[4, 6]$. A level $U < \bar{C} < \mathcal{B}_{1/2}$ thus gives a common shell and clock duration H for all competitive curves, independently of τ. As $\tau \to \tau_0 > 0$ the profiles converge in $C^1([0, H])$; compactness gives lower semicontinuity of the value, and lifting a fixed minimizing spatial curve with the nearby clock gives upper semicontinuity, preserving the target exactly. Hence $m_\pm$ is continuous.

The frozen reference at $A = 1$ is unique and non-degenerate: $\delta_{\mathrm{ring}} \ge 128/(291\sqrt{194})$, which exceeds $1/50$. In the example the frozen launch has non-zero radial momentum and β has only an angular component, so the canonical field stays uniformly regular for $\alpha \in [1/2, 1]$ in a tube around the frozen geodesic. For short windows, rescaling $u = \eta/\tau$ and using this field gives $m_\pm = C_0 - \tau \int_0^1 \mathcal{H}_{\alpha(u)}(x_0, p_\pm)\,\mathrm{d}u + O(\tau^2)$, where $\alpha(u) = 1 - \frac{1}{2}\sin^2 \pi u$. The reflected frozen costates have the same radial component and norm, $|p_\pm|^2 = N_1\omega_1^2$, and $p_\varphi = \pm L$. Put $B = p \cdot \beta^\sharp$ and $h = \mathcal{H}_\alpha$. For $\alpha < 1$, $N_\alpha\omega_\alpha^2 < N_1\omega_1^2$, the selected root is positive and

$$(N - |\beta|^2)h^2 + 2(N\omega + B)h + N\omega^2 - |p|^2 = 0, \qquad \partial_B h = -\frac{h}{(N - |\beta|^2)h + N\omega + B} < 0,$$

the denominator being positive on the selected root. Since $B_+ < B_-$, $m_+ < m_-$ for sufficiently small τ. For long windows, on the common competitive interval, $A_\tau(\eta) = 1 - \pi^2\eta^2/(2\tau^2) + O(\tau^{-4})$ in C^1, and the local-window form of Corollary XII.6 gives

$$m_\pm = C_0 - \frac{\pi^2}{2\tau^2} I_\pm + o(\tau^{-2}), \qquad I_\pm = \int_0^{C_0} (N + 1)\,\eta_\pm(\ell)^2 \,\mathrm{d}\ell,$$

with ℓ the frozen proper length along the reflected geodesics and $\eta_\pm(0) = 0$. Their clocks satisfy $\mathrm{d}\eta_\pm/\mathrm{d}\ell = 1/\omega \pm \beta_\varphi \varphi'_+(\ell)$, so $\eta_- > \eta_+$ for $\ell > 0$, $I_- > I_+$ and $m_- < m_+$ for sufficiently large τ. At least one equality of the two target costs follows by continuity; this is not a Maxwell tie in one problem.

Floating-point diagnostics, not certified: multi-start free-arrival shooting returned one accepted solution at each of twelve sampled values of $\tau \in [0.12, 0.20]$ in each sense. The reported arrival times lie on opposite sides of $T = \tau$ at $\tau = 0.160, 0.165$ (prograde) and at $\tau = 0.170, 0.175$ (retrograde). At every sampled parameter the reported prograde candidate has lower cost than the retrograde candidate. The proof of Theorem 30 shows that an exact zero of the shooting map in the rectangle Q_s of that proof is unique and coincides with the global minimizer, by existence and localization of all minimizers. The numerical shooting outputs and their apparent brackets remain uncertified diagnostics; no rigorous error bound is assigned to them here. A unique transversal sector change, with rational enclosures for the global minimizer, is proved analytically in Theorem 31.

D. Global uniqueness in the example

We keep the data of Remark XII.7: $M = 1$, $a = 1/2$, $\hat{E} = \sqrt{2}$, $x_0 = (6, 0)$, the target $x_f = (6, s/50)$ with $s = \pm 1$, and $A_\tau(\eta) = 1 - \frac{1}{2}\sin^2(\pi\eta/\tau)$ on $[0, \tau]$, $A_\tau = 1$ afterwards; A_τ is of class C^1 and takes values in $[1/2, 1]$. On the equator $\Delta = r^2 - 2r + \frac{1}{4}$, $h_r = r^2/(f\Delta)$, $h_\varphi = \Delta/f^2$, $\beta_\varphi = -1/(r-2)$ and $|\beta|^2_{h_0} = 1/(r^2\Delta)$. Write $p = (p_r, J)$ for the costate, $J = p_\varphi$, $j = |J|$, $g = A^2/(\sqrt{2}r)$, so that the angular component of $p + \omega\beta$ is $\xi = J - g$, and put

$$d = N - |\beta|^2_{h_0}, \quad a_H = \frac{1}{dh_r}, \quad b_H = \frac{N}{d^2 h_\varphi}, \quad k_H = \frac{\beta_\varphi}{dh_\varphi}, \quad S = \sqrt{a_H p_r^2 + b_H \xi^2}.$$

The support function in (32) gives the Hamiltonian $\mathcal{H} = -\omega + S - k_H\xi$, with $\dot{r} = a_H p_r/S$ and $\dot{\varphi} = b_H\xi/S - k_H$. All derivatives in r are taken at fixed (A, p_r, J). We use the shell $\mathcal{S} = \{23/4 \le r \le 25/4\}$; on it h_r and $|\beta|^2_{h_0}$ decrease, h_φ, f and N increase (with $\Delta < (r-1)^2$, $(\log h_r)' = 3/r - 1/(r-2) - 2(r-1)/\Delta < (6-4r)/(r(r-1)(r-2)) < 0$ and $(\log h_\varphi)' = 2(r-1)/\Delta - 4/(r(r-2)) > 2(r^2-4r+2)/(r(r-1)(r-2)) > 0$ for $r > 2+\sqrt{2}$), and $184/169 \le N \le 50/33$, $h_r \ge 15625/7293$, $184621/3600 \le h_\varphi \le 268125/4624$, $|\beta|^2_{h_0} \le 256/(23^2 \cdot 349)$. At $r = 6$ write $c_r(A) = \omega\sqrt{Nh_r}$ and $c_\varphi(A) = \omega\sqrt{Nh_\varphi}$; they increase with A, with $c_r^2 \in [36/1067, 72/97]$ and $c_\varphi^2 \in [291/352, 291/16]$ on $[1/2, 1]$, and $c_r/c_\varphi = \kappa_6$ with $\kappa_6^2 = 3456/84681$, $1/5 < \kappa_6 < 203/1000$.

Proposition XII.8 (localization)**.** *Let $\tau > 0$ and $s = \pm 1$. Every relaxed curve from x_0 to any lift of the target with cost at most $7/80$ satisfies $23/4 < r < 25/4$, $\operatorname{Var}\varphi < 7/64$ and $T < 35/44$; since $m_s \le \sqrt{291}/200 < 7/80$, every global minimizer ends at the principal lift $\varphi_f = s/50$. Every global minimizer is a normal rail extremal, with φ monotone, and*

$$\tfrac{3}{25} < T < \tfrac{21}{100}, \quad |r-6| < \tfrac{1}{50}, \quad |p_r| < \tfrac{63}{2000}, \quad \operatorname{sign} J = s,$$
$$j > \tfrac{8}{9}, \quad |p + \omega\beta|_{h_0^{-1}} > \tfrac{1}{10}, \quad |\psi(T) - s\tfrac{\pi}{2}| < \tfrac{1}{5}.$$

Here the representative of the terminal heading is chosen around $s\pi/2$ modulo 2π.

Structure of the proof. Leaving the shell and returning costs more than $7/80$, by the radial coefficient of the frozen length at $A = 1/2$, while the circle $r = 6$ run with the actual clock costs less; inside the shell the angular coefficient bounds the angle, and ω bounds the duration. A minimizer is a normal extremal; since its radius returns to 6, the radial costate vanishes somewhere, and a force bound on the shell keeps it small. The terminal ellipse given by transversality then bounds j from below, so the angular speed is bounded below and φ is monotone; this bounds T, and repeating the steps sharpens the bounds on p_r, j, r and the terminal heading. The constants are in section S10 of the supplementary material. □

Proposition XII.9 (late sector)**.** *For $\tau \le \tau_{\mathrm{no}} := 1425548384/9363983275 \approx 0.15224$ no relaxed curve reaches a lift of the target by the clock τ; every minimizer crosses $\eta = \tau$. For $\tau \ge 1/5$ every minimizer arrives before τ.*

Proof. For $\tau \le 1/5$, $|\dot r| < 1$ keeps $r \ge 29/5$ up to the clock τ, where the largest angular speed is the prograde circular one, $|\dot\varphi| \le 1/D_+$ with $D_+ = \sqrt{N}\sqrt{\Delta}/f - 1/(r-2)$. On $r \ge 29/5$, $\sqrt{\Delta}/f > 36/5$, $f \ge 19/29$ and $\sqrt{N} \ge 1 + (19/116)A^2$, so $D_+ > b_* + k_* A^2$ with $b_* = 659/95$, $k_* = 171/145$. The function $z \mapsto 1/(b_* + k_* z)$ is convex, hence below its chord on $[1/4, 1]$, and $\int_0^1 (1 - \frac{1}{2}\sin^2 \pi u)^2 \, \mathrm{d}u = 19/32$; therefore $\operatorname{Var} \varphi < \tau K \le 1/50$ with $K = 374559331/2851096768 = 1/(50\tau_{\mathrm{no}})$. The second statement is Remark XII.7 for $1/5 \le \tau \le 1$, and Proposition XII.8 ($T < 21/100$) for $\tau \ge 21/100$. □

Lemma XII.10 (a triangular criterion)**.** *Let $Q = [T_1, T_2] \times [-\theta_0, \theta_0]$ and $F = (F_1, F_2) \in C^1(Q, \mathbb{R}^2)$. Suppose that $\partial_\theta F_2 > 0$ on Q, that $F_2(T, -\theta_0) < 0 < F_2(T, \theta_0)$ for every T, and that $\det DF > 0$ on $\{F_2 = 0\}$. Then F has at most one zero in Q.*

Proof. For every T there is a unique $\theta^*(T) \in (-\theta_0, \theta_0)$ with $F_2(T, \theta^*(T)) = 0$, and θ^* is C^1 with $\theta^{*\prime} = -\partial_T F_2 / \partial_\theta F_2$. The zeros of F are the points $(T, \theta^*(T))$ at which $\Gamma(T) = F_1(T, \theta^*(T))$ vanishes, and $\Gamma' = \det DF / \partial_\theta F_2 > 0$. □

Theorem 30 (global uniqueness in the example)**.** *For every $\tau > 0$ and each sense $s = \pm 1$, the global minimizer of the example is unique, as a curve parametrized by the clock up to its arrival.*

Structure of the proof. For (T, θ) in a rectangle of arrival clocks and terminal headings, integrate the canonical system backwards from the target and let $F_s(T, \theta) = (-s\varphi(0), s(r(0) - 6))$; by Proposition XII.8 every minimizer is a zero. The whole family stays regular in the shell, and F_2 has opposite signs on the two sides $\theta = \mp 1/5$. On the returning shots $F_2 = 0$ the radial costate is small and has zero mean against a weight of bounded ratio. Explicit bounds on the coefficients of the variational system then give $\partial_\theta F_2 > 0$ on the rectangle and $\det DF_s > 0$ on $F_2 = 0$, and Lemma XII.10 allows at most one zero. The constants are in section S10 of the supplementary material. □

The uniqueness argument uses, in addition to the existence and localization premises, only that $A \in C^1$ takes values in $[1/2, 1]$ and that $T|(\log c_\varphi(A(T)))'| \le 46/7$ on $[3/25, 21/100]$; no bound on A' along the trip is needed. The constants used in the proof are obtained by rational comparisons. Floating-point checks, not certified, give $D_{2\theta}/T \ge 0.52$ on the sampled shots in Q_s, and $\Gamma' \ge 0.10$ on the separately sampled radial-return curves $F_2 = 0$. These sampled values are diagnostics, not uniform bounds used in the proof.

Corollary XII.11 (the sector change)**.** *The arrival time $T_s(\tau)$ of the minimizer is continuous in $\tau > 0$; $T_s(\tau) > \tau$ for $\tau \le \tau_{\mathrm{no}}$ and $T_s(\tau) < \tau$ for $\tau \ge 1/5$. In particular the minimizer changes sector at least once, at a clock τ with $T_s(\tau) = \tau$ in $(\tau_{\mathrm{no}}, 1/5)$.*

Proof. Fix $\tau_0 > 0$ and the unique zero (T_0, θ_0) of the shooting map. Write $A_\tau(\eta) = \alpha(\eta/\tau)$, where $\alpha(\nu) = 1 - \frac{1}{2}\sin^2(\pi\nu)$ for $0 \le \nu \le 1$ and $\alpha(\nu) = 1$ for $\nu \ge 1$. Since $\alpha'(1) = 0$, the profile is jointly C^1 in (η, τ) for $\tau > 0$, with $\partial_\tau A_\tau = -(\eta/\tau^2)\alpha'(\eta/\tau)$. The regular canonical flow and its terminal data therefore make F_s jointly C^1 in (T, θ, τ) near this zero. The zero is in the interior of Q_s by Proposition XII.8, and $\det D_{(T,\theta)} F_s > 0$ by the proof of Theorem 30. The implicit-function theorem gives a local C^1 branch of zeros. For each nearby τ, a global minimizer exists, lies in Q_s and is a zero; the same proof shows that there is at most one zero in Q_s. Thus the local branch is precisely the branch of global minimizers. In particular T_s is continuous at τ_0 (indeed C^1). The signs follow from Proposition XII.9, and the intermediate-value theorem gives a sector change in $(\tau_{\mathrm{no}}, 1/5)$. □

Theorem 31 (a single transversal sector change)**.** *For each sense s, $T_s \in C^1((0, \infty))$ and there is exactly one $\tau_s^* > 0$ with $T_s(\tau_s^*) = \tau_s^*$: the minimizer crosses $\eta = \tau$ for $\tau < \tau_s^*$ and arrives before τ for $\tau > \tau_s^*$. The change is transversal, $T_s'(\tau_s^*) - 1 < -49/71$, and*

$$\tfrac{159}{1000} < \tau_+^* < \tfrac{161}{1000} < \tfrac{169}{1000} < \tau_-^* < \tfrac{172}{1000}.$$

Structure of the proof. Corollary XII.11 gives $T_s \in C^1$ and at least one zero of $\Theta_s = T_s - \tau$ in $(\tau_{\mathrm{no}}, 1/5)$. At a zero $A(T) = 1$, which sharpens all bounds. Along the diagonal $T = \tau$ a dilation of the clock leaves the profile and the terminal datum unchanged, and gives the derivative $\Gamma_T + \Gamma_\tau$ of the angular miss on the returning shots, bounded below, while Γ_T is bounded above; hence $\Theta_s' < -49/71$ at every zero, which is therefore unique and transversal. At the zero $1/50 = \tau \int_0^1 s\dot\varphi \,\mathrm{d}\nu$; chord and Jensen bounds for the mean angular speed, concave in A^2, enclose $\tau_\pm^*$. The constants are in section S10 of the supplementary material. □

The proof uses the self-similar form $A_\tau = \alpha(\eta/\tau)$ with $\alpha(1) = 1$, $\alpha'(1) = 0$, and the mean $\int_0^1 \alpha^2 = 19/32$; it is not claimed for other families of profiles. Between τ_+^* and τ_-^* the prograde minimizer already arrives early while the retrograde one still crosses the pulse; this is not a statement about the order of the two costs. Floating-point root-finding (not certified) gives $\tau_+^* \approx 0.160195$ and $\tau_-^* \approx 0.170226$, with $T_s' - 1 \approx -1.10$ and -1.09, consistent with the theorem.

Not proved here: global uniqueness for general time-dependent profiles, the regime $1 < q < 3/2$, and the comparison with three-dimensional competitors, for which the clock of the paths near the poles enters the history of A.

XIII. THE PROPER-TIME BRANCH OF INGOING VAIDYA

For ingoing Vaidya, $g = -f\,\mathrm{d}v^2 + 2\,\mathrm{d}v\,\mathrm{d}r + r^2\mathrm{d}\Omega^2$ with $f = 1 - 2m(v)/r$, the mass enters the seed itself rather than a conformal factor, and the Kodama selector ∂_v is the rail selector. Let $\hat E > 1$, let $m \in C^1$ be positive, write $s = \mathrm{d}r/\mathrm{d}v$, $\zeta = \mathrm{d}\Omega/\mathrm{d}v$ and $w = \hat E^2 - f$, and work on $r > 2m(v)$. For constant mass m_0 the static change $v = t + r_*(r)$, $r_*' = 1/f$, gives $\mathrm{d}\tau = f\,\mathrm{d}t/\hat E$: the proper-time problem is that of section X with $A = 1$ and $M = m_0$, and we write $R(r)$ for the function of proposition X.2 with these values. Explicitly,

$$R(r) = \int_{2m_0}^{r} \frac{\mathrm{d}s}{\sqrt{\hat E^2 - 1 + 2m_0/s}}.$$

Lemma XIII.1 (rail, cost and velocity body)**.** *Along a rail curve* $\mathrm{d}\tau/\mathrm{d}v = (f - s)/\hat E$ *and* $r^2|\zeta|^2 = w - (s + w)^2/\hat E^2$*. The relaxed velocity body*

$$K(v, r) = \{(s, \zeta) :\ (s + w)^2/\hat E^2 + r^2|\zeta|^2 \le w\}$$

is a strictly convex ellipsoid whose boundary is the rail; it contains the origin in its interior exactly when $0 < f < \hat E^2$*, and on it* $f - s \ge \hat E(\hat E - \sqrt{w}) > 0$*. For every relaxed curve with radial end points* r_0*,* r_f*, the convexified cost is*

$$\int \frac{f(v, r(v))}{\hat E}\,\mathrm{d}v - \frac{r_f - r_0}{\hat E},$$

a potential $\omega = f/\hat E$ *with* $\partial_r\omega = 2m/(\hat E r^2) > 0$*, plus a boundary term. Every radial passage, inward or outward, costs at least* $|\mathrm{d}r|/\sqrt{\hat E^2 - f}$*.*

Proof. The rail condition gives $\hat E = f\dot v - \dot r$, and the normalization of u gives the ellipsoid; the cost identity is $\int (f - s)/\hat E\,\mathrm{d}v$ integrated by parts. The radial cost per unit radius, $(f - s)/(\hat E|s|)$, is monotone in $|s|$ in both directions and equals $1/\sqrt{w}$ at the extreme speeds $s = -w \pm \hat E\sqrt{w}$. □

The density $(f - s)/\hat E$ is affine in s, so on K it is the convexification of the rail cost, not the physical proper time of a slower curve. As in lemma XI.1, the cost depends only on

the radial profile, and the maximal angular speed $\sqrt{w-(s+w)^2/\hat{E}^2}/r$ of the rail with the same profile allows any angular length not smaller than that of a relaxed competitor. Every relaxed curve therefore has a rail curve of equal cost. In the following PMP formulas, p_r, p_Ω and p_v are the costates for the potential cost $\omega = f/\hat{E}$, after removing the fixed boundary term $-\Delta r/\hat{E}$; p_v is conjugate to the clock v. Set $S = (\hat{E}^2 p_r^2 + |p_\Omega|^2/r^2)^{1/2}$; the support function of K is $h = -wp_r + \sqrt{w}\,S$, positive for $p \neq 0$ because $0 \in \operatorname{int} K$, although the body is translated. As in proposition XI.2, this excludes the abnormal case for a relaxed minimizer attained with free final time inside the domain. For a normal extremal, on the set where the spatial costate vanishes, the radial adjoint equation gives $p_r' = \partial_r \omega > 0$ almost everywhere; since an absolutely continuous function has zero derivative almost everywhere on its zero set, that set has measure zero. At every other time strict convexity makes the maximizing velocity a boundary point of K, so the minimizer is a rail curve almost everywhere.

Theorem 32 (existence with an arrival deadline)**.** *Assume $m' \geq 0$ on $[v_0, v_1]$, $m(v_0) > 0$, $r_0 > 2m(v_0)$ and $r_f > 2m(v_1)$. For every angular separation, if the class of relaxed curves arriving at the target by v_1 is non-empty, the minimum exists and is attained by a rail curve. Every non-trivial minimizer is a rail curve almost everywhere.*

Proof. For $x = r - 2m(v)$ every relaxed curve has $\dot{x} \leq x/(4m(v_0)) - 2m'$, since the outward speed is at most $f/2$. Integrating backwards from the arrival, $x(v) \geq (r_f - 2m(v_1))\, e^{-(v_1-v_0)/(4m(v_0))} > 0$, and $r \leq r_0 + (v_1 - v_0)/2$. Speeds are bounded; extending each curve by rest at x_f after arrival, which is admissible and does not enter the cost, Arzelà–Ascoli gives a uniform limit. In local coordinates along the curve (or with S^2 embedded in $\mathbb{R}^3$), for every constant covector a one has $a \cdot y_n' \leq \sigma_{K(v,y_n)}(a)$; integrating against non-negative test functions and using the continuity of the support functions on the compact region, the limit satisfies $y' \in K(v, y)$ almost everywhere. The cost of lemma XIII.1 contains no derivative and passes to the limit. The radial reduction gives a rail minimizer. A minimizer with $T < v_1$ is a normal extremal and a rail almost everywhere. If $T = v_1$, apply the fixed-time PMP. In the normal case the same zero-set argument applies. In the abnormal case the spatial adjoint system is homogeneous, so its non-trivial solution cannot vanish; strict convexity again puts the maximizing velocity on ∂K almost everywhere. Normality of such minimizers is not claimed. □

With a finite deadline existence holds already at constant mass; it does not decide attainment for unrestricted arrival time, where for constant mass and $\delta > \delta_c$ the infimum is not attained.

Theorem 33 (global selection near constant mass)**.** *Let $m_0 > 0$, $r_0, r_f > 2m_0$, $0 < \delta < \pi$, and let the arrival time be unrestricted. Suppose that the frozen problem has a unique interior global minimizer γ_0 of cost $C_0 < R(r_0) + R(r_f)$ with non-degenerate planar shooting. There is $\varrho > 0$ such that every positive $m \in C^1$ with $\|m - m_0\|_{C^1_b([v_0,\infty))} < \varrho$ has a unique global proper-time minimizer among all regular three-dimensional rail curves, the continuation of γ_0. Its shooting data depend continuously on m, and $|p_{v,\mathrm{in}}| \leq C\|m'\|_{L^\infty([v_0, v_0+H])}$ for uniform C, H. For $m_\epsilon = m_0 + \epsilon\beta$, $\beta \in C^1_b$,*

$$\tau_{\min,\epsilon} = C_0 + \epsilon \int_{v_0}^{T_0} \Big[\frac{p_r}{r} - \frac{\hat{E}^2 + w}{\hat{E}\, r\, w}\Big]_0 \beta(v)\, \mathrm{d}v + o(\epsilon),$$

the bracket being evaluated along γ_0, where it is negative. No sign condition on m' is required.

Proof. Choose $C_0 < C^* < R(r_0) + R(r_f)$; first $r_1 > 2m_0$ below γ_0 such that the radial cost of descending to r_1 and returning, with weight $(\hat{E}^2 - 1 + 2m_0/r)^{-1/2}$, exceeds C^* with a margin; then $m_+ > m_0$ with $2m_+ < r_1$ for which the same holds with m_+. For small ϱ, $m \leq m_+$ on the whole future, so by lemma XIII.1 every competitor of cost at most C^*

stays in $r > r_1$, in a bounded shell, and has duration at most $H = (\hat{E}C^* + r_f - r_0)/f_{\min}$ with $f_{\min} = 1 - 2m_+/r_1$. The frozen minimizer slowed by a factor $\lambda < 1$ has velocities in the interior of the frozen body K_0 with a uniform margin, since $\lambda K_0 + (1-\lambda)B_c \subset K_0$ when K_0 contains the ball B_c of radius c about the origin; it is admissible for small ϱ, with cost tending to $(C_0 + \Delta r/\hat{E})/\lambda - \Delta r/\hat{E}$. Existence, convergence of all minimizers to γ_0, normality and planarity follow as in theorems 20 and 22, with Proposition XIII.4 for the plane and the short arc. Since h is positively homogeneous and $h \geq c|p|$, the condition $\mathcal{H} = p_v + h - \omega = 0$ gives $|p| \leq C(\omega + |p_v|)$; the clock costate solves $p_v' = -m' \, \partial_m(h - \omega)$, $p_v(T) = 0$, so Grönwall gives $\|p_v\|_\infty \leq C_H \|m'\|$, after which $h = \omega - p_v$ bounds $|p|$ below. The shooting map $(p_{r0}, L, T_v) \mapsto (r(T_v) - r_f, \varphi(T_v) - \delta, p_v(T_v))$, with $p_v(v_0)$ fixed by $\mathcal{H} = 0$, involves m and m' only. At constant mass it is non-degenerate: the launch variables $p_{r0} = (S_0/\hat{E})\cos\psi$, $L = S_0 r_0 \sin\psi$ give $\det \partial(p_{r0}, L)/\partial(\psi, p_{v,\mathrm{in}}) = r_0 S_0/(\hat{E} b(\psi)) > 0$ with $w_0 = \hat{E}^2 - f(v_0, r_0)$, $b = \sqrt{w_0} - (w_0/\hat{E})\cos\psi$ and $S_0 = (\omega_0 - p_{v,\mathrm{in}})/b(\psi)$, positive near the reference $p_{v,\mathrm{in}} = 0$; the costates of the v and t conventions are related by $p_t = p_v$, $L^{(t)} = L^{(v)}$, $p_r^{(t)} = p_r^{(v)} + p_v/f - 1/\hat{E}$, and the arrival clocks, measured from the same launch, by $T_v = T_t + r_*(r(T_t)) - r_*(r_0)$, with $\partial T_v/\partial T_t = 1 + (\mathrm{d}r/\mathrm{d}t)/f > 0$. Hence non-degeneracy is that of the frozen Schwarzschild problem. The preceding bounds separate the spatial costate from zero and make the initial shooting data of all minimizers compact. Every limiting extremal generates the unique frozen minimizer, and the shooting data converge to its data. Thus all minimizers lie in the same local shooting neighborhood for sufficiently small ϱ, and the uniform implicit function theorem identifies every global minimizer with the unique nearby zero. For $m_\epsilon = m_0 + \epsilon\beta$ the shooting map depends on ϵ only through the values m_ϵ and m_ϵ', which are affine in ϵ, and smoothly on them on the interior tube, so the shooting data are C^1 in ϵ without using β''. For the first variation, with the adjoint equation and the free final time, the derivative is $-\int \partial_\epsilon \mathcal{H} \, \mathrm{d}v$ plus $p_v(T_0) T_0' = 0$; at $\epsilon = 0$, $p_v = 0$ and $h = \omega$, which gives the bracket. If $p_r \geq 0$, $h \geq (-w + \hat{E}\sqrt{w})p_r$ gives $p_r \leq 1/\hat{E} + 1/\sqrt{w}$ and the bracket is at most $(\sqrt{w} - \hat{E})/(rw) < 0$; if $p_r < 0$ it is negative at once. □

Corollary XIII.2. *The frozen hypotheses of theorem 33 hold for $\hat{E}^2 \geq 3/2$, $r_0 \geq r_f > 2m_0$ and $0 < \delta < \delta_c$, with δ_c from theorem 17 or 19 for $A = 1$; the tangent separation is included whenever $\delta_{\tan} < \delta_c$.*

Proof. At constant mass m_0 the frozen problem is the proper-time problem of conformal Schwarzschild with $M = m_0$, $A = 1$ and $q = \hat{E}^2$, written in the coordinate v. Uniqueness and $C_0 < R(r_0) + R(r_f)$ are theorems 17 and 19; non-degeneracy is that of corollary XI.6, transferred to the v convention as in the proof of theorem 33. □

At fixed end points, launch clock and charge, a non-negative perturbation of the mass along the frozen minimizer therefore lowers the minimal proper time to first order, strictly if it is positive on a set of positive measure along the way; this is an infinitesimal statement, not a monotonicity for finite changes of mass. For a conformal factor the corresponding weight in theorem 22 is positive.

A. Unrestricted arrival time and the escape price

Let $\mathcal{A}(\delta)$ be the class of relaxed curves from (v_0, r_0) that stay in $r > 2m(v)$ and reach the target, at angular separation δ, at some finite time. Let $m^*(\delta)$ be the infimum of the cost of lemma XIII.1 over $\mathcal{A}(\delta)$ ($+\infty$ if the class is empty), and define the escape price with the apparent horizon $r = 2m(v)$, the marginally trapped tube of the ingoing metric[41–43],

$$B_V(\delta) = \liminf_{\epsilon \downarrow 0} \Big\{ C(\gamma) : \ \gamma \in \mathcal{A}(\delta), \ \min_v [r - 2m(v)] \leq \epsilon \Big\}.$$

By the radial reduction both quantities are the same over rail curves, since the profile, and hence the gap $r - 2m(v)$, is preserved.

Theorem 34 (existence below the escape price)**.** *Let $\hat{E} > 1$, let $m \in C^1([v_0, \infty))$ be positive and bounded, and let $r_0 > 2m(v_0)$, $r_f > 2\sup m$, excluding the trivial pair $r_0 = r_f$, $\delta = 0$. Then $m^*(\delta) \le B_V(\delta)$, and if $m^*(\delta) < B_V(\delta)$ the minimum is attained by a rail curve with finite arrival and positive gap, which is a normal free-time extremal. Equivalently, if no minimizer exists then $m^*(\delta) = B_V(\delta)$; the converse fails, since a minimizer may tie with the escape price, as at $\delta = \delta_c$ for constant mass. The value $m^*(\delta)$ is non-decreasing in δ.*

Proof. Let γ_n be minimizing. If their minimal gaps tended to zero along a subsequence, their costs would have $\liminf \ge B_V > m^*$; so the gaps are bounded below by some $\xi > 0$. Since $w < \hat{E}^2$, every radial passage costs at least $|\mathrm{d}r|/\hat{E}$, which bounds r by some $r_{\max}$; then $f \ge \xi/r_{\max}$, and $(T_n - v_0)\xi/r_{\max} \le \int f \,\mathrm{d}v = \hat{E}C_n + r_f - r_0$ bounds the arrival times. Extending each curve by rest at r_f, which is admissible because $r_f > 2\sup m$, the closure argument of theorem 32 gives a minimizer with gap at least $\min(\xi, r_f - 2\sup m)$; the radial reduction makes it a rail curve and the state constraint is inactive, so the free-time argument applies. Monotonicity in δ: the rail curve with the profile of a competitor at separation δ' has angular length at least δ', and for $\delta \le \delta'$ the excess is spent by a back-and-forth arc on a great circle, with the same profile, duration, cost and gap. □

For non-decreasing mass the price can be computed. Let m be non-decreasing with $m \to m_\infty$, $h_\infty = 2m_\infty < r_f$, and write $s_\pm = -w \pm \hat{E}\sqrt{w}$ for the extreme radial speeds; the radial projection of K is $[s_-, s_+]$, and the cost density is $c_- = \hat{E} + \sqrt{w}$ on a maximal fall and $c_+ = \hat{E} - \sqrt{w}$ on a maximal rise, both $1/\sqrt{w}$ per unit radius. Let b_T solve $b' = s_+(v, b)$ backwards from $b_T(T) = r_f$. Then $b_T(v)$ decreases in T to a limit $b_\infty(v) \ge 2m(v)$, which solves the same equation, is non-decreasing and tends to h_∞; a curve can reach r_f in finite time from (v, r) if and only if $r > b_\infty(v)$ (comparison for $r' \le s_+$, with the exponential barrier of theorem 32 keeping b_T above $2m$). On $\Omega = \{r > b_\infty(v)\}$ define the field cost

$$W(v, r) = \int_v^{v_c} \big(\hat{E} + \sqrt{w(t, \rho)}\big)\mathrm{d}t + \int_{v_c}^{\infty} \big(\hat{E} - \sqrt{w(t, b_\infty(t))}\big)\mathrm{d}t,$$

where ρ is the maximal fall $\rho' = s_-$ from (v, r) and $(v_c, \rho(v_c))$ its first contact with b_∞. The contact exists and is transversal, since $s_- \le -(\hat{E}^2 - 1)$ while $b'_\infty = s_+ \ge 0$; the second integral equals the radial integral of $1/\sqrt{w}$ along b_∞ and lies between 0 and $(h_\infty - b_\infty(v_c))/\sqrt{\hat{E}^2 - 1}$. Let $R_\infty(r) = \int_{h_\infty}^{r} \mathrm{d}s/\sqrt{\hat{E}^2 - 1 + 2m_\infty/s}$.

Lemma XIII.3 (calibration)**.** *W is C^1 on Ω, and with $\xi = \partial\rho/\partial r$ along the fall,*

$$\partial_r W - \frac{1}{\hat{E}} = \frac{1}{\hat{E}} \int_v^{v_c} \frac{2m(t)}{\rho^2}\, \xi \,\mathrm{d}t > 0.$$

Consequently $W_v + sW_r + (f - s)/\hat{E} \ge 0$ for every $s \in [s_-, s_+]$, and every relaxed curve in Ω costs at least $W(\mathit{start}) - W(\mathit{end})$ between two instants.

Proof. Regularity follows from the implicit function theorem at the transversal contact. Differentiating, with $\partial_r\sqrt{w} = -m/(r^2\sqrt{w})$, the derivative of the second integral at v_c is $-c_+$, so the contact contributes $(c_- + (-c_+))\,\partial_r v_c = 2\sqrt{w_c}\,\partial_r v_c$; from $\xi(v_c) + s_-\partial_r v_c = s_+\partial_r v_c$ this equals $\xi(v_c)/\hat{E}$. Hence $\partial_r W = \xi(v_c)/\hat{E} - \int m\xi/(\rho^2\sqrt{w})\,\mathrm{d}t$. Since $\partial_r s_- = (2m/r^2)(1 + \hat{E}/(2\sqrt{w}))$, $\xi(v_c) - 1 = \int (2m/\rho^2)(1 + \hat{E}/(2\sqrt{w}))\xi \,\mathrm{d}t$, and the terms in $\hat{E}/\sqrt{w}$ cancel. Along the fall $W_v + s_-W_r + c_- = 0$, and the density is affine in s, so $W_v + sW_r + (f - s)/\hat{E} = (s - s_-)(\partial_r W - 1/\hat{E}) \ge 0$; integrate along absolutely continuous curves. □

The sign does not use m'; for constant mass the identity reduces to $\partial_r W - 1/\hat{E} = 1/\sqrt{w} - 1/\hat{E}$.

Theorem 35 (escape price for Vaidya)**.** *Let $\hat{E} > 1$, let $m \in C^1$ be positive and non-decreasing with $m \to m_\infty$, $h_\infty = 2m_\infty < r_f$ and $r_0 > b_\infty(v_0)$. Then for every $\delta \in [0, \pi]$*

$$B_V(\delta) = W(v_0, r_0) + R_\infty(r_f) = D_c + \int_{v_c}^{\infty} \big(\hat{E} - \sqrt{w(v, b_\infty(v))}\big)\mathrm{d}v + R_\infty(r_f),$$

where D_c is the cost of the maximal fall from the launch point to its first contact with b_∞ at v_c. In words: fall as fast as possible to the return frontier, follow it to the final horizon, and climb out. For constant mass this is $R(r_0) + R(r_f)$.

Proof. ($\geq$) Every competitor stays in Ω until it arrives. Split it at a point (v_n, r_n) of gap at most $\epsilon_n \to 0$. Before it, lemma XIII.3 bounds the cost below by $W(v_0, r_0) - W(v_n, r_n)$; after it, the radial weight with $m \leq m_\infty$ gives at least $R_\infty(r_f) - R_\infty(r_n)$. Here $r_n > b_\infty(v_n) \geq 2m(v_n)$, and the backward barrier gives $b_\infty(v) - 2m(v) \geq 2\int_v^\infty e^{-(t-v)/(4m(v_0))} m'(t)\,\mathrm{d}t$; so $r_n \to h_\infty$ and $m(v_n) \to m_\infty$: if $v_n \to \infty$ directly, and if $v_n \to v_* < \infty$ because a vanishing gap forces $m' = 0$ after v_*, hence $m(v_*) = m_\infty$. Moreover $W(v_n, r_n) \leq [(r_n - b_\infty(v_n)) + (h_\infty - b_\infty(v_n))]/\sqrt{\hat{E}^2 - 1} \to 0$. ($\leq$) Fall maximally to $b_\infty + \eta$, then rise maximally; the curve stays above b_∞ and reaches r_f. In the radial parametrization its passage time through a radius z tends, as $\eta \to 0$, to that of b_∞ for $z < h_\infty$ and to infinity for $z > h_\infty$; the weight $1/\sqrt{w}$ is bounded by $1/\sqrt{\hat{E}^2 - 1}$, so by dominated convergence the radial cost tends to $W(v_0, r_0) + R_\infty(r_f)$. For the angle, fix $R_\epsilon = h_\infty + \epsilon < r_f$ and choose $\eta = \eta(\epsilon)$ so small that R_ϵ is first reached on the way up at a time V with $m_\infty - m(V) \leq \epsilon/2$; afterwards $\epsilon/R_\epsilon \leq f(v, R_\epsilon) \leq 2\epsilon/R_\epsilon$, an arc of angle δ at radius R_ϵ costs $O(\delta\sqrt{\epsilon})$, the final rise costs at most $\int_{R_\epsilon}^{r_f} \mathrm{d}z/\sqrt{\hat{E}^2 - 1 + 2(m_\infty - \epsilon/2)/z}$, and the apparent gap at R_ϵ is at most 2ϵ. The choice $\eta = \epsilon$ would not do: the level could be reached too early. □

The price depends on the history of the mass only through the fall and the frontier. The field describes the limit that computes the price; it does not assert a unique structure of all minimizing sequences. As an illustration (numerical, not certified), for $m = 1 + \frac{1}{2}\tanh(v/5)$, $\hat{E} = \sqrt{2}$, $r_0 = r_f = 6$, $v_0 = 0$, the frontier starts at $b_\infty(0) \approx 2.6014$, strictly outside the apparent horizon, and $B_V \approx 5.21295$, whereas the cruder bound $R_\infty(r_0) + R_\infty(r_f) \approx 4.62260$, which weighs the whole descent with the final mass, is lower. The continued interior extremal is an explicit rail curve below B_V up to $\delta \approx 0.522\pi$, so theorem 34 gives attained minima there; for constant mass $m = 1$ the same criterion reproduces the certified δ_c. Beyond the crossing it remains open whether another minimum lies below B_V, a minimum ties with it, or the infimum is not attained. Finite variations of the mass are treated in section XIII B; for this profile the alternatives beyond the displayed crossing remain open. The arrival branch is treated separately by Theorems 5 and 8.

B. Finite variations of the mass

Theorem 33 selects the minimizer only for masses close to a constant. For finite accretion, the question is decided, as for the conformal factor, by comparing the cost of the interior candidates with the cost of escaping: late accretion leaves the frozen minimizer in place (Theorem 36), early accretion beyond the frozen critical separation leaves no minimizer (Theorem 37), and in between there is a last contact with the escape price (Theorem 38). Throughout, $\hat{E} > 1$, $m \in C^1$ is non-decreasing with $0 < m_0 \leq m \leq m_1$, $r_0, r_f > 2m_1$ and $0 < \delta < \pi$; $V_m(\delta)$ denotes the infimum and B_m the escape price of section XIII A.

Proposition XIII.4 (budget and plane, Vaidya)**.** *Let $0 < \delta < \pi$. Every global proper-time minimizer attained at a finite time saturates its angular budget*

$$\int r^{-1}\sqrt{w - (s + w)^2/\hat{E}^2}\,\mathrm{d}v = \delta,$$

runs along the short great-circle arc between the end directions, and has a non-zero constant angular momentum J.

Proof. As for Proposition XI.3: the cost depends only on the radial profile, and a velocity with the same s and strictly smaller angular speed lies in the interior of the translated body K. A surplus of budget would give a minimizer that is not a rail on a set of positive measure. □

Lemma XIII.5 (the clock costate under accretion)**.** *If $m' \geq 0$, then $0 \leq p_v < \omega$ along every minimizer.*

Proof. $p_v' = -m'\,\partial_m(h-\omega)$ with $\partial_m(h-\omega) = (2/r)B$, $B = -p_r/2 + h/(2w) + 1/\hat{E}$, $x = \sqrt{w} \in (0, \hat{E})$. Where $p_v \geq 0$, $0 < h \leq \omega$ ($h > 0$ because $p \neq 0$, since $J \neq 0$ by Proposition XIII.4). If $p_r \leq 0$, $B > 0$. If $p_r > 0$, $h \geq x(\hat{E}-x)p_r$ gives $p_r \leq (\hat{E}+x)/(\hat{E}x)$ and $B \geq 1/\hat{E}$ for $2x \leq \hat{E}$, $B \geq (\hat{E}-x)/(2x^2) > 0$ otherwise. Hence $p_v' \leq 0$ wherever $p_v \geq 0$, and $p_v(T) = 0$ keeps $p_v \geq 0$ backwards (Grönwall for the negative part). □

Lemma XIII.6 (turning points under accretion)**.** *At a point with $\dot{r} = 0$ along a minimizer, $\ddot{r} = h_{pp}\,\dot{p}_r - m'/r$, where*

$$h_{pp}\dot{p}_r = \frac{(r-2m)P(p_v)}{\hat{E}^2 r^4\,[r(1-\hat{E}p_v) - 2m]} > 0,$$

$P(p_v) = 8m^3 + 2m^2(\hat{E}^2 - 6 + 4\hat{E}p_v)r + 3(\hat{E}^2-2)m(\hat{E}p_v - 1)r^2 - (\hat{E}^2-1)(\hat{E}p_v-1)r^3$. If $m' < r\,h_{pp}\dot{p}_r$ at every turning point, the minimizer has at most one periastron and no apoastron.

Proof. $\dot{r} = 0$ exactly at $p_r = \sqrt{w}J/(\hat{E}r\sqrt{f})$, where $\partial_m \dot{r} = -1/r$; the constraint $h = \omega - p_v$ fixes J. P is affine in p_v, with $P(0)/m^3 = au^3 + 3(a+1)u^2 + 2(a+1)u > 0$, $a = \hat{E}^2 - 1$, $u = (r-2m)/m$, and $P(\omega) = 2m(r-2m)[(\hat{E}^2-1)r + 2m] > 0$. By Lemma XIII.5, $0 \leq p_v < \omega$, so $P(p_v) > 0$ and $r(1-\hat{E}p_v) - 2m > r(1-\hat{E}\omega) - 2m = 0$. Under the condition on m' every turning point is a strict local minimum of r, so there is at most one, a periastron. □

At $p_v = 0$ the turning threshold tends to zero at the horizon; $m' \geq 0$ alone does not enforce the inequality. For the cubic step from 1 to 3/2 of duration 1/2, at its midpoint ($m = 5/4$, $m' = 3/2$), the equations admit a local apoastron with $r = 3$, $p_v = \omega/2$ and $\ddot{r} = -23/54$. Theorem 36 does not use this lemma.

Theorem 36 (late accretion does not matter)**.** *Let $\hat{E} > 1$, $0 < m_0 \leq m_1$, $r_0, r_f > 2m_1$ and $0 < \delta < \pi$. Suppose that the frozen problem at mass m_0 has a unique global minimizer γ_0, of cost C_0 and duration T_0. With $w(r,m) = \hat{E}^2 - 1 + 2m/r$ put*

$$\Lambda(s) = \int_s^{r_0} \frac{\mathrm{d}r}{\sqrt{w(r,m_0)}} + \int_s^{r_f} \frac{\mathrm{d}r}{\sqrt{w(r,m_1)}}, \qquad 2m_0 < s < \min(r_0, r_f),$$

and suppose $\Lambda(s_*) > C_0$ for some s_*. Then for every non-decreasing $m \in C^1$ with $m = m_0$ on $[v_0, v_0 + d]$ and $m \leq m_1$, and every

$$d \;\geq\; D = \max\Big\{T_0,\; \frac{\hat{E}C_0 + r_f - r_0}{1 - 2m_0/s_*}\Big\},$$

γ_0 is the unique proper-time global minimizer among all three-dimensional rail curves.

Proof. By Lemma XIII.1 the cost of a relaxed curve is $\int f/\hat{E}\,\mathrm{d}v - (r_f - r_0)/\hat{E}$, and along it $(f-s)/\hat{E} \geq |s|/\sqrt{w}$, where $1/\sqrt{w}$ decreases in m. A curve arriving by $v_0 + d$ lives at mass m_0, so it costs at least C_0, with equality only for γ_0, which arrives at $v_0 + T_0 \leq v_0 + d$. Let

a curve arrive later, and let s_d be its least radius on $[v_0, v_0 + d]$. If $s_d \geq s_*$, the part on $[v_0, v_0 + d]$ alone gives a cost of at least $d(1 - 2m_0/s_*)/\hat{E} - (r_f - r_0)/\hat{E}$, since $f > 0$, and this exceeds C_0 for $d > D$; at $d = D$ the inequality is still strict, because $r(v_0) = r_0 > s_*$. If $s_d < s_*$, the curve descends from r_0 to s_* at mass m_0, at a cost of at least $\int_{s_*}^{r_0} \mathrm{d}r/\sqrt{w(r, m_0)}$, and later climbs from s_* to r_f with $m \leq m_1$, at a cost of at least $\int_{s_*}^{r_f} \mathrm{d}r/\sqrt{w(r, m_1)}$: in total at least $\Lambda(s_*) > C_0$. Hence γ_0 is the unique global minimizer. □

The condition $\Lambda(s_*) > C_0$ compares C_0 with a lower bound for the cost of descending near the initial horizon and subsequently returning to the target; Λ is a comparison weight, not the escape price of Theorem 35, which contains the return frontier.

Theorem 37 (early accretion beyond the critical separation)**.** *Suppose that the frozen problem at mass m_1 has no minimizer at the separation δ. Then there is $h_* > 0$ such that every m as above with $m = m_1$ on $[v_0 + h, \infty)$, $0 \leq h \leq h_*$, has $V_m(\delta) = B_m$, and no rail curve attains it. No bound on m' is assumed.*

Proof. Idea. A minimizer for a vanishing accretion layer either keeps its angular momentum away from zero, and then converges to a minimizer of the frozen final problem, which does not exist; or loses it, and then it cannot sweep the angle δ.

Put $v_0 = 0$ and $a = \hat{E}^2 - 1$. Every minimizer costs at most $B_m \leq \bar{C} = (r_0 + r_f - 4m_0)/\sqrt{a}$, by the three legs of Theorem 35; hence $2m_0 < r \leq r_0 + \hat{E}\bar{C}$. The rail speeds obey $|\dot{r}| < 2\hat{E}^2$ and $|\dot{\varphi}| \leq \hat{E}/r$. For h small the full fall does not meet the frontier during $[0, h]$, the frontier is $2m_1$ after h, and the formula of Theorem 35 reads

$$B_m = \int_0^h \big(\hat{E} + \sqrt{w(v, r_{\mathrm{fall}}(v))}\big)\mathrm{d}v + R_1(r_{\mathrm{fall}}(h)) + R_1(r_f),$$

with R_1 the radial function at mass m_1; since $|r_{\mathrm{fall}}(h) - r_0| \leq 2\hat{E}^2 h$, $B_m \to B_1 = R_1(r_0) + R_1(r_f)$ uniformly over all admissible m. At mass m_1 a frozen segment with Clairaut constant J has at most two monotone radial legs, and $|\mathrm{d}\varphi| = J w_{\mathrm{warp}}\,\mathrm{d}\rho/(\rho\sqrt{\rho^2 - J^2})$ with $w_{\mathrm{warp}} = \mathrm{d}R/\mathrm{d}\rho < b(r) = r\sqrt{f}/(r - m_1)$, and $b \to 0$ at the horizon; splitting at a radius r_s gives $\Delta\varphi \leq \pi b(r_s) + 2\arcsin(J/\rho(r_s))$. Fixing r_s close to the horizon and then letting $J \to 0$ makes the angle as small as desired, uniformly and without equal radial end points.

Suppose $h_n \to 0$ and minimizers γ_n with costs $C_n \leq B_{m_n}$. The layer alone cannot reach the target, so $T_n > h_n$. If $J_n \to 0$ along a subsequence, the angle of the layer, at most $\hat{E}h_n/(r_0 - 2\hat{E}^2 h_n)$, and that of the frozen tail both tend to zero: a contradiction. Otherwise $J_n \geq j_0 > 0$, the tail stays in $\rho \geq j_0$, hence at a uniform distance from $2m_1$ and from the frontier, and $\int f\,\mathrm{d}v = \hat{E}C_n + r_f - r_0$ bounds the durations. By Arzelà–Ascoli and the closure of the velocity constraint through support functions, a subsequence converges to a relaxed curve of the frozen problem at m_1 with cost at most B_1, the frozen infimum: a frozen minimizer, which does not exist. The argument uses only $C_n \leq B_{m_n}$, so it also excludes ties with the escape price. Theorem 34 gives $V_m = B_m$. □

Lemma XIII.7 (compactness of minimizers)**.** *Let $m_d(v) = \mu(v - v_0 - d)$ with $\mu \in C^1$ non-decreasing, $\mu = m_0$ for $v \leq 0$ and $\mu = m_1$ for $v \geq \epsilon$, and let $0 \leq d \leq D_0$. There is $J_* > 0$ such that every minimizer has $J \geq J_*$; all minimizers stay at a uniform distance from the return frontier, have uniformly bounded duration, and form a compact family of clocked curves.*

Proof. On the region $2m_0 \leq r \leq r_0 + \hat{E}\bar{C}$, $h_r = (2m/r^2)p_r - mS/(r^2\sqrt{w}) - \sqrt{w}J^2/(r^3 S)$ is bounded by $K_p|p_r| + K_J J$, with no m', while $\omega_r \geq \kappa > 0$. For small J, in the strip $|p_r| \leq \sqrt{J}$ one has $\dot{p}_r = \omega_r - h_r \geq \kappa/2$, so the costate crosses the strip at most once, in a clock $O(\sqrt{J})$, and outside it $\dot{\varphi} \leq \sqrt{J}/r^2$. Up to the clock $H = D_0 + \epsilon$ the angle is $O(\sqrt{J})$, and after H the frozen tail sweeps an angle that tends to zero with J, as in the proof of Theorem 37; so J is bounded below. The zero of p_r is not a periastron in Vaidya; the strip only controls the angular speed. Put $\xi_0 = \min\{\rho^{-1}(J_*) - 2m_1,\, r_f - 2m_1\} > 0$. If $T > H$,

the frozen tail gives $r(H) - b(H) \geq \xi_0$; if $T \leq H$, the target gives $r_f - b(T) \geq \xi_0$. Since $\dot{r} \leq s_+(v, r)$ and $b' = s_+(v, b)$, the gap $g = r - b$ obeys $g' \leq L_s g$, with L_s a bound of $|\partial_r s_+|$, so $g \geq \xi_0 e^{-L_s H}$ up to $\min(T, H)$; after H the tail keeps its frozen gap. Costs then bound the durations, and Arzelà–Ascoli with the closure of the constraint through support functions, as in Theorem 32, gives compactness. □

Theorem 38 (the last contact)**.** *In the setting of Lemma XIII.7, assume the hypotheses of Theorem 37 with $0 < \epsilon \leq h_*$, and those of Theorem 36 with threshold D. Then V_d and B_d are continuous in d, the set $\{d \geq 0 : V_d = B_d\}$ is non-empty and compact, and its largest element d_* satisfies $h_* - \epsilon < d_* < D$. For $d > d_*$ a minimizer exists, for $d \geq D$ it is the unique frozen minimizer γ_0, and at d_* at least one interior minimizer ties with the escape price.*

Proof. The fall and the frontier solve scalar equations whose coefficients depend continuously on d, and the contact of the fall with the frontier is transversal, $s_- - s_+ = -2\hat{E}\sqrt{w} < 0$; so B_d is continuous. For V_d no nesting of the velocity bodies is assumed: they are not ordered in general as the mass varies. Set $v_0 = 0$. On a compact neighborhood of a finite ϵ_0-optimal competitor of duration T, let $B_c \subset K$ uniformly, and let L_K be a Lipschitz constant of the support functions in the clock and in d, per unit covector. Choose $k \geq \max\{1, 4L_K/c\}$ and $\beta > 0$ with $\beta e^{kT} \leq 1$ and $L_K|d - d_0| \leq c\beta/4$, and follow the same spatial curve at the clock $t(s) = s + (\beta/k)(e^{ks} - 1)$. The factor $1/t'(s)$ leaves a margin $c\beta e^{ks}/2$ inside the body, while $L_K|t(s) - s| \leq c\beta e^{ks}/4$ and the change of parameter costs at most another quarter; so the retimed curve is admissible for d, with the same end points. The radial reduction turns it into a rail of equal cost, and letting $\beta \to 0$ as $d \to d_0$ gives $\limsup_{d \to d_0} V_d \leq V_{d_0}$. Lower semicontinuity follows from Lemma XIII.7 when minimizers exist, and from $V_d = B_d \to B_{d_0} \geq V_{d_0}$ otherwise. Hence the set $\{V_d = B_d\}$ is closed; it contains $[0, h_* - \epsilon]$, where no minimizer exists, by Theorem 37. For $d \geq D$, the proof of Theorem 36 gives $B_d > C_0 = V_d$ with a margin: a curve whose prefix reaches below s_* pays at least $\Lambda(s_*) - C_0$ more than C_0, and one staying above s_* on $[v_0, v_0 + d]$ pays a fixed surplus already in an initial interval of length $\min\{d, (r_0 - s_*)/(4\hat{E}^2)\}$; the sequences computing the escape price arrive after $v_0 + d$, since curves with that deadline keep a uniform gap by Theorem 32. Thus the equality set is a non-empty compact subset of $[0, D)$; write d_* for its maximum. For $d > d_*$, $V_d < B_d$ and Theorem 34 gives a minimizer. Choosing $d_n \downarrow d_*$, Lemma XIII.7 gives a subsequence of minimizers converging to a minimizer at d_*, of cost B_{d_*}. Since no minimizer exists for $d \leq h_* - \epsilon$, $h_* - \epsilon < d_* < D$. □

Proposition XIII.8 (structure in the separation)**.** *For a fixed profile with bounded mass, $r_0, r_f > 2 \sup m$ and $\delta \in [0, \pi]$, $V(\delta)$ is non-decreasing and Lipschitz, $|V(\delta_2) - V(\delta_1)| \leq r_f|\delta_2 - \delta_1|/\sqrt{\hat{E}^2 - 1}$, and $V(\delta_1) < V(\delta_2)$ for $\delta_1 < \delta_2$ whenever a minimizer exists at δ_2. Under the hypotheses of Theorem 35, the escape price $B = B_V$ does not depend on δ and $V \leq B$; if $\{\delta : V(\delta) = B\}$ is non-empty, it is an interval $[\delta_c, \pi]$, and no minimizer exists beyond δ_c.*

Proof. The radial reduction gives monotonicity; a short rail arc at r_f, of cost per radian $r_f\sqrt{f}/\sqrt{w} \leq r_f/\sqrt{\hat{E}^2 - 1}$, gives the Lipschitz bound. Equality $V(\delta_1) = V(\delta_2)$ with a minimizer at δ_2 would, by compressing its angular arc, give a relaxed minimizer at δ_1 interior to K on a set of positive measure, which is not a rail; at $\delta_2 = \pi$ the same argument uses a chosen half great circle. The last statement follows by continuity. □

The set $\{V = B\}$ may be empty for a given profile; monotonicity alone does not produce a critical separation.

Remark XIII.9 (an example with three phases)**.** Let $\hat{E} = \sqrt{2}$, $r_0 = r_f = 6$, $\delta = \pi/2$, $m_0 = 1$, $m_1 = 3/2$, and let μ be the cubic step of duration ϵ. At the constant mass $3/2$ one has $r_0 = 4m_1$ and $\delta_c < 39/25 < \pi/2$, so the frozen final problem has no minimizer. With the constants of section S11 of the supplementary material, Theorem 37 holds with $h_* = 1/1000$ for every profile, Theorem 36 holds with $D = 91$ for every shape of the accretion, and for

$\epsilon = 1/2000$ Theorem 38 gives $1/2000 < d_* < 91$: no minimizer for $d \le 1/2000$, a last contact with the escape price, and the frozen minimizer at mass 1 for $d \ge 91$. Numerically, for a step of duration $1/2$ the continued shallow extremal crosses the escape price near $d \approx 0.6$ and coincides with the frozen minimizer for $d \ge 14$, while a second, deep family of extremals stays above the escape price; these numbers are not certified.

Here the obstruction to attainment is escape to the return frontier; no assertion about conjugate points is needed or made.

Not proved: that for intermediate d the minimizer is the continued shallow extremal, the uniqueness of the contact, and monotonicity of $V_d - B_d$. In particular, between the last contact and the threshold D the existence of a global minimizer is proved, but not its identification with a specific extremal.

XIV. CONCLUSIONS

Global minimality of controlled-rail brachistochrones has been settled, in explicit regimes, for the two costs that matter physically, and the two costs turned out to be governed by different mechanisms.

a. Arrival time: the cut point. The rails are the null curves of the auxiliary metric g_{opt}, so first arrival is a Fermat problem, and wherever g_{opt} is globally hyperbolic an extremal is a global minimum among all three-dimensional competitors exactly up to its null cut point. For conformal Schwarzschild and for Vaidya with non-decreasing mass the cut point is at swept angle π when $q \ge 3/2$, with no assumption on how fast the factor or the mass changes; below $3/2$ the rule holds within an explicit radius, and for constant data it genuinely fails far out, where the Fermat geometry acquires a repulsive first-order mass. The thresholds come from sign conditions on curvature along null rays. For conformal Schwarzschild the difference of the two screen modes is independent of time derivatives of the factor; for Vaidya the mass derivative enters with a favorable sign when the mass does not decrease; on the Kerr equator, curvature certificates give the constant-data result and explicit bounds on the time derivatives of the factor in the variable-data bands. For Vaidya below $3/2$ one further ingredient is needed, a bound on the accretion rate, because a moving velocity body can turn an outgoing ray back.

b. Proper time: escape. The proper-time cost is a potential integrated in conformal time under a speed limit, and the boundary of the regular domain lies at finite cost. Minimality is lost when escaping toward it becomes cheaper than every interior candidate. For constant data this happens beyond a critical separation, and below it the minimizer is selected by a convex-envelope rule, which can produce ties between interior candidates. For time-dependent data the escape price is explicit, the selection persists near constant data, and finite variations are decided by comparing the frozen minimum with the cost of escaping: a late drop of the conformal factor destroys the minimum under the cost-gap and transition-duration conditions of Theorem 25, late accretion leaves the frozen minimizer in place, early accretion beyond the frozen critical separation prevents a minimizer, and in between there is a last parameter at which an interior minimizer ties with the escape price. Both mechanisms rest on two elementary facts, that waiting above a radius costs in proportion to the time spent and that every radial passage costs a fixed weight per unit radius.

c. Methods. The arrival branch uses the causal theory of globally hyperbolic spacetimes, a Sturm comparison of the screen modes and a count of light rays in the universal cover of a plane; the proper-time branch uses relaxation and radial reduction, the sign of the clock costate, propagated backwards from the arrival, barriers and calibrations for the escape price, and compactness for the contact parameters. Where specific numbers are needed, the statements are reduced to finitely many explicit inequalities, certified in exact or interval arithmetic with independent verifiers; closed forms are used wherever they exist.

d. Open problems.

- Conformal Kerr in three dimensions at finite spin, beyond the excursion barrier, including the selection at the antipodal tie, which at small spin is decided by the sign of a second-order quantity; and the state-constrained problem near the stationary limit.

- On the equator: a time-dependent factor with $2 \le q < 5/2$ or with spin above the stated band, and the physical reading of the rail inside the ergoregion.

- The regime outside the uniform bound $\sup A < \hat{E}$, including inward-moving freezing fronts, which shrink the regular domain; there global hyperbolicity has to be established separately.

- The optimal radius below $q = 3/2$, and whether the accretion-rate bound for Vaidya below $3/2$ can be removed under the same radius restriction.

- On the proper-time branch: the identification of the minimizer with a specific extremal between the regimes covered by the theorems; ties between interior candidates for time-dependent data; the three-dimensional problem with spin.

- For Vaidya, decreasing masses beyond the stated bounds. In the ingoing form they violate the null energy condition; a radiating star is described by outgoing Vaidya[28,44].

e. Outlook. The regimes treated here are relativistic: energies with q bounded away from 1, or radii within a few gravitational radii, and the surface $r = 2M$ of the seed, or an apparent horizon, as the boundary. A body traveling in the solar system has $q - 1$ of order 10^{-9}, moves at radii of order 10^8–10^9 gravitational radii, and meets a stellar surface instead; its global problem is a weak-field one, with a state constraint at the surface and a slow mass loss. A central mass in an expanding universe in McVittie's form[45], which is not a conformal rescaling[46], is a second natural continuation.

SUPPLEMENTARY MATERIAL

The supplementary material contains the alternative convex-front argument, the descriptions of the computer-assisted certificates, and the detailed proofs and constants of the examples.

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AUTHOR DECLARATIONS

Conflict of Interest

The author has no conflicts to disclose.

Author Contributions

Iman Rosignoli: Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Software; Validation; Visualization; Writing – original draft; Writing – review & editing.

DATA AVAILABILITY

The certificate files, the independent verifier with its tampering tests, the interval-arithmetic and Riemann-sum scripts and the symbolic verification scripts supporting the findings of this study will be openly available, with the preprint, in the folder `paper3` of the repository NonStationaryBrachistochrone2, `https://github.com/ImanetorX98/NonStationaryBrachistochrone2`, archived on Zenodo[52]. Each computer-assisted step of the paper is listed in its file `RESULTS_MAP.md` with the script that checks it and the log of our run.

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